A Conditional Direct-Sum Theorem for Private Learning of Cartesian VC-One Products

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1 Theoretical Setup

All logarithms denoted by log are natural. For $t \geq 1$, define

$$\log^* t = \min\{r \in \mathbb{N}_0 : \log_2^{(r)}(t) \leq 1\},$$

where $\log_2^{(0)}(t) = t$ and $\log_2^{(r)}$ is the r-fold iterate of the base-two logarithm for $r \geq 1$.

Let (X, Σ) be a measurable instance space and $C \subseteq \{0,1\}^X$ a binary concept class. A partition $\mathcal{P}$ of X is *Cartesian for C* if, after writing $C_B = \{c|_B : c \in C\}$, the restriction map

$$c \longmapsto (c|_B)_{B \in \mathcal{P}}$$

is a bijection from C onto $\prod_{B \in \mathcal{P}} C_B$. A Cartesian partition is finest if it refines every Cartesian partition; such a partition is canonical up to relabeling.

Assumption 1 (Canonical disjoint Cartesian factorization)**.** *The class C admits a finite finest Cartesian partition*

$$X = \bigsqcup_{i=1}^{k} X_i, \qquad 1 \leq k < \infty,$$

of the whole domain, and the restriction map identifies C with the full product

$$C = \prod_{i=1}^{k} C_i, \qquad C_i = \{c|_{X_i} : c \in C\}.$$

This is a static property of C, independent of any learner, dataset, distribution, event, or generated object.

Assumption 2 (Nonconstant finite-Littlestone VC-one factors)**.** *Every C_i is nonconstant, has* $\mathrm{VC}(C_i) = 1$*, and has finite*

$$d_i = \mathrm{LD}(C_i).$$

No ordering, finite-domain, finite-cardinality, properness, or computational assumption is imposed.

Under Assumptions 1 and 2, define

$$s_i = 1 + \log^*(d_i + 1), \qquad M_\oplus(C) = \sum_{i=1}^{k} s_i, \qquad \omega_i = \frac{s_i}{M_\oplus(C)}. \tag{1.1}$$

For a candidate sample size $n \in \mathbb{N} = \{1, 2, \ldots\}$, define the factor budget

$$m_{n,i} = \max\{8, \lceil 4n\omega_i \rceil\}. \tag{1.2}$$

For $x, x' \in X_i$, write

$$x \equiv_i x' \iff c_i(x) = c_i(x') \text{ for every } c_i \in C_i.$$

Let $Q_i = X_i/{\equiv_i}$ and let $\kappa_i : X_i \to Q_i$ be the quotient map. Give Q_i its discrete sigma-field 2^{Q_i}, and let

$$\Sigma_i = \{A \cap X_i : A \in \Sigma\}.$$

Assumption 3 (Finite-or-countable measurable evaluation quotient)**.** *Every X_i belongs to Σ, every evaluation quotient Q_i is finite or countable, and every quotient cell $\kappa_i^{-1}(\{q\})$, $q \in Q_i$, belongs to Σ_i. Equivalently, $\kappa_i : (X_i, \Sigma_i) \to (Q_i, 2^{Q_i})$ is measurable. This is a primitive static condition on the evaluation structure. It does not assume any data-dependent version space, core, depth, selector, learner, or output kernel to exist or be measurable, and it does not require X_i to be finite, countable, countably generated, or standard Borel.*

Each $c_i \in C_i$ has a unique representative $\bar{c}_i : Q_i \to \{0, 1\}$ satisfying $c_i = \bar{c}_i \circ \kappa_i$. Put

$$\bar{C}_i = \{\bar{c}_i : c_i \in C_i\}, \qquad \mathcal{H}_i = \{0, 1\}^{Q_i},$$

where $\mathcal{H}_i$ has the product Borel sigma-field $\mathscr{H}_i$ generated by finite evaluation cylinders. Also put

$$\mathcal{H}^{\oplus} = \prod_{i=1}^{k} \mathcal{H}_i, \qquad \mathscr{H}^{\oplus} = \bigotimes_{i=1}^{k} \mathscr{H}_i.$$

For $\bar{h} = (\bar{h}_1, \ldots, \bar{h}_k) \in \mathcal{H}^{\oplus}$, define the decoded hypothesis

$$h_{\bar{h}}(x) = \bar{h}_i(\kappa_i(x)) \qquad (x \in X_i). \tag{1.3}$$

Assumption 3 makes every target and every decoded hypothesis Σ-measurable.

Assumption 4 (Approximate-DP parameter range)**.** *The privacy parameters satisfy*

$$0 < \varepsilon \le \frac{1}{10}, \qquad 0 < \delta < 1.$$

The upper theorem has no further restriction on δ.

The lower argument uses the following primitive numerical condition only at the particular candidate to which it is applied.

Assumption 5 (Candidate-wise lower-bound budget)**.** *For the candidate $n \in \mathbb{N}$,*

$$0 < \delta \le \min\left\{\frac{1}{n\log(n+1)}, \min_{1 \le i \le k} \frac{c_\delta}{m_{n,i}^2 \log(m_{n,i}+1)}\right\}, \tag{1.4}$$

where $c_\delta > 0$ is the universal constant in the unrestricted one-factor lower interface of Alon et al. (2019). This condition is checked separately at the actual candidate; it is not a uniform-in-n schedule or a condition on a realized object.

Our objective is a conditional two-sided direct-sum theorem for this measurable Cartesian subclass. The upper result will hold throughout Assumption 4; the lower result will be candidate-wise under Assumption 5. The theorem therefore does not purport to characterize arbitrary finite-Littlestone classes or classes with uncountably many evaluation types.

2 Preliminaries

Risk and unrestricted learners. A target is any $c = (c_1, \dots, c_k) \in C$. For an arbitrary probability measure D on (X, Σ), let $\rho_i = D(X_i)$. When $\rho_i > 0$, let

$$D_i = D(\,\cdot \mid X_i), \qquad \bar{D}_i = (\kappa_i)_{\#} D_i.$$

There is no balance, positive-mass, product-distribution, finite-support, or public-knowledge condition on these objects. For a Σ-measurable binary hypothesis h, define

$$R_D(h, c) = \Pr_{x \sim D}[h(x) \neq c(x)].$$

For every quotient tuple $\bar{h}$, the exact risk identity is

$$\begin{aligned} R_D(h_{\bar{h}}, c) &= \sum_{i:\rho_i>0} \rho_i \Pr_{q \sim \bar{D}_i}[\bar{h}_i(q) \neq \bar{c}_i(q)] \\ &= \sum_{i=1}^{k} \sum_{q \in Q_i} D(\kappa_i^{-1}(\{q\})) \mathbf{1}\{\bar{h}_i(q) \neq \bar{c}_i(q)\}. \end{aligned} \tag{2.1}$$

Thus the quotient output preserves raw factor and global risk exactly.

Let $Z = X \times \{0, 1\}$, with sigma-field $\Sigma \otimes 2^{\{0,1\}}$. Datasets have fixed size, and $S, S' \in Z^n$ are adjacent when one record is replaced. An unrestricted randomized learner may use any measurable output space $(\Omega, \mathscr{F})$, any Markov kernel $A_n : Z^n \rightsquigarrow \Omega$, and any decoder $\omega \mapsto h_\omega$ into the measurable binary hypotheses, provided all finite-evaluation cylinders are in $\mathscr{F}$ and every asserted risk event is measurable. It may be randomized, joint across factors, improper, and computationally unbounded. It is replacement-(ε, δ)-DP if, for all adjacent S, S', including nonrealizable inputs, and every $E \in \mathscr{F}$,

$$A_n(S, E) \leq e^{\varepsilon} A_n(S', E) + \delta. \tag{2.2}$$

Let D_c be the law of $(x, c(x))$ for $x \sim D$. With accuracy and failure probability fixed at

$$\alpha_0 = \beta_0 = \frac{1}{16},$$

define $\mathrm{SC}_{\varepsilon,\delta}(C)$ as the least $n \in \mathbb{N}$ for which an unrestricted private learner satisfies, for every $c \in C$ and every probability measure D,

$$\Pr_{\substack{S \sim D_c^n \\ \Omega \sim A_n(S,\cdot)}} \left[R_D(h_\Omega, c) \leq \frac{1}{16} \right] \geq \frac{15}{16}. \tag{2.3}$$

The routed quotient-first learner. Fix

$$\alpha_{\mathrm{fac}} = \frac{1}{64}, \qquad \beta_{\mathrm{fac}} = \frac{1}{4096}, \qquad \varepsilon_{\mathrm{fac}} = \frac{\varepsilon}{2}, \qquad \delta_{\mathrm{fac}} = \frac{\delta}{2}.$$

Let $K_Y > 0$ be the universal constant obtained from the audited VC-one private-median and choosing construction of Yan (2025). Define the exact factor quotas

$$q_i = \left\lceil K_Y \frac{s_i}{\varepsilon} \log^2\!\left(\frac{e s_i}{\varepsilon \delta}\right) \right\rceil, \qquad Q_{\oplus} = \sum_{i=1}^{k} q_i. \tag{2.4}$$

Independently of the target and data, fix a reference $\bar{f}_i^\circ \in \bar{C}_i$, an enumeration of Q_i, and a padding record $\bar{z}_i^\circ = (\xi_i^\circ, b_i^\circ) \in Q_i \times \{0,1\}$.

Given $S = ((x_j, y_j))_{j=1}^n$, let

$$J_i(S) = |\{j : x_j \in X_i\}|.$$

Form $\bar{T}_i(S) \in (Q_i \times \{0,1\})^{q_i}$ from the first q_i routed records $(\kappa_i(x_j), y_j)$ with $x_j \in X_i$, in sample order, padding any shortage with $\bar{z}_i^\circ$. On $\bar{T}_i(S)$, run the fixed permutation-symmetrized and fully totalized quotient-first VC-one transition rule

$$\bar{A}_i^{\mathrm{Yan}} : (Q_i \times \{0,1\})^{q_i} \rightsquigarrow \mathcal{H}_i$$

at privacy $(\varepsilon_{\mathrm{fac}}, \delta_{\mathrm{fac}})$. The fixed enumeration and reference specify tie-breaking and fallback outputs on every inconsistent or otherwise partial branch. With independent internal randomness across factors, define

$$\bar{A}_n^{\oplus,Q}(S) = \left(\bar{A}_i^{\mathrm{Yan}}(\bar{T}_i(S))\right)_{i=1}^k \in \mathcal{H}^\oplus \tag{2.5}$$

and decode it by (1.3). Factor utility is invoked only when $J_i(S) \geq q_i$, where no padding occurs.

3 Main Theorem

Theorem 3.1 (Conditional private direct sum for Cartesian VC-one products)**.** *Under Assumptions 1, 2, 3, and 4, there are simultaneous universal constants*

$$C_{\mathrm{up}} = 65536, \qquad C_{\mathrm{quota}} = \max\left\{1, K_Y + \frac{1}{20}\right\}, \qquad c_{\mathrm{low}} > 0,$$

with the following properties.

1. *For every integer*

$$n \geq \lceil C_{\mathrm{up}} Q_\oplus \rceil,$$

the routed rule $\bar{A}_n^{\oplus,Q}$ in (2.5) is a measurable Markov kernel into $(\mathcal{H}^\oplus, \mathscr{H}^\oplus)$, is replacement-$(\varepsilon,\delta)$-DP on every pair of adjacent labeled datasets, and satisfies, for every target $c \in C$ and every probability measure D on (X, Σ),

$$\Pr_{\substack{S \sim D_c^n \\ \bar{H} \sim \bar{A}_n^{\oplus,Q}(S,\cdot)}} \left[R_D(h_{\bar{H}}, c) \leq \frac{1}{16} \right] \geq \frac{15}{16}. \tag{3.1}$$

Consequently,

$$\mathrm{SC}_{\varepsilon,\delta}(C) \leq \lceil C_{\mathrm{up}} Q_\oplus \rceil, \qquad Q_\oplus \leq C_{\mathrm{quota}} \frac{M_\oplus(C)}{\varepsilon} \log^2\left(\frac{e M_\oplus(C)}{\varepsilon\delta}\right). \tag{3.2}$$

This clause holds for every $0 < \delta < 1$ allowed by Assumption 4.

2. *Fix $n \geq 1$ and suppose Assumption 5 holds at this candidate. Every unrestricted measurable replacement-(ε,δ)-DP learner satisfying the universal guarantee (2.3) at this n obeys*

$$n \geq c_{\mathrm{low}} M_\oplus(C). \tag{3.3}$$

Equivalently, if $n < c_{\mathrm{low}} M_\oplus(C)$, then for every such learner there are a full-product target $c \in C$ and a legal probability measure D with $D(X_i) = \omega_i$ for all i such that

$$\Pr_{\substack{S \sim D_c^n \\ \Omega \sim A_n(S,\cdot)}} \left[R_D(h_\Omega, c) > \frac{1}{16} \right] > \frac{1}{16}. \tag{3.4}$$

The learner in this clause may be improper, joint across factors, and computationally unbounded.

3. *Let*

$$n_* = \mathrm{SC}_{\varepsilon,\delta}(C).$$

If both

$$0 < \delta \le \frac{1}{n_* \log(n_* + 1)} \tag{3.5}$$

and

$$\delta \le \frac{c_\delta}{m_{n_*,i}^2 \log(m_{n_*,i} + 1)} \qquad (1 \le i \le k) \tag{3.6}$$

hold, then

$$c_{\mathrm{low}} M_\oplus(C) \le n_* \le \lceil C_{\mathrm{up}} Q_\oplus \rceil \le C_{\mathrm{up}} C_{\mathrm{quota}} \frac{M_\oplus(C)}{\varepsilon} \log^2\!\left(\frac{e M_\oplus(C)}{\varepsilon\delta}\right). \tag{3.7}$$

If either candidate condition fails, the upper bounds (3.2) remain valid, but no lower bound at n_* is asserted.

4. *If $k = 1$, then $M_\oplus(C) = s_1$, $Q_\oplus = q_1$, and the upper learner is exactly the unpadded measurable quotient-first factor rule. It has zero quotient-risk residual and satisfies the stronger guarantee*

$$\Pr\left[R_D(h_{\bar{H}_1}, c) \le \frac{1}{64}\right] \ge \frac{4095}{4096}. \tag{3.8}$$

For an admissible lower candidate,

$$\omega_1 = 1, \qquad m_{n,1} = \max\{8, 4n\},$$

the hidden-factor simulation has zero overflow, and the unrestricted one-factor VC/Littlestone lower conclusion is

$$n \ge c_{\mathrm{low}}\big(1 + \log^*(d_1 + 1)\big), \tag{3.9}$$

with the strict failure event (3.4) on the contradicted branch.

All utility probabilities are over the iid sample and learner randomness; privacy is pointwise for adjacent fixed-size datasets and measurable output events; and every risk is exact distributional binary 0-1 risk. The upper horizon is fixed-sample and the lower horizon is one checked candidate. The exposed quantities are $k, (d_i, s_i, q_i, m_{n,i})_i, M_\oplus(C), Q_\oplus, n, \varepsilon, \delta$. The fixed quantities are the accuracy and confidence levels $1/16, 1/64, 1/4096$, the equal factor privacy split, the logarithm convention, and the universal source constants K_Y, c_δ. The hidden constants are universal and independent of the class, factor or quotient cardinalities, distribution supports, block masses, learner, candidate, and privacy parameters. No balance, finite-support, properness, factorwise-output, or computational restriction is hidden.

Corollary 3.1 (Explicit-rate specialization)**.** *Under Assumptions 1, 2, 3, and 4, choose the integer quotas $(q_i)_i$ exactly as in (2.4), $C_{\mathrm{up}} = 65536$, and $C_{\mathrm{quota}} = \max\{1, K_Y + 1/20\}$. Then, for every $0 < \delta < 1$,*

$$\mathrm{SC}_{\varepsilon,\delta}(C) \le C_{\mathrm{up}} C_{\mathrm{quota}} \frac{M_\oplus(C)}{\varepsilon} \log^2\!\left(\frac{e M_\oplus(C)}{\varepsilon\delta}\right). \tag{3.10}$$

If (3.5) and (3.6) hold at the attained sample complexity n_*, then the left side is also at least $c_{\mathrm{low}} M_\oplus(C)$. The probability, horizon, risk, fixed quantities, and universal hidden-constant dependence are exactly those declared in Theorem 3.1.

Proof. Proposition A.15 verifies the exact quota choices and absorbs the k ceiling terms using $M_\oplus(C) \geq 2k$, $\varepsilon \leq 1/10$, and the displayed logarithmic scale. Proposition A.32 performs the PAC conversion at $C_{\mathrm{up}}Q_\oplus$. Because C_{up} and $Q_\oplus$ are integers, Proposition A.34 proves $\lceil C_{\mathrm{up}}Q_\oplus \rceil = C_{\mathrm{up}}Q_\oplus$, multiplies the quota bound by C_{up}, and, only after checking both candidate conditions at n_*, supplies the lower inequality. These are precisely the technical checks and probability conversions behind (3.10). □

4 Proof Sketch

The quotient map preserves every finite factor labeling pattern and every finite Littlestone tree, while the full Cartesian restriction map preserves independent factor choices. The structural results in Proposition A.2 therefore give the exact quotient dimensions, the product VC and Littlestone identities, measurable decoded outputs, and the risk identity (2.1).

For the upper bound, each countable quotient supports a fully totalized measurable VC-one factor kernel. A common random permutation turns a one-record change into at most one factor-summary replacement, and Proposition A.9 shows that routing affects at most two factors. Their two half-budget privacy costs compose to (ε, δ), including on nonrealizable inputs. The utility proof does not ask every factor to receive its quota. Instead, Proposition A.11 controls the total distribution mass of short factors, and Proposition A.14 combines this with the unpadded factor guarantee to obtain the exact global PAC event. Proposition A.15 then sums the heterogeneous quotas and absorbs every ceiling into the public rate.

For the lower bound, the additive VC obstruction first pays for the total mass of factors with small iterated-log complexity. Each remaining factor has a subcritical exact budget and satisfies the one-factor delta cap. The compact finite game in Proposition A.22 fixes a hard task prior before seeing the global learner. A one-use simulator embeds each factor input row into at most one global row and is therefore private on all factor inputs. The only discrepancy between this simulator and the ideal product experiment is sample-count overflow, whose marginal probability is bounded by $\eta_0 = e^7(2/9)^9 < 3/2048$.

All active factor risks are marginals of one common product-prior experiment. Lemma A.35 gives an exact pointwise weighted risk identity, and Proposition A.28 charges each overflow residual once by its factor weight. The resulting expected global risk is separated from the PAC expectation ceiling by the exact rational calculation in Lemma A.37. Finite conditioning then selects one deterministic full-product target and legal mixture with strict PAC failure, yielding Proposition A.31.

Finally, Proposition A.32 establishes the arbitrary-δ upper theorem and finiteness of the least successful sample size. Proposition A.33 retains both lower delta conditions at one fixed candidate. Proposition A.34 combines the two only after those conditions are checked at the attained sample complexity. Propositions A.35 and A.36 separately verify the unpadded upper rule and zero-overflow unrestricted lower baseline when $k = 1$.

References

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A Proof Details

A.1 Quotient and Cartesian-product structure

Lemma A.1 (Evaluation-quotient invariance)**.** *Under Assumptions 2 and 3, define*

$$\mathsf{P}_i : \mathcal{H}_i \longrightarrow \{0,1\}^{X_i}, \qquad \mathsf{P}_i(\bar{h}_i) = \bar{h}_i \circ \kappa_i.$$

Then $\mathsf{P}_i : \bar{C}_i \to C_i$ is a bijection. For every finite tuple $x_1, \dots, x_m \in X_i$,

$$\begin{aligned}\{(\bar{c}_i(\kappa_i(x_1)), \dots, \bar{c}_i(\kappa_i(x_m))) : \bar{c}_i \in \bar{C}_i\} \\ = \{(c_i(x_1), \dots, c_i(x_m)) : c_i \in C_i\}.\end{aligned} \tag{A.1}$$

The analogous equality holds for quotient tuples after choosing raw representatives. Raw and quotient classes shatter the same finite VC patterns and the same finite depths of Littlestone trees. Consequently,

$$\mathrm{VC}(\bar{C}_i) = \mathrm{VC}(C_i) = 1, \qquad \mathrm{LD}(\bar{C}_i) = \mathrm{LD}(C_i) = d_i.$$

Proof. By the definition of $\equiv_i$, every $c_i \in C_i$ is constant on each fiber of κ_i. It therefore has a unique representative $\bar{c}_i : Q_i \to \{0,1\}$ satisfying $c_i = \bar{c}_i \circ \kappa_i$, which proves surjectivity of the displayed pullback on $\bar{C}_i$. If $\bar{c}_i \neq \bar{c}'_i$, the two functions differ at some $q \in Q_i$. Since κ_i is onto, a point $x \in \kappa_i^{-1}(\{q\})$ witnesses that their pullbacks differ. Thus the pullback is injective.

Equation (A.1) follows from this bijection. For a finite quotient tuple, choose one raw representative for each occurrence and apply the same equality. This is only finite set-theoretic choice and does not construct a measurable section. Quotient collisions cause no loss: if $\kappa_i(x_j) = \kappa_i(x_\ell)$, every concept gives the same label at x_j and x_ℓ. A pattern assigning different labels to those occurrences is absent on both sides, and every consistent pattern is present on one side precisely when it is present on the other. Hence the VC dimensions agree.

For Littlestone dimension, replace every raw node label x in a finite complete binary instance tree by $\kappa_i(x)$. Every root-to-leaf bit sequence remains realized by the quotient representative of the raw concept that realized it. A repeated quotient label with conflicting bits on one path cannot occur in a shattered raw tree because equivalent points have identical labels under every concept. Conversely, a finite quotient tree has finitely many nodes; choose a raw representative for each node label. Pullback then realizes every representative path. Thus the two classes shatter the same finite tree depths, and Assumption 2 gives the stated dimensions. □

Lemma A.2 (VC dimension and finite cardinality of the full product)**.** *Under Assumptions 1 and 2,*

$$\mathrm{VC}(C) = k, \qquad |C| = \prod_{i=1}^{k} |C_i|.$$

If every factor is finite, then $\log |C| = \sum_{i=1}^{k} \log |C_i|$. If at least one factor is infinite, then C is infinite and no finite-real logarithmic identity is asserted.

Proof. For each factor choose $x_i \in X_i$ whose singleton is shattered. Given $b = (b_1, \dots, b_k) \in \{0,1\}^k$, choose a factor concept taking value b_i at x_i. Assumption 1 combines these factor choices into a global concept, so $\{x_1, \dots, x_k\}$ is shattered. Conversely, any $k+1$ distinct points contain two in a common block. Shattering the whole set would shatter those two points in the corresponding factor, contradicting its VC dimension one. This proves $\mathrm{VC}(C) = k$.

The restriction bijection in Assumption 1 gives the cardinal product. For finite factors, taking the fixed natural logarithm gives the sum identity. If one factor is infinite, fixing one concept in every other nonempty factor injects it into C, so C is infinite. Constant-valued concepts are allowed. In particular, when a quotient has one cell and the factor contains both constant labels, that cell is still shattered. A genuinely singleton factor is excluded by Assumption 2. $\square$

Lemma A.3 (Rank predictor for a finite-Littlestone class)**.** *Let $H \subseteq \{0,1\}^Y$ be nonempty with $\mathrm{LD}(H) = d < \infty$. There is a deterministic online predictor making at most d mistakes on every sequence realizable by one member of H. The construction requires neither an order nor a cardinality assumption on Y or H, and it selects no hypothesis.*

Proof. Set $\mathrm{LD}(\varnothing) = -1$. At a query x, let $V \neq \varnothing$ be the current version class and write

$$V_b(x) = \{h \in V : h(x) = b\}, \qquad b \in \{0,1\}.$$

If $a = \mathrm{LD}(V)$, both children cannot have Littlestone dimension at least a, since adjoining the query x above depth- a trees for both children would give a depth-$a+1$ tree shattered by V. Hence

$$\min_{b \in \{0,1\}} \mathrm{LD}(V_b(x)) \leq a - 1. \tag{A.2}$$

Predict a bit whose child has larger rank, breaking ties in favor of zero. On a mistake, the realized child is a smaller-ranked child; in a tie both children satisfy the minimum bound. Thus every mistake lowers the rank by at least one. Correct predictions pass to a subclass and cannot increase rank. Starting at rank d, at most d mistakes occur. At rank zero, two nonempty children would shatter a depth-one tree, so the realizable next label cannot cause another mistake. $\square$

Proposition A.1 (Additivity of Littlestone dimension)**.** *Under Assumptions 1 and 2,*

$$\mathrm{LD}(C) = \sum_{i=1}^{k} d_i.$$

Proof. For each i, choose a complete depth-d_i tree shattered by C_i. Attach a copy of the factor-two tree at every leaf of the factor-one tree, then the factor-three tree at every new leaf, and continue through factor k. The resulting depth is $\sum_i d_i$. Along a path, its segment in factor i is realized by some $c_i \in C_i$, and Assumption 1 combines these choices into one global concept. Therefore

$$\mathrm{LD}(C) \geq \sum_i d_i. \tag{A.3}$$

For the reverse inequality, run one copy of the predictor in Lemma A.3 for each factor, updating only the copy corresponding to the block containing the current instance. A sequence realized by $c \in C$ induces a factor subsequence realized by $c|_{X_i}$, so copy i makes at most d_i mistakes and the global predictor makes at most $\sum_i d_i$. If C shattered a tree of depth $\sum_i d_i + 1$, traverse it by taking at each node the edge opposite the global prediction. Shattering supplies one concept realizing the resulting path, on which the predictor makes a mistake at every level, a contradiction. Thus

$$\mathrm{LD}(C) \leq \sum_i d_i. \tag{A.4}$$

The two bounds prove the claim. A zero-dimensional singleton factor would contribute neither tree depth nor mistakes, while the present VC-one factors have $d_i \geq 1$. At $k = 1$, both bounds are the one-factor identity. $\square$

Lemma A.4 (Log-star and aggregate-size comparisons)**.** *Under Assumption 2, let*

$$r_i = \log_2^* d_i, qquad s_i = 1 + \log_2^*(d_i + 1), qquad M = \sum_{i=1}^{k} s_i = M_\oplus(C).$$

Then, for every i,

$$s_i \geq 2, qquad s_i \leq r_i + 2, qquad M \geq 2k.$$

Proof. VC dimension one gives a depth-one Littlestone tree, so $d_i \geq 1$. Consequently $d_i + 1 \geq 2$, at least one base-two logarithm is needed to reach a value at most one, and $s_i \geq 2$. The stopping time $\log_2^* t$ is nondecreasing on $[1, \infty)$. For each integer $d \geq 1$,

$$\log_2(d + 1) \leq d, \tag{A.5}$$

because $2^d \geq d + 1$, with equality at $d = 1$ and induction thereafter. The log-star recursion and monotonicity yield

$$\log_2^*(d_i + 1) = 1 + \log_2^*\big(\log_2(d_i + 1)\big) \leq 1 + \log_2^* d_i = r_i + 1. \tag{A.6}$$

Adding one proves $s_i \leq r_i + 2$, and summing $s_i \geq 2$ proves $M \geq 2k$. At the smallest allowed dimension,

$$d_i = 1, \qquad r_i = 0, \qquad s_i = 2,$$

so the endpoint is retained exactly. □

Lemma A.5 (Standard-Borel output and measurable decoding)**.** *Under Assumptions 1 and 3, every $(\mathcal{H}_i, \mathscr{H}_i) = (\{0,1\}^{Q_i}, \mathscr{H}_i)$ and their finite product $(\mathcal{H}^\oplus, \mathscr{H}^\oplus)$ are standard Borel. The map*

$$E : \mathcal{H}^\oplus \times X \to \{0, 1\}, \qquad E(\bar{h}, x) = \bar{h}_i(\kappa_i(x)) \quad (x \in X_i),$$

is jointly measurable. Hence every decoded $h_{\bar{h}} = E(\bar{h}, \cdot)$ is Σ-measurable, and decoding is measurable for every finite evaluation cylinder specified in the learning model.

Proof. If Q_i is finite, $\mathcal{H}_i$ is finite discrete. If it is countably infinite, enumerate it as $q_{i,1}, q_{i,2}, \ldots$ and define

$$\Delta_i(\bar{h}, \bar{g}) = \sum_{m=1}^{\infty} 2^{-m} \mathbf{1}\{\bar{h}(q_{i,m}) \neq \bar{g}(q_{i,m})\}. \tag{A.7}$$

This metric induces the product topology: a finite coordinate cylinder is clopen, while agreement on a sufficiently long initial segment makes the remaining metric tail arbitrarily small. It is complete because a Cauchy sequence stabilizes coordinatewise and converges to its coordinatewise limit. It is separable because functions vanishing beyond a finite initial segment form a countable dense subset. Thus the topology is Polish, and its Borel sigma-field is exactly the finite-evaluation-cylinder sigma-field $\mathscr{H}_i$.

The disjoint coordinate set $\widetilde{Q} = \bigsqcup_{i=1}^{k} (\{i\} \times Q_i)$ is finite or countable. Coordinate relabeling identifies $\mathcal{H}^\oplus$ with $\{0,1\}^{\widetilde{Q}}$, and the two sigma-fields have the same coordinate generators. The preceding argument therefore makes the finite product standard Borel.

For joint evaluation,

$$E^{-1}(\{1\}) = \bigcup_{i=1}^{k} \bigcup_{q \in Q_i} \{\bar{h} : \bar{h}_i(q) = 1\} \times \kappa_i^{-1}(\{q\}). \tag{A.8}$$

The output sets are cylinders, the raw cells are measurable, and the union is countable. For fixed $\bar{h}$,

$$h_{\bar{h}}^{-1}(\{1\}) = \bigcup_{i=1}^{k} \bigcup_{q:\bar{h}_i(q)=1} \kappa_i^{-1}(\{q\}) \in \Sigma.$$

Finally, for fixed $x_1, \ldots, x_m$ and $b \in \{0,1\}^m$, the decoder preimage of the corresponding evaluation cylinder is

$$\bigcap_{j=1}^{m} \{\bar{h} : \bar{h}_{i(j)}(\kappa_{i(j)}(x_j)) = b_j\}, \tag{A.9}$$

where $i(j)$ is the unique block containing x_j. This proves exactly the required decoder measurability without imposing a sigma-field on the entire raw hypothesis collection. □

Lemma A.6 (Exact measurable quotient-to-raw risk pullback)**.** *Under Assumptions 1 and 3 and Lemma A.1, fix $c \in C$, a probability measure D on (X, Σ), and $\bar{h} \in \mathcal{H}^{\oplus}$. For $\rho_i = D(X_i) > 0$,*

$$\begin{aligned}\Pr_{x \sim D_i}[\mathsf{P}_i(\bar{h}_i)(x) \neq c_i(x)] &= \Pr_{q \sim \bar{D}_i}[\bar{h}_i(q) \neq \bar{c}_i(q)] \\ &= \sum_{q \in Q_i} D_i(\kappa_i^{-1}(\{q\}))\mathbf{1}\{\bar{h}_i(q) \neq \bar{c}_i(q)\}.\end{aligned} \tag{A.10}$$

Globally,

$$\begin{aligned}R_D(h_{\bar{h}}, c) &= \sum_{i:\rho_i>0} \rho_i \Pr_{q \sim \bar{D}_i}[\bar{h}_i(q) \neq \bar{c}_i(q)] \\ &= \sum_{i=1}^{k} \sum_{q \in Q_i} D(\kappa_i^{-1}(\{q\}))\mathbf{1}\{\bar{h}_i(q) \neq \bar{c}_i(q)\}.\end{aligned} \tag{A.11}$$

These risks are measurable functions of the quotient output. The identities hold for arbitrary-support D, and a block of mass zero contributes zero without requiring D_i to be defined.

Proof. Lemma A.1 gives $c_i(x) = \bar{c}_i(\kappa_i(x))$ and $\mathsf{P}_i(\bar{h}_i)(x) = \bar{h}_i(\kappa_i(x))$. The raw mistake indicator is therefore the pullback of the quotient mistake indicator. Pushforward by κ_i gives the first equality in (A.10); discreteness and countable additivity give the second.

The measurable blocks partition X, so the global mistake set is the disjoint union of its block intersections. Countable additivity across the finite blocks and then their quotient cells gives the second line of (A.11). For positive block mass,

$$D(\kappa_i^{-1}(\{q\})) = \rho_i D_i(\kappa_i^{-1}(\{q\})),$$

which gives the first line. If $\rho_i = 0$, every cell in X_i has zero mass, so the contribution vanishes without a conditional law.

Each summand is a fixed nonnegative coefficient times the indicator of an evaluation cylinder $\{\bar{h} : \bar{h}_i(q) \neq \bar{c}_i(q)\}$. Finite partial sums are measurable and increase pointwise to the displayed risks. Their limits are measurable, as is the finite sum over factors. Hence every risk threshold event used below is measurable, and no approximation term is introduced by quotient decoding. □

Proposition A.2 (Quotient-product structural certificate)**.** *Under Assumptions 1, 2, and 3, quotient pullback preserves every factor's VC and Littlestone dimensions; the global class satisfies*

$$\mathrm{VC}(C) = k, qquad\, \mathrm{LD}(C) = \sum_{i=1}^{k} d_i, qquads_i \geq 2, quads_i \leq \log_2^* d_i + 2, quadM_{\oplus}(C) \geq 2k,$$

and, for finite factors, $\log|C| = \sum_i \log|C_i|$*. Moreover* $\mathcal{H}^{\oplus}$ *is standard Borel, decoding yields measurable raw hypotheses, and quotient, raw-factor, and global risks agree exactly as in* (A.10)*–*(A.11)*.*

Proof. The quotient and factor-dimension statements are Lemma A.1. Product VC and the qualified cardinality formula are Lemma A.2. The rank argument in Lemma A.3 and Proposition A.1 gives product Littlestone additivity. Lemma A.4 supplies every scalar comparison, including $d_i = 1$. Output legality follows from Lemma A.5, and Lemma A.6 gives the exact measurable risk identities, including countably infinite quotients, arbitrary-support distributions, and zero-mass blocks. At $k = 1$, each product, output, and risk identity is precisely its sole-factor version. □

A.2 A measurable totalized private learner for one factor

Lemma A.7 (Reference-transformed quotient order geometry)**.** *Under Assumptions 2 and 3 and Lemma A.1, fix a factor* i *and the pre-data reference* $\bar{f}_i^\circ \in \bar{C}_i$*. There is a finite-height partially ordered active transformed quotient* $(U_i^+, \preceq_i)$ *with the following properties. The transformed class has VC dimension one and Littlestone dimension* d_i*; every nonempty transformed positive set and every nonempty intersection of such sets is a finite principal upset; if* H_i *is the maximum strict-chain length, then*

$$1 \le H_i \le \mathrm{TD}(\bar{C}_i \triangle \bar{f}_i^\circ) \le 2^{d_i+1}; \tag{A.12}$$

and each principal upset contains exactly one point of every integer depth between one and the depth of its minimum.

Proof. For $\bar{c} \in \bar{C}_i$, set

$$g_{\bar{c}}(q) = \bar{c}(q) \oplus \bar{f}_i^\circ(q), \qquad \mathcal{G}_i = \{g_{\bar{c}} : \bar{c} \in \bar{C}_i\}. \tag{A.13}$$

XOR by a fixed function is a bijection on every finite label vector and on every binary instance tree, with edge bits XORed nodewise. Hence

$$\mathrm{VC}(\mathcal{G}_i) = 1, qquad\, \mathrm{LD}(\mathcal{G}_i) = d_i, qquad 0 = g_{\bar{f}_i^\circ} \in \mathcal{G}_i. \tag{A.14}$$

The transform may identify coordinates, so define

$$q \sim_i q' \quad \Longleftrightarrow \quad g(q) = g(q') \text{ for every } g \in \mathcal{G}_i,$$

let $U_i = Q_i/{\sim_i}$, and write $\tau_i : Q_i \to U_i$. This quotient is finite or countable. The two-way finite-representative argument in Lemma A.1 shows that quotienting preserves all finite patterns and finite Littlestone trees, so (A.14) remains true on U_i.

Let

$$U_i^+ = \{u \in U_i : \text{some } g \in \mathcal{G}_i \text{ has } g(u) = 1\}.$$

It is nonempty because the factor is nonconstant and the reference is one of its concepts. Put

$$u \preceq_i v \quad \Longleftrightarrow \quad \big(g(u) = 1 \Longrightarrow g(v) = 1\big) \quad \text{for every } g \in \mathcal{G}_i. \tag{A.15}$$

This relation is reflexive and transitive. Antisymmetry follows from the secondary evaluation quotient: mutual implication means all transformed concepts agree at the two coordinates.

If u, v are incomparable, failures of the two implications provide the patterns $(1,0)$ and $(0,1)$, while the zero concept provides $(0,0)$. A concept taking value one at both points would shatter the pair, contrary to VC dimension one. Therefore every positive set

$$P_g = \{u \in U_i^+ : g(u) = 1\} \tag{A.16}$$

is both a chain and an upset.

Every strict chain $u_1 \prec_i \cdots \prec_i u_r$ yields a threshold pattern of length r. Active membership supplies the all-one suffix starting at u_1; for $j \geq 2$, strictness supplies a concept zero at u_{j-1} and one at u_j, and implication makes it zero before j and one from j onward. The threshold-dimension comparison $\mathrm{TD}(H) \leq 2^{\mathrm{LD}(H)+1}$ for a binary class (Yan, 2025) proves (A.12). Thus every chain is finite and the largest integer chain length exists. No tree-construction or countable-set selector is used.

For $u \in U_i^+$, define

$$I_i(u) = \{v \in U_i^+ : u \preceq_i v\}, \qquad \ell_i(u) = |I_i(u)|. \tag{A.17}$$

A transformed concept positive at u is positive throughout $I_i(u)$, so the incomparability argument makes $I_i(u)$ a chain of at most H_i points. It is the longest chain starting at u, and successive points in its finite total order have suffix sizes $\ell_i(u), \ell_i(u) - 1, \ldots, 1$. This proves uniqueness at every depth.

Finally, for nonempty $V \subseteq \mathcal{G}_i$, set

$$B(V) = \bigcap_{g \in V} P_g. \tag{A.18}$$

It is an upset contained in the finite chain P_{g_0} for any $g_0 \in V$. If nonempty it has a unique minimum u_V, and upward closure gives $B(V) = I_i(u_V)$. This proves every asserted structural property. ☐

Lemma A.8 (VC dimension of the improper version-core class)**.** *Under Lemma A.7, define, with every function extended by zero on $U_i \setminus U_i^+$,*

$$\mathcal{K}_i = \{\mathbf{1}_{B(V)} : \varnothing \neq V \subseteq \mathcal{G}_i\} \cup \{0\}. \tag{A.19}$$

Then $\mathrm{VC}(\mathcal{K}_i) \leq 1$, every transformed target is in $\mathcal{K}_i$, and the totalized version-core indicator of every realizable labeled block belongs to $\mathcal{K}_i$ and has zero empirical error against its target.

Proof. For two distinct active points u, v, if $u \preceq_i v$, no upset realizes $(1, 0)$. If they are incomparable, every nonempty core is contained in one defining positive set, and the argument in Lemma A.7 shows that no such set contains both points; hence no core realizes $(1, 1)$. Adding zero cannot restore the missing pattern. A pair containing an inactive coordinate also cannot be shattered, so the VC dimension is at most one.

For $g_* \in \mathcal{G}_i$, its positive set is $B(\{g_*\})$, unless empty, in which case $g_* = 0$. Thus every target is in the class. For a finite transformed block T, let

$$V(T) = \{g \in \mathcal{G}_i : g \text{ agrees with every record of } T\}. \tag{A.20}$$

Set its core to $B_T = \varnothing$ when $V(T) = \varnothing$, and otherwise to $B(V(T))$. On a realizable block, $g_* \in V(T)$. A positive record belongs to every consistent concept's positive set and therefore to B_T; a negative record is excluded by the consistent target. Hence $\mathbf{1}_{B_T}$ agrees with every record. The argument never asserts that this core indicator is a proper concept. ☐

Lemma A.9 (Countable positive-support replacement choosing rule)**.** *Let F be finite or countable, let $\mathsf{O} = F \cup \{\perp\}$ have the discrete sigma-field, and let $D = (v_1, \ldots, v_t) \in \mathsf{O}^t$. For $0 < b < 1$, $0 < \varepsilon_c \leq 2$, and $0 < \delta_c < 1$, define*

$$q(D, f) = \sum_{j=1}^{t} \mathbf{1}\{v_j = f\}, \qquad \mathrm{OPT}(D) = \sup_{f \in F} q(D, f),$$

with optimum zero when $F = \varnothing$. Apply the thresholded choosing kernel of Bun et al. (2015) with bounded-growth parameter one: set $\widetilde{\mathrm{OPT}} = operatorname OPT + operatorname Lap(4/\varepsilon_c)$, return $\perp$ if

$$\widetilde{\mathrm{OPT}} < \frac{8}{\varepsilon_c}\log\left(\frac{4}{b\varepsilon_c\delta_c}\right),$$

and otherwise use its $\varepsilon_c/2$-exponential stage on $G(D) = \{f : q(D,f) > 0\}$. If $G(D) = \varnothing$, return $\perp$ immediately. The resulting rule is a replacement- $(\varepsilon_c, \delta_c)$-DP Markov kernel on the countable spaces. Defining $q(D, \perp) = 0$, with probability at least $1 - b$,

$$q(D, \hat{f}) \geq \mathrm{OPT}(D) - \frac{16}{\varepsilon_c}\log\left(\frac{4t}{b\varepsilon_c\delta_c}\right) \tag{A.21}$$

whenever the right side is positive; then $\hat{f} \neq \perp$.

Proof. Under addition adjacency, one added non-sentinel summary increments exactly one score, while a sentinel changes none, so the score is one-bounded-growth. For fixed-size replacement, neighboring databases share $t - 1$ entries and have one leaving and one entering entry, whence

$$q(D', f) - q(D, f) = \mathbf{1}\{f = v_{\mathrm{in}}\} - \mathbf{1}\{f = v_{\mathrm{out}}\}. \tag{A.22}$$

Every score and the optimum have sensitivity one, and each directional set difference between $G(D)$ and $G(D')$ has size at most one. The privacy proof for the cited choosing mechanism uses exactly these facts and sums only over the positive supports, each of which contains at most t points (Bun et al., 2015). Its finite sums therefore remain valid when the ambient solution set is countable. The same observation preserves the cited utility bound (A.21).

For each fixed database, the thresholded and finite-support stages define a probability measure. Since the database and output spaces are countable discrete, every transition-probability map is measurable.

It remains to check the newly totalized empty-support branch. Suppose $G(D_0) = \varnothing$ and D_0, D_1 are replacement neighbors. If the second support is also empty, both laws are $\delta_\perp$. Otherwise (A.22) gives $G(D_1) = \{f_0\}$ and $q(D_1, f_0) = \mathrm{OPT}(D_1) = 1$. Since

$$1 < \frac{4}{\varepsilon_c}\log\left(\frac{4}{b\varepsilon_c\delta_c}\right),$$

passing the noisy threshold requires a Laplace fluctuation exceeding the right side. Thus, for $p = \Pr[A(D_1) \neq \perp]$,

$$p \leq \frac{1}{2}\exp\left(-\log\frac{4}{b\varepsilon_c\delta_c}\right) = \frac{b\varepsilon_c\delta_c}{8} \leq \delta_c. \tag{A.23}$$

If $\perp \notin E$, the D_1-probability of E is at most δ_c and the D_0-probability is zero, proving both DP directions. If $\perp \in E$, one direction follows from $1 \leq e^{\varepsilon_c} + \delta_c$; for the other, $\Pr[A(D_1) \in E] \geq 1 - \delta_c$ and

$$1 \leq e^{\varepsilon_c}(1 - \delta_c) + \delta_c \leq e^{\varepsilon_c}\Pr[A(D_1) \in E] + \delta_c. \tag{A.24}$$

Relabeling the neighbors covers a transition into empty support. Hence the total rule has the stated privacy for every event and sentinel transition. If the right side of (A.21) is positive, output $\perp$, whose score is zero, cannot satisfy it. ☐

Lemma A.10 (Explicit internal calibration and quota domination)**.** *Under Assumptions 2 and 4, Lemma A.4, and Lemma A.7, let $C_{PM} > 0$ be the universal constant in the following fixed-accuracy private-median consequence of Yan (2025): on a finite ordered domain of size $m \geq 2$, a* *replacement-*$(\varepsilon_m, \delta_m)$*-DP kernel returns a* $1/3$*-median except with probability* η *whenever its input count is strictly larger than*

$$C_{PM}\frac{\log_2^* m}{\varepsilon_m}\log^2\left(\frac{e\log_2^* m}{\eta\delta_m}\right).$$

Set

$$\eta = \frac{1}{16384}, \quad \gamma = \frac{1}{48}, \quad \varepsilon_m = \varepsilon_c = \frac{\varepsilon}{4}, quad\delta_m = \delta_c = \frac{\delta}{4}, \tag{A.25}$$

and

$$L_* = \left\lceil \frac{48}{1/64}\left(10\log\frac{48e}{1/64} + \log\frac{5}{\gamma}\right)\right\rceil. \tag{A.26}$$

There are universal constants C_T, K_* *such that, for*

$$\Lambda_i = \log\frac{es_i}{\varepsilon\delta}, \qquad \Xi_i = \frac{s_i}{\varepsilon}\Lambda_i^2, \qquad t_i = \lceil C_T\Xi_i\rceil, qquadN_i = L_*t_i, \tag{A.27}$$

the count t_i *exceeds that private-median threshold on* $\{0,\ldots,H_i\}$*, and*

$$e^{-t_i/128} \le \eta, qquad\frac{16}{\varepsilon_c}\log\left(\frac{4t_i}{\eta\varepsilon_c\delta_c}\right) \le \frac{t_i}{12}. \tag{A.28}$$

If the setting constant satisfies $K_Y \ge K_*$*, then*

$$N_i \le \left\lceil K_Y\frac{s_i}{\varepsilon}\log^2\left(\frac{es_i}{\varepsilon\delta}\right)\right\rceil = q_i. \tag{A.29}$$

All statements include $d_i = 1$*.*

Proof. Let

$$\Lambda_0 = \log(20e), \qquad c_1 = 1 + \frac{\log((3/5)/\eta)}{\Lambda_0}.$$

Choose C_T as the least positive integer satisfying

$$\begin{aligned} C_T &> 6C_{PM}c_1^2, \\ 20C_T\Lambda_0^2 &\ge 128\log(1/\eta), \\ C_T &\ge \frac{384}{\Lambda_0}\left(3 + \frac{\log(64(C_T+1)/(\eta e))}{\Lambda_0}\right). \end{aligned} \tag{A.30}$$

Such an integer exists because the last right side grows logarithmically while the left side grows linearly. Set

$$K_* = L_*(C_T + 1). \tag{A.31}$$

Writing $u_i = \log_2^*(H_i+1)$, the height bound gives $H_i + 1 \le 2^{d_i+1} + 1$. Log-star recursion and monotonicity, $d_i + 2 \le 2(d_i+1)$, and $\log_2^*(2x) \le 1 + \log_2^* x$ for $x \ge 2$ imply

$$u_i \le 1 + \log_2^*(d_i+2) \le 2 + \log_2^*(d_i+1) = s_i + 1 \le \frac{3}{2}s_i. \tag{A.32}$$

At $d_i = 1$, this says $u_i \le 3$, while $s_i = 2$ and $H_i + 1 \le 5$; no zero-domain endpoint is used.

The privacy range and $s_i \ge 2$ give

$$\Lambda_i \ge \Lambda_0, qquad\Xi_i \ge 20\Lambda_0^2 > 1. \tag{A.33}$$

Moreover,

$$
\begin{aligned}
\log \frac{eu_i}{\eta\delta_m} &= \log \frac{4eu_i}{\eta\delta} = \Lambda_i + \log \frac{4u_i\varepsilon}{\eta s_i} \\
&\leq \Lambda_i + \log \frac{3/5}{\eta} \leq c_1\Lambda_i.
\end{aligned}
\tag{A.34}
$$

The private-median threshold is therefore at most

$$
C_{PM}\frac{u_i}{\varepsilon_m}\log^2 \frac{eu_i}{\eta\delta_m} \leq 6C_{PM}c_1^2\Xi_i < C_T\Xi_i \leq t_i, \tag{A.35}
$$

including its strict sample requirement. The first two conditions in (A.30) and (A.33) give $t_i \geq 128\log(1/\eta)$, proving the first inequality in (A.28).

Since $t_i \leq (C_T+1)\Xi_i$, $\log(1/\varepsilon) \leq \Lambda_i$, and $2\log\Lambda_i \leq \Lambda_i$ for $\Lambda_i \geq \Lambda_0$,

$$
\begin{aligned}
\log \frac{4t_i}{\eta\varepsilon_c\delta_c} &\leq \log \frac{64(C_T+1)s_i\Lambda_i^2}{\eta\varepsilon^2\delta} \\
&\leq \left(3 + \frac{\log(64(C_T+1)/(\eta e))}{\Lambda_0}\right)\Lambda_i.
\end{aligned}
\tag{A.36}
$$

Because $s_i\Lambda_i \geq 2\Lambda_0$, the final condition in (A.30) yields

$$
\frac{t_i}{12} \geq \frac{C_T s_i\Lambda_i^2}{12\varepsilon} \geq \frac{64}{\varepsilon}\log\frac{4t_i}{\eta\varepsilon_c\delta_c} = \frac{16}{\varepsilon_c}\log\frac{4t_i}{\eta\varepsilon_c\delta_c}.
$$

Finally, $\Xi_i > 1$ gives

$$
N_i = L_*t_i \leq L_*(C_T\Xi_i + 1) \leq L_*(C_T+1)\Xi_i = K_*\Xi_i. \tag{A.37}
$$

Together with $K_Y \geq K_*$, this proves (A.29) including the exact ceiling. □

Proposition A.3 (Totalized quotient transition kernel)**.** *Under Assumptions 2, 3, and 4, Lemmas A.5 and A.4, and Lemmas A.7–A.10, the rule defined below is a fixed permutation-invariant Markov kernel*

$$
\bar{A}_i^{\mathrm{Yan}} : (Q_i \times \{0,1\})^{q_i} \rightsquigarrow (\mathcal{H}_i, \mathscr{H}_i)
$$

on every input, including inconsistent and nonrealizable inputs.

Proof. Freeze the reference $\bar{f}_i^\circ$, the enumeration of Q_i, the induced first-occurrence enumeration of U_i, implementations of the private-median and choosing kernels, and every tie and fallback convention before observing data. On $\bar{T} = ((q_j,b_j))_{j=1}^{q_i}$, draw a uniform permutation, retain the first $N_i \leq q_i$ records, and split them into t_i consecutive blocks of size L_*. Transform a record by

$$
(q,b) \longmapsto (\tau_i(q), b \oplus \bar{f}_i^\circ(q)). \tag{A.38}
$$

For each block T_b, form the version space (A.20), totalized core B_b, and depth

$$
a_b = \begin{cases} 0, & B_b = \varnothing, \\ \ell_i(u_b), & B_b = I_i(u_b) \neq \varnothing. \end{cases} \tag{A.39}
$$

Apply the private median on the fixed ordered set $\{0,\ldots,H_i\}$ to obtain z. At $z=0$, output $\bar{f}_i^\circ$. For fixed $z \geq 1$, define

$$
p_z(T_b) = \begin{cases} \text{the unique } u \in B_b \text{ with } \ell_i(u) = z, & a_b \geq z, \\ \perp, & a_b < z. \end{cases} \tag{A.40}
$$

Use Lemma A.9 on the t_i summaries with solution set $F_z = \{u \in U_i^+ : \ell_i(u) = z\}$. If it returns $\perp$, output the reference; if it returns $\hat{u}$, output

$$\bar{h}_{\hat{u}}(q) = \bar{f}_i^\circ(q) \oplus \mathbf{1}\{\tau_i(q) \in I_i(\hat{u})\}. \tag{A.41}$$

Every branch is total: an inconsistent or empty version space has empty core, an empty core has depth zero, a missing layer has summary $\perp$, and empty choosing support or a failed selection returns the fixed reference. The input space is finite or countable discrete. Hence for every $E \in \mathscr{H}_i$, the transition-probability function is measurable, regardless of the set-theoretic form of a version space. For fixed input, the finite chain of kernels and deterministic maps gives a probability measure on the standard-Borel output space from Lemma A.5. Finally, precomposing the input with a fixed permutation merely composes two permutations while the sampled one remains uniform, proving permutation invariance. ☐

Proposition A.4 (All-input replacement privacy of the factor kernel). *Under the assumptions and construction of Proposition A.3, $\bar{A}_i^{\mathrm{Yan}}$ is replacement-$(\varepsilon/2, \delta/2)$-DP on every adjacent pair of length-q_i inputs, including every nonrealizable and totalization branch.*

Proof. Couple the data-independent uniform permutation on two inputs differing in one record. If the changed position is not retained, all used blocks agree; otherwise exactly one block changes by one replacement. Consequently the depth vector (A.39) changes in at most one coordinate. The calibrated private-median kernel is replacement- $(\varepsilon_m, \delta_m)$-DP on arbitrary depth vectors.

Condition on the released depth. At $z = 0$, the second-stage output is fixed. At $z \geq 1$, only the changed block's summary can change, and (A.22) covers a leaving and entering point, a transition to or from $\perp$, and a change between two ordinary points. Lemma A.9 supplies replacement- $(\varepsilon_c, \delta_c)$-privacy even with empty support. Sequential composition and postprocessing therefore give

$$(\varepsilon_m + \varepsilon_c, \delta_m + \delta_c) = \left(\frac{\varepsilon}{2}, \frac{\delta}{2}\right). \tag{A.42}$$

Integrating the inequality over the common data-independent permutation preserves it. No consistency, realizability, successful private event, or utility condition has been used, so the result holds on the full input space and for every measurable output event. ☐

Lemma A.11 (Fixed-marginal core control and bad-block concentration). *Under Assumptions 2 and 3, fix $\bar{c}_i \in \bar{C}_i$ and a probability measure $\bar{D}_i$ on Q_i. For iid realizable input and the symmetrized blocks of Proposition A.3, let $h_b^\circ = \mathbf{1}_{B_b}$, let $g_* = \bar{c}_i \oplus \bar{f}_i^\circ$, and call block b bad if its transformed population risk exceeds $1/64$. Then*

$$\Pr[b \textit{ is bad}] \leq \frac{1}{48}, \tag{A.43}$$

the bad indicators are independent, and

$$\Pr\left[\#\{b : b \textit{ is bad}\} \geq \frac{t_i}{12}\right] \leq e^{-t_i/128} \leq \eta. \tag{A.44}$$

Proof. A uniform permutation and restriction of an iid sequence is again iid; the disjoint length-L_* blocks are mutually independent. The transform (A.38) maps them to a common distribution on countable U_i labeled by g_*. By Lemma A.8, the target and each data-dependent core indicator lie in one fixed class of VC dimension at most one and have zero empirical disagreement. The VC-one generalization theorem of Yan (2025), with accuracy $1/64$, confidence $1/48$, and block size (A.26), gives (A.43). Its event is measurable because the quotient is countable.

Each indicator depends on only one independent block. Applying Hoeffding's inequality (Hoeffding, 1963) with mean sum at most $t_i/48$ and threshold $t_i/12$ gives

$$\exp\left(-2t_i\left(\frac{1}{12}-\frac{1}{48}\right)^2\right)=e^{-t_i/128}.$$

The first inequality in (A.28) proves the final bound. This is a fixed marginal estimate followed by concentration, not a union bound over blocks. $\square$

Proposition A.5 (Unpadded fixed-confidence factor utility). *Under Assumptions 2, 3, and 4, for every target $\bar{c}_i \in \bar{C}_i$ and probability measure $\bar{D}_i$ on Q_i, if all q_i input records to $\bar{A}_i^{\mathrm{Yan}}$ are iid and labeled by $\bar{c}_i$, then*

$$\Pr\left[\Pr_{q\sim\bar{D}_i}[\bar{A}_i^{\mathrm{Yan}}(\bar{T})(q)\neq\bar{c}_i(q)]\leq\frac{1}{64}\right]\geq 1-\frac{1}{4096}. \tag{A.45}$$

The probability includes the sample, symmetrization, and internal randomness. No utility claim is made for padded or nonrealizable input.

Proof. On a realizable transformed block, $g_*\in V(T_b)$, so

$$B_b\subseteq P_{g_*},\qquad h_b^\circ\leq g_*\quad\text{pointwise.} \tag{A.46}$$

Every nonempty core is a principal upset whose minimum lies on the one finite chain P_{g_*}, and depth is one-to-one and order-reversing along that chain.

Intersect three events: the private median is a 1/3-median; fewer than $t_i/12$ cores are bad; and, when $z\geq 1$, the choosing rule satisfies (A.21). Each event fails with probability at most η, by (A.35), Lemma A.11, and the choosing guarantee. On the median event, at least $t_i/6$ block depths lie on each side of z.

If $z=0$, at least $t_i/6$ blocks have empty core. Since fewer than $t_i/12$ blocks are bad, one such core is good. The output is transformed zero, equal to that core indicator, and has risk at most $1/64$. This also handles $g_*=0$, for which every realizable core is empty.

Suppose $z\geq 1$. Every block of depth at least z has a unique layer-z point. Since all cores lie in the target chain, these points are one common point u_z^*. Its histogram score is at least $t_i/6$, and (A.21) together with (A.28) leaves selected score at least

$$\frac{t_i}{6}-\frac{t_i}{12}=\frac{t_i}{12}>0. \tag{A.47}$$

The selection is not $\perp$. Positive score and uniqueness at depth force it to equal u_z^*, so the transformed output is

$$h_{\mathrm{out}}^\circ=\mathbf{1}_{I_i(u_z^*)}\leq g_*. \tag{A.48}$$

Among the at least $t_i/6$ blocks of depth at most z, one, say b_0, is good. If its core is empty, its indicator is zero. Otherwise the core minimum has depth at most z, so its principal upset is contained in $I_i(u_z^*)$. Thus

$$h_{b_0}^\circ\leq h_{\mathrm{out}}^\circ\leq g_*. \tag{A.49}$$

For binary functions under this pointwise order, disagreement with g_* is reverse monotone, and hence

$$\Pr[h_{\mathrm{out}}^\circ\neq g_*]\leq\Pr[h_{b_0}^\circ\neq g_*]\leq\frac{1}{64}. \tag{A.50}$$

Reference XOR cancels pointwise:

$$\mathbf{1}\{\bar{f}_i^\circ(q)\oplus h_{\mathrm{out}}^\circ(\tau_i(q))\neq\bar{c}_i(q)\}=\mathbf{1}\{h_{\mathrm{out}}^\circ(\tau_i(q))\neq g_*(\tau_i(q))\}. \tag{A.51}$$

The secondary quotient preserves this indicator, while Lemma A.6 further pulls it to raw factor risk without residual. Finally,

$$3\eta = \frac{3}{16384} < \frac{1}{4096}, \tag{A.52}$$

which proves (A.45). The confidence $1/48$ was used only as a fixed block marginal; the concentration step avoids any confidence fixed point or additional logarithm in the quota. □

Proposition A.6 (Measurable private factor certificate). *Under Assumptions 2, 3, and 4, for $K_Y \geq K_*$, the fixed permutation-symmetrized quotient-first rule $\bar{A}_i^{\mathrm{Yan}}$ is a Markov kernel on exactly q_i records, is replacement-$(\varepsilon/2, \delta/2)$-DP on every input, and satisfies the unpadded iid utility bound (A.45). Its hidden constants depend only on the fixed numerical accuracy, confidence, privacy split, and universal constants in the cited private-median, VC-generalization, and choosing results, never on the class, quotient, distribution, d_i, ε, or δ.*

Proof. Lemma A.7 derives the finite order and all principal-core properties, including the positive endpoint. The improper core class and its empirical consistency are Lemma A.8. The exact leaving-and-entering histogram, countable extension, and every empty-support privacy direction are proved in Lemma A.9. The source thresholds, Hoeffding confidence, choosing slack, and exact ceiling-safe bound $N_i \leq q_i$, including $d_i = 1$, are Lemma A.10. Proposition A.3 then supplies the total measurable transition, and Proposition A.4 supplies all-input factor privacy. Lemma A.11 and Proposition A.5 provide the unpadded fixed- confidence risk guarantee. The construction remains the same one-factor quotient-first rule when $k = 1$; no cross-factor restriction, finite-support condition, or properness condition has been introduced. □

A.3 Routed product kernel and support-two privacy

Lemma A.12 (Measurable quotient routing and padded prefixes). *Under Assumptions 1 and 3, fix $n \in \mathbb{N}$. For every factor i, the count $J_i : Z^n \to \{0, \dots, n\}$, the map*

$$\bar{T}_i : Z^n \longrightarrow (Q_i \times \{0,1\})^{q_i}$$

that takes the first q_i routed records and pads a shortage, and the finite product $(\bar{T}_1, \dots, \bar{T}_k)$ are measurable on every labeled input, independently of realizability.

Proof. Let $Y_i = Q_i \times \{0,1\}$ be countable discrete, adjoin $\dagger_i \notin Y_i$, and define

$$\varphi_i(x, y) = \begin{cases} (\kappa_i(x), y), & x \in X_i, \\ \dagger_i, & x \notin X_i. \end{cases} \tag{A.53}$$

For any subset of the countable codomain, the preimage is a countable union of measurable sets $\kappa_i^{-1}(\{q\}) \times \{b\}$, together, when appropriate, with $(X \setminus X_i) \times \{0,1\}$. Thus φ_i is measurable and

$$J_i(S) = \sum_{j=1}^{n} \mathbf{1}\{\varphi_i(z_j) \neq \dagger_i\} \tag{A.54}$$

is measurable.

For $1 \leq r \leq q_i$ and $1 \leq j \leq n$, let

$$F_{i,r,j} = \left\{ S : \varphi_i(z_j) \neq \dagger_i,\ \sum_{\ell < j} \mathbf{1}\{\varphi_i(z_\ell) \neq \dagger_i\} = r - 1 \right\}. \tag{A.55}$$

This is the event that position j is the r-th routed record. The r-th padded-prefix coordinate equals that routed record on the unique such event and equals $\bar{z}_i^\circ$ when $J_i < r$. Hence, for $u \in Y_i$,

$$\{\bar{T}_{i,r} = u\} = \big(\{J_i < r\} \cap \{\bar{z}_i^\circ = u\}\big) \cup \bigcup_{j=1}^{n} \big(F_{i,r,j} \cap \{\varphi_i(z_j) = u\}\big), \tag{A.56}$$

where the first set is empty if $u \neq \bar{z}_i^\circ$. This fiber is measurable. Countable discreteness gives coordinate and tuple measurability, and finite products preserve it. No target, selector, or consistency event is used. $\square$

Lemma A.13 (One-edit stability of a padded prefix multiset)**.** *Let Y be a set, $q \geq 1$, and $p \in Y$. For a finite list $w = (w_1, \ldots, w_J)$, define*

$$\mathrm{Pad}_{q,p}(w) = (w_1, \ldots, w_{\min\{J,q\}}, \underbrace{p, \ldots, p}_{q-\min\{J,q\}\ \textit{copies}}).$$

If w' is obtained by one replacement, deletion, or insertion, the two padded-prefix multisets are equal or differ by one multiset replacement. Consequently, the two tuples can be permuted to become equal or to differ in exactly one coordinate.

Proof. The complete calculation is as follows:

edit	position/length	sole multiset change
$w_r \mapsto v$	$r > q$	none
$w_r \mapsto v$	$r \leq q$	$w_r \mapsto v$
delete w_r	$r > q$	none
delete w_r	$r \leq q,\ J \geq q+1$	$w_r \mapsto w_{q+1}$
delete w_r	$r \leq q,\ J \leq q$	$w_r \mapsto p$
insert v at r	$r > q$	none
insert v at $r \leq q,\ J \geq q$	$w_q \mapsto v$	
insert v at $r \leq q,\ J < q$	$p \mapsto v$	

For instance, after deleting w_r with $r \leq q < J$, shifted entries $w_{r+1}, \ldots, w_q$ occur in both multisets; only w_r leaves and w_{q+1} enters. If the list ends within the prefix, padding enters. Insertion is the reverse calculation. Coinciding entering and leaving values can only turn the nominal replacement into equality.

The multiset intersection of two size-q multisets differing by at most one replacement has multiplicity at least $q-1$. Align these common occurrences in the first $q-1$ positions and put the possible unmatched occurrences last. The resulting permutations give equality or ordered one-coordinate replacement, including $q = 1$. $\square$

Proposition A.7 (Support-two adjacency of routed inputs)**.** *Under Assumption 1, Lemma A.13, and Proposition A.3, if $S, S' \in Z^n$ differ by one replacement, there is $I(S,S') \subseteq \{1, \ldots, k\}$, $|I(S,S')| \leq 2$, such that unaffected factor-input multisets agree; for each affected factor the two inputs can be permuted into one-coordinate replacement neighbors; and the corresponding factor output laws are unchanged by those permutations. This includes arbitrary labels, changed rows outside the selected prefixes, same-block and cross-block moves, and every full/short padding boundary.*

Proof. Let the changed row be $z = (x,y)$ in S and $z' = (x', y')$ in S', and let $x \in X_a$, $x' \in X_b$. Write $R_i(S)$ for the ordered list of quotient records routed to factor i; then $\bar{T}_i(S) = \mathrm{Pad}_{q_i, \bar{z}_i^\circ}(R_i(S))$.

If $a = b$, only R_a changes, by one replacement at the same local rank. Lemma A.13 makes the selected multiset equal if the rank exceeds q_a, and otherwise one-replacement different. Thus $I \subseteq \{a\}$.

If $a \neq b$, every other routed list is unchanged. The leaving list deletes one record and the entering list inserts one record at the rank determined by the unchanged global position. The lemma gives at most one multiset replacement in each: a deletion from a full prefix brings in the next real record, or padding if none remains; deletion from a short prefix brings in padding; insertion into a full prefix replaces the former last selected record; insertion into a short prefix replaces padding; and an edit after the selected real prefix changes nothing. Hence $I \subseteq \{a, b\}$.

The same lemma supplies permutations σ_i, σ_i' aligning all common multiplicities. Permutation invariance from Proposition A.3 yields, for every measurable $E_i \subseteq \mathcal{H}_i$,

$$\begin{aligned} \bar{A}_i^{\mathrm{Yan}}(\bar{T}_i(S), E_i) &= \bar{A}_i^{\mathrm{Yan}}(\sigma_i \bar{T}_i(S), E_i), \\ \bar{A}_i^{\mathrm{Yan}}(\bar{T}_i(S'), E_i) &= \bar{A}_i^{\mathrm{Yan}}(\sigma_i' \bar{T}_i(S'), E_i). \end{aligned} \tag{A.57}$$

The permutations are only a coupling for this fixed adjacent pair; the learner need not choose them measurably. Padding and arbitrary labels lie in the total factor input space. At $k = 1$, necessarily $a = b = 1$, so at most the sole factor changes. $\square$

Proposition A.8 (Measurable routed product kernel and decoder)**.** *Under Assumptions 1 and 3, Lemmas A.5 and A.6, Proposition A.3, and Lemma A.12,*

$$\bar{A}_n^{\oplus,Q}(S, d\bar{h}) = \bigotimes_{i=1}^{k} \bar{A}_i^{\mathrm{Yan}}(\bar{T}_i(S), d\bar{h}_i) \tag{A.58}$$

defines a Markov kernel into $(\mathcal{H}^{\oplus}, \mathscr{H}^{\oplus})$, with independent factor randomness. Piecewise decoding is measurable under the finite-evaluation- cylinder convention, so its pushforward is a decoded Markov kernel. For every fixed target and distribution, every displayed risk event is measurable in the tuple output.

Proof. For $E_i \in \mathscr{H}_i$,

$$K_i(S, E_i) = \bar{A}_i^{\mathrm{Yan}}(\bar{T}_i(S), E_i) \tag{A.59}$$

is measurable in S by Lemma A.12 and the factor-kernel property. For fixed S, the finite product of these probability measures exists, and for every measurable rectangle,

$$\bar{A}_n^{\oplus,Q}\left(S, \prod_{i=1}^{k} E_i\right) = \prod_{i=1}^{k} K_i(S, E_i), \tag{A.60}$$

which is measurable in S.

Let $\mathcal{D}$ be the events in $\mathscr{H}^{\oplus}$ whose probability under (A.58) is measurable in S. It contains the whole space, is closed under complements because the complementary probability is one minus the original probability, and is closed under countable disjoint unions by pointwise convergence of partial sums. It is therefore a Dynkin system. Measurable rectangles form a pi-system contained in $\mathcal{D}$ by (A.60) and generate $\mathscr{H}^{\oplus}$. The pi-lambda theorem gives $\mathcal{D} = \mathscr{H}^{\oplus}$, establishing kernel measurability for arbitrary joint events.

Let $\mathcal{H}_\Sigma$ denote the Σ-measurable binary hypotheses and let $\mathscr{G}_{\mathrm{ev}}$ be generated by their finite evaluation cylinders. Lemma A.5 makes

$$\mathrm{Dec} : \mathcal{H}^{\oplus} \to \mathcal{H}_\Sigma, \qquad \mathrm{Dec}(\bar{h}) = h_{\bar{h}},$$

measurable. Thus

$$\bar{A}_n^{\oplus,\mathrm{dec}}(S,G) = \bar{A}_n^{\oplus,Q}(S,\mathrm{Dec}^{-1}(G)) \tag{A.61}$$

is a kernel for $G \in \mathscr{G}_{\mathrm{ev}}$. Finally, Lemma A.6 gives

$$\{\bar{h} : R_D(h_{\bar{h}}, c) \le t\} \in \mathscr{H}^{\oplus} \tag{A.62}$$

for fixed c, D, t. This is the required output convention without a standard-Borel assumption on the raw instance space. □

Proposition A.9 (Support-two composition for joint output events)**.** *Under Assumption 4, Proposition A.4, and Propositions A.7 and A.8, for every n, adjacent $S, S' \in Z^n$, and $E \in \mathscr{H}^{\oplus}$,*

$$\bar{A}_n^{\oplus,Q}(S,E) \le e^{\varepsilon}\bar{A}_n^{\oplus,Q}(S',E) + \delta. \tag{A.63}$$

The same inequality holds for every measurable deterministic postprocessing, including decoding. At most two factor privacy costs are used, with no further dependence on k.

Proof. Fix S, S', let $I = I(S,S')$, and put $m = |I| \le 2$. Define

$$\mu_i = \bar{A}_i^{\mathrm{Yan}}(\bar{T}_i(S), \cdot), \qquad \nu_i = \bar{A}_i^{\mathrm{Yan}}(\bar{T}_i(S'), \cdot).$$

For $i \notin I$, permutation invariance gives $\mu_i = \nu_i$. For $i \in I$, (A.57) and factor privacy give, for every $B \in \mathscr{H}_i$,

$$\mu_i(B) \le a\nu_i(B) + d, qquad \nu_i(B) \le a\mu_i(B) + d, qquad a = e^{\varepsilon/2}, \quad d = \delta/2. \tag{A.64}$$

We prove composition on the full product sigma-field. For $i \in I$, let $\lambda_i = \mu_i + \nu_i$, with densities $u_i = d\mu_i/d\lambda_i$ and $v_i = d\nu_i/d\lambda_i$. The elementary hockey-stick identity

$$\sup_B(\mu_i(B) - a\nu_i(B)) = \int (u_i - av_i)_+ \, d\lambda_i$$

follows by taking the positive-density set, and (A.64) in both directions gives

$$\int (u_i - av_i)_+ \, d\lambda_i \le d, qquad \int (v_i - au_i)_+ \, d\lambda_i \le d. \tag{A.65}$$

Define subprobability densities

$$u_i^0 = \min\{u_i, av_i\}, qquad v_i^0 = \min\{v_i, au_i\}. \tag{A.66}$$

Their measures μ_i^0, ν_i^0 satisfy

$$\mu_i^0(\mathcal{H}_i) \ge 1 - d, qquad \nu_i^0(\mathcal{H}_i) \ge 1 - d, \tag{A.67}$$

and pointwise

$$u_i^0 \le av_i^0, qquad v_i^0 \le au_i^0. \tag{A.68}$$

For example, if $v_i^0 = v_i$, the first inequality follows from $u_i^0 \le av_i$; if $v_i^0 = au_i$, then $av_i^0 = a^2 u_i \ge u_i \ge u_i^0$. The other direction is symmetric.

For unaffected indices set $\mu_i^0 = \nu_i^0 = \mu_i$, and form

$$\mu = \bigotimes_i \mu_i, quad \nu = \bigotimes_i \nu_i, quad \mu^0 = \bigotimes_i \mu_i^0, quad \nu^0 = \bigotimes_i \nu_i^0. \tag{A.69}$$

Product densities under the product dominating measure show $\mu^0 \le \mu$, $\nu^0 \le \nu$, and $\mu^0 \le a^m \nu^0$. Moreover,

$$\mu^0(\mathcal{H}^{\oplus}) = \prod_{i \in I} \mu_i^0(\mathcal{H}_i) \ge (1-d)^m \ge 1 - md. \tag{A.70}$$

Thus every joint event E satisfies

$$\begin{aligned} \mu(E) &\le \mu^0(E) + 1 - \mu^0(\mathcal{H}^{\oplus}) \\ &\le a^m \nu^0(E) + md \\ &\le e^{m\varepsilon/2} \nu(E) + m\delta/2 \\ &\le e^{\varepsilon} \nu(E) + \delta. \end{aligned} \tag{A.71}$$

These are exactly the output laws in (A.58), proving (A.63). For $m = 0$ there is equality; for $m = 1$ the stronger factor parameters remain; and for $m = 2$ each factor budget is used once. No unchanged factor is composed and no rectangle restriction appears. Reversing S, S' proves the other ordered direction. For a measurable map g, apply the inequality to $g^{-1}(G)$; taking $g = \mathrm{Dec}$ proves decoded privacy. Since factor privacy held on all inputs, the proof includes inconsistent labels and padding. □

Proposition A.10 (Measurable routed privacy certificate)**.** *Under Assumptions 1, 3, and 4, the routed quotient-first rule $\bar{A}_n^{\oplus,Q}$ is a measurable Markov kernel for every fixed sample size, its decoded risk events are measurable, and it is replacement- (ε, δ)-DP on every labeled input pair. A replacement affects at most two factor laws. When $k = 1$, the rule is the sole routed factor kernel and in fact retains the stronger $(\varepsilon/2, \delta/2)$ privacy parameters.*

Proof. Lemma A.12 proves measurable raw-to- quotient preprocessing. Lemma A.13 accounts for every replacement, deletion, insertion, and padding transition. Proposition A.7 applies it to a global record replacement and invokes permutation invariance only after aligning the multisets. Proposition A.8 gives the full joint kernel, decoder, and risk-event convention. Proposition A.9 then proves arbitrary- event privacy with exact cost at most $(2\varepsilon/2, 2\delta/2) = (\varepsilon, \delta)$. Its $m = 1$ branch is the claimed one-factor specialization. □

A.4 Weighted occupancy shortages

Lemma A.14 (Marginal occupancy and weighted expectation identity)**.** *Under Assumption 1 and Lemmas A.1 and A.5, fix $n \in \mathbb{N}$, $c \in C$, a probability measure D on (X, Σ), and an iid labeled sample $S = ((x_j, c(x_j)))_{j=1}^n \sim D_c^n$. With*

$$\rho_i = D(X_i), \qquad J_i = \left|\{j : x_j \in X_i\}\right|, \qquad W_{\text{short}} = \sum_{i=1}^k \rho_i \mathbf{1}\{J_i < q_i\},$$

one has

$$J_i \sim \mathrm{Bin}(n, \rho_i) \quad \textit{for every } i. \tag{A.72}$$

If $\rho_i = 0$, then $\rho_i \mathbf{1}\{J_i < q_i\} = 0$ for every sample realization and $J_i = 0$ almost surely. Moreover, W_{short} is integrable and

$$\mathbb{E} W_{\text{short}} = \sum_{i=1}^k \rho_i \Pr(J_i < q_i). \tag{A.73}$$

Neither conclusion requires independence among the counts for distinct factors.

Proof. For each factor define $\bar{b}^{(i)} \in \mathcal{H}^{\oplus}$ by

$$\bar{b}^{(i)}_{\ell}(q) = \mathbf{1}\{\ell = i\}, \qquad 1 \le \ell \le k, \quad q \in Q_\ell.$$

Lemma A.5 makes the decoded hypothesis measurable, and the whole-domain partition in Assumption 1 gives $h_{\bar{b}^{(i)}} = \mathbf{1}_{X_i}$. Thus $X_i \in \Sigma$. Lemma A.1 gives quotient representatives $\bar{c}_i$ of the restrictions of c; their tuple decodes exactly to c, so Lemma A.5 also makes c measurable. Consequently D_c, D_c^n, the masses ρ_i, and the block indicators below are well defined.

For fixed i, put

$$I_{j,i} = \mathbf{1}\{x_j \in X_i\}, \qquad 1 \le j \le n.$$

The iid sampling assumption makes $I_{1,i}, \dots, I_{n,i}$ independent Bernoulli variables with success probability $D(X_i) = \rho_i$. Since $J_i = \sum_{j=1}^{n} I_{j,i}$, this proves (A.72). Counts for distinct factors can be dependent because each observation belongs to exactly one block, but no subsequent argument uses their joint independence.

When $\rho_i = 0$, multiplication by the deterministic weight gives

$$\rho_i \mathbf{1}\{J_i < q_i\} = 0$$

for every realization. Also $I_{j,i} = 0$ almost surely for each of the finitely many sample indices, hence $J_i = 0$ almost surely. This handles zero mass without forming a conditional law on X_i.

The finite measurable partition of all of X gives

$$0 \le W_{\text{short}} \le \sum_{i=1}^{k} \rho_i = D(X) = 1.$$

Thus all summands and their sum are integrable. Finite linearity of expectation now yields

$$\mathbb{E}W_{\text{short}} = \sum_{i=1}^{k} \rho_i \mathbb{E}\mathbf{1}\{J_i < q_i\} = \sum_{i=1}^{k} \rho_i \Pr(J_i < q_i),$$

which is (A.73). This is a finite-sum identity, not a limit interchange or a union bound. $\square$

Lemma A.15 (Explicit heavy-factor shortage tail)**.** *Let $J \sim \mathrm{Bin}(n, p)$, put $\mu = np$, and let $q \in \mathbb{N}$ satisfy $q \ge 1$. If $\mu \ge 128q$, then*

$$\Pr(J < q) \le \exp\left(-\frac{16129}{256} q\right) \le e^{-16}. \tag{A.74}$$

Proof. For any $u \in (0, 1)$, exponential Markov gives

$$\begin{aligned} \Pr\big(J \le (1-u)\mu\big) &= \Pr\big(e^{-uJ} \ge e^{-u(1-u)\mu}\big) \\ &\le e^{u(1-u)\mu}\, \mathbb{E}e^{-uJ}. \end{aligned} \tag{A.75}$$

The binomial transform and $1 + v \le e^v$ give

$$\mathbb{E}e^{-uJ} = (1 - p + pe^{-u})^n \le \exp\big(\mu(e^{-u} - 1)\big). \tag{A.76}$$

For $u \ge 0$,

$$e^{-u} \le 1 - u + \frac{u^2}{2}. \tag{A.77}$$

Indeed, for $f(u) = 1 - u + u^2/2 - e^{-u}$, one has $f(0) = f'(0) = 0$ and $f''(u) = 1 - e^{-u} \geq 0$. Substitution of (A.76) and (A.77) into (A.75) yields

$$\Pr\big(J \leq (1-u)\mu\big) \leq \exp\left(\mu\left[u(1-u) - u + \frac{u^2}{2}\right]\right) = \exp\left(-\frac{\mu u^2}{2}\right). \tag{A.78}$$

Under the stated hypothesis, $q \leq \mu/128$, and hence

$$\{J < q\} \subseteq \left\{J \leq \frac{\mu}{128}\right\}. \tag{A.79}$$

Apply (A.78) with $u = 127/128$. Then

$$\frac{\mu}{2}\left(\frac{127}{128}\right)^2 \geq \frac{128q}{2}\frac{127^2}{128^2} = \frac{16129}{256}q. \tag{A.80}$$

Since $q \geq 1$,

$$\frac{16129}{256}q \geq \frac{16129}{256} > \frac{4096}{256} = 16. \tag{A.81}$$

Equations (A.78)– (A.81) prove (A.74). $\square$

Proposition A.11 (Weighted shortage certificate without balance)**.** *Under Assumption 1, Lemmas A.1, A.5, A.14, and A.15, fix $n \in \mathbb{N}$, $c \in C$, a probability measure D, and $S \sim D_c^n$. For the setting-defined masses, counts, quotas, and shortage mass, suppose $q_i \geq 1$ and put $Q_\oplus = \sum_{i=1}^k q_i$. Then, for every mass vector induced by D,*

$$\mathbb{E}W_{\text{short}} \leq \frac{128Q_\oplus}{n} + e^{-16}. \tag{A.82}$$

No balance or positive-mass condition, factor-count independence, or union bound over factors is required. If $k = 1$ and $n \geq q_1$, then

$$W_{\text{short}} = 0 \quad \textit{for every sample realization.} \tag{A.83}$$

Proof. Put $\mu_i = n\rho_i$ and split the finite factor set as

$$\mathcal{L} = \{i : \mu_i < 128q_i\}, \qquad \mathcal{H} = \{i : \mu_i \geq 128q_i\}. \tag{A.84}$$

For a light factor, $\rho_i < 128q_i/n$, and a shortage probability is at most one. Therefore the universally valid non-strict bound, including the empty-light case, is

$$\sum_{i\in\mathcal{L}} \rho_i \Pr(J_i < q_i) \leq \sum_{i\in\mathcal{L}} \rho_i \leq \frac{128}{n}\sum_{i\in\mathcal{L}} q_i \leq \frac{128Q_\oplus}{n}. \tag{A.85}$$

A zero-mass factor is light because $q_i \geq 1$, and its summand is identically zero by Lemma A.14; no conditional factor law is needed.

For $i \in \mathcal{H}$, Lemma A.14 gives $J_i \sim \mathrm{Bin}(n, \rho_i)$, while $n\rho_i \geq 128q_i$. Lemma A.15 thus gives $\Pr(J_i < q_i) \leq e^{-16}$. Because the measurable blocks partition all of X, $\sum_i \rho_i = 1$, so

$$\sum_{i\in\mathcal{H}} \rho_i \Pr(J_i < q_i) \leq e^{-16}\sum_{i\in\mathcal{H}} \rho_i \leq e^{-16}. \tag{A.86}$$

This is a weighted average of marginal shortage probabilities, not a bound on an all-factor union event.

Splitting the exact identity (A.73) according to (A.84) and using (A.85)–(A.86) proves (A.82).

Finally, if $k = 1$, Assumption 1 gives $X_1 = X$, so $\rho_1 = 1$ and every record is routed to factor one. Hence $J_1 = n$ pointwise. When $n \geq q_1$, the shortage event is empty and

$$W_{\text{short}} = \rho_1 \mathbf{1}\{J_1 < q_1\} = 0,$$

proving (A.83) exactly rather than through the generic residual bound. $\square$

A.5 From factor utility to global PAC utility

Lemma A.16 (Measurability of routed bad-factor mass)**.** *Under Assumptions 1 and 3 and Proposition A.3, fix $n \in \mathbb{N}$, $c \in C$, and a probability measure D on (X, Σ). In the joint experiment consisting of $S \sim D_c^n$ and the routed factor outputs, every J_i, routed input $\bar{T}_i(S)$, factor risk, and event B_i is measurable. Consequently W_{bad} is measurable and integrable, with*

$$0 \le W_{\mathrm{bad}} \le 1. \tag{A.87}$$

Proof. Assumption 3 gives $X_i \in \Sigma$ and makes $\kappa_i : X_i \to Q_i$ measurable into a finite or countable discrete quotient. Hence

$$J_i(S) = \sum_{j=1}^{n} \mathbf{1}\{x_j \in X_i\} \tag{A.88}$$

is measurable on the labeled sample space Z^n.

For $1 \le r \le q_i$ and $1 \le j \le n$, let

$$E_{i,r,j} = \left\{ x_j \in X_i,\ \sum_{\ell < j} \mathbf{1}\{x_\ell \in X_i\} = r - 1 \right\}. \tag{A.89}$$

For fixed i, r, these measurable events are disjoint in j. On $E_{i,r,j}$, the r-th routed record is $(\kappa_i(x_j), y_j)$, while on the measurable complement $\{J_i < r\}$ it is the fixed padding record $\bar{z}_i^{\circ}$. Finite measurable pasting makes every routed coordinate measurable into the countable discrete space $Q_i \times \{0,1\}$. Thus

$$\bar{T}_i : Z^n \longrightarrow (Q_i \times \{0,1\})^{q_i} \tag{A.90}$$

is measurable as an ordered prefix, not merely as a multiset.

Write

$$K_i(t, E) = \bar{A}_i^{\mathrm{Yan}}(t, E), \qquad E \in \mathscr{H}_i. \tag{A.91}$$

Proposition A.3 and (A.90) make $S \mapsto K_i(\bar{T}_i(S), E)$ measurable. The prescribed independent factor randomizations define the finite product kernel

$$K^{\oplus}(S, d\bar{h}) = \bigotimes_{i=1}^{k} K_i(\bar{T}_i(S), d\bar{h}_i) \tag{A.92}$$

on $(\mathcal{H}^{\oplus}, \mathscr{H}^{\oplus})$. For every measurable rectangle $E_1 \times \cdots \times E_k$, its transition probability is $\prod_i K_i(\bar{T}_i(S), E_i)$, a measurable function of S. Measurable rectangles form a generating pi-system. The events whose transition probabilities are measurable form a Dynkin system: they contain the whole space and are closed under complements and countable disjoint unions. The pi-lambda theorem therefore extends measurability to all of $\mathscr{H}^{\oplus}$. For fixed S, (A.92) is the ordinary finite product measure. Consequently

$$\mathbb{P}_{c,D,n}(dS, d\bar{h}) = D_c^n(dS) K^{\oplus}(S, d\bar{h}) \tag{A.93}$$

is a well-defined joint law. If $\bar{H}_i$ is the i-th output coordinate, then

$$\mathbb{P}_{c,D,n}(\bar{H}_i \in E \mid S) = K_i(\bar{T}_i(S), E). \tag{A.94}$$

Only this marginal identity, not independence of failure events, is used below.

Fix i with $\rho_i > 0$. Enumerate the finite or countable Q_i as $q_{i,1}, q_{i,2}, \ldots$, stopping in the finite case, and define

$$
\begin{aligned}
r_i(\bar{h}_i) &= \Pr_{q \sim \bar{D}_i}[\bar{h}_i(q) \neq \bar{c}_i(q)] \\
&= \sum_m \bar{D}_i(\{q_{i,m}\})\mathbf{1}\{\bar{h}_i(q_{i,m}) \neq \bar{c}_i(q_{i,m})\}.
\end{aligned} \tag{A.95}
$$

Its finite partial sums are $\mathscr{H}_i$-measurable because coordinate evaluations generate $\mathscr{H}_i$, and the series is their nondecreasing pointwise limit. Thus r_i and the strict event $\{r_i(\bar{H}_i) > 1/64\}$ are measurable. This uses countability only in the quotient, not in the raw support of D_i.

On the joint space (A.93), for $\rho_i > 0$,

$$
\mathsf{B}_i = \{J_i < q_i\} \cup \big(\{J_i \geq q_i\} \cap \{r_i(\bar{H}_i) > 1/64\}\big), \tag{A.96}
$$

and $\mathsf{B}_i = \varnothing$ when $\rho_i = 0$. Every bad event is therefore measurable. The finite measurable partition gives $\sum_i \rho_i = 1$, so

$$
W_{\mathrm{bad}} = \sum_{i=1}^{k} \rho_i \mathbf{1}_{\mathsf{B}_i} \tag{A.97}
$$

is measurable and satisfies (A.87), proving integrability over both sample and learner randomness. $\square$

Lemma A.17 (Conditional iid law of every unpadded first prefix)**.** *Under Assumptions 1 and 3, Proposition A.5, and Lemma A.16, fix n, c, D and the joint experiment above. For every i with $\rho_i > 0$, conditional on any positive-probability block-position vector containing at least q_i occurrences of i, the unpadded input $\bar{T}_i$ has law $((\bar{D}_i)_{\bar{c}_i})^{q_i}$, and*

$$
\Pr\left[r_i(\bar{H}_i) > \frac{1}{64} \;\middle|\; \textit{that block-position vector}\right] \leq \frac{1}{4096}. \tag{A.98}
$$

If the vector has fewer than q_i occurrences, the prefix is padded and no factor-utility assertion is made. If $\rho_i = 0$, neither D_i nor $\bar{D}_i$ is formed and $\mathsf{B}_i = \varnothing$.

Proof. Let $G_j \in \{1, \ldots, k\}$ be the unique block index with $x_j \in X_{G_j}$, and put $G = (G_1, \ldots, G_n)$. This is a measurable finite-valued random vector. For $a = (a_1, \ldots, a_n) \in \{1, \ldots, k\}^n$, iid sampling gives

$$
\Pr(G = a) = \prod_{j=1}^{n} \rho_{a_j}. \tag{A.99}
$$

Suppose this probability is positive. Then all $\rho_{a_j} > 0$. For measurable $E_j \subseteq X_{a_j}$,

$$
\begin{aligned}
&\Pr(x_1 \in E_1, \ldots, x_n \in E_n \mid G = a) \\
&= \frac{\prod_{j=1}^{n} D(E_j)}{\prod_{j=1}^{n} \rho_{a_j}} = \prod_{j=1}^{n} D_{a_j}(E_j).
\end{aligned} \tag{A.100}
$$

For general measurable coordinate sets, intersect them with their prescribed blocks. Equality on measurable rectangles identifies the full conditional product law. This conditions on one atom of a finite partition and needs no regular conditional distribution on the raw space.

Fix i with $\rho_i > 0$, and define the labeled quotient law

$$
\nu_i = \big(q \mapsto (q, \bar{c}_i(q))\big)_{\#}\bar{D}_i =: (\bar{D}_i)_{\bar{c}_i} \quad \text{on } Q_i \times \{0, 1\}. \tag{A.101}
$$

On a positive-probability atom $\{G = a\}$, (A.100), deterministic target labels, and measurable quotient pushforward show that the records at positions with $a_j = i$ are independent with common law ν_i. When there are at least q_i such positions, their first q_i positions are deterministic functions of a, and hence

$$\mathcal{L}(\bar{T}_i \mid G = a) = \nu_i^{q_i}. \tag{A.102}$$

Conditioning on positions has not conditioned on the within-block data or on learner randomness.

By (A.94), after $\bar{T}_i = t$ is fixed the factor output has law $K_i(t, \cdot)$. Therefore (A.102) gives

$$\begin{aligned} &\Pr\left(r_i(\bar{H}_i) > \frac{1}{64} \mid G = a\right) \\ &\quad = \int K_i\left(t, \left\{\bar{h}_i : r_i(\bar{h}_i) > \frac{1}{64}\right\}\right) \nu_i^{q_i}(dt) \le \frac{1}{4096}, \end{aligned} \tag{A.103}$$

where the last inequality is Proposition efprop:step-002-factor-utility. It includes the iid factor prefix and all internal factor randomness; no other factor is conditioned to be good.

If a has fewer than q_i occurrences of i, at least one padding record is inserted, so the factor-utility premise is unavailable. This entire branch is contained in $\{J_i < q_i\} \subseteq \mathsf{B}_i$. If $\rho_i = 0$, every atom containing i has probability zero by (A.99); the definition instead sets $\mathsf{B}_i = \varnothing$, with zero weight in every risk sum. $\square$

Proposition A.12 (Weighted bad-mass expectation without factor unions)**.** *Under Assumptions 1, 3, and 4, Propositions A.5 and A.11, and Lemmas A.16 and A.17, for every $n \in \mathbb{N}$, $c \in C$, and probability measure D,*

$$\mathbb{E}W_{\text{bad}} \le \frac{128 Q_{\oplus}}{n} + e^{-16} + \frac{1}{4096}. \tag{A.104}$$

The expectation is over the sample and all learner randomness. No independence among output or bad events and no factor union bound is used.

Proof. For $\rho_i > 0$, let

$$F_i = \left\{J_i \ge q_i,\ r_i(\bar{H}_i) > \frac{1}{64}\right\}. \tag{A.105}$$

The events $\{J_i < q_i\}$ and F_i are disjoint, and (A.96) gives

$$\Pr(\mathsf{B}_i) = \Pr(J_i < q_i) + \Pr(F_i). \tag{A.106}$$

Summing (A.103) over the finitely many positive-probability position atoms with at least q_i occurrences gives

$$\begin{aligned} \Pr(F_i) &= \sum_{\substack{a:\Pr(G=a)>0 \\ |\{j:a_j=i\}|\ge q_i}} \Pr(G = a) \Pr\left(r_i(\bar{H}_i) > \frac{1}{64} \mid G = a\right) \\ &\le \frac{1}{4096} \sum_{\substack{a:\Pr(G=a)>0 \\ |\{j:a_j=i\}|\ge q_i}} \Pr(G = a) \\ &= \frac{1}{4096} \Pr(J_i \ge q_i) \le \frac{1}{4096}. \end{aligned} \tag{A.107}$$

This is a one-factor marginal calculation, not a bound on a union or a factorization of joint probabilities.

For $\rho_i = 0$, $\mathsf{B}_i = \varnothing$ and its weighted probability is exactly zero. Finite linearity, integrability from Lemma A.16, (A.106)– (A.107), and $\sum_{i:\rho_i>0} \rho_i = 1$ yield

$$\begin{aligned}\mathbb{E}W_{\text{bad}} &= \sum_{i=1}^{k} \rho_i \Pr(\mathsf{B}_i) \\ &\le \sum_{i=1}^{k} \rho_i \Pr(J_i < q_i) + \frac{1}{4096} \sum_{i:\rho_i>0} \rho_i \\ &= \mathbb{E}W_{\text{short}} + \frac{1}{4096}.\end{aligned} \tag{A.108}$$

Proposition A.11 supplies

$$\mathbb{E}W_{\text{short}} \le \frac{128 Q_{\oplus}}{n} + e^{-16},$$

which proves (A.104). Arbitrarily many small-mass factors add no unweighted probability term. □

Proposition A.13 (Pointwise domination of exact global risk)**.** *Under Assumptions 1 and 3 and Lemma A.16, for every fixed n, c, D and every sample/output pair in the joint experiment,*

$$R_D(h_{\bar{H}}, c) \le \frac{1}{64} + W_{\text{bad}}. \tag{A.109}$$

The global risk is measurable. The inequality holds on shortages and for arbitrary factor outputs, without a simultaneous all-factor good event.

Proof. For $\rho_i > 0$, if B_i does not occur, then $J_i \ge q_i$ and $r_i(\bar{H}_i) \le 1/64$; if it occurs, binary risk is at most one. Pointwise,

$$r_i(\bar{H}_i) \le \frac{1}{64}\mathbf{1}_{\mathsf{B}_i^c} + \mathbf{1}_{\mathsf{B}_i} = \frac{1}{64} + \frac{63}{64}\mathbf{1}_{\mathsf{B}_i} \le \frac{1}{64} + \mathbf{1}_{\mathsf{B}_i}. \tag{A.110}$$

This is a consequence of the bad-event definition, not a pointwise use of the probabilistic utility theorem.

For every fixed quotient tuple $\bar{h}$, block disjointness, the positive-mass conditional laws, deterministic target labels, and quotient pushforward give the exact identity

$$\begin{aligned}R_D(h_{\bar{h}}, c) &= \sum_{i:\rho_i>0} D\big(\{x \in X_i : h_{\bar{h}}(x) \ne c(x)\}\big) \\ &= \sum_{i:\rho_i>0} \rho_i \Pr_{q\sim\bar{D}_i}[\bar{h}_i(q) \ne \bar{c}_i(q)] \\ &= \sum_{i:\rho_i>0} \rho_i r_i(\bar{h}_i).\end{aligned} \tag{A.111}$$

This is exact for arbitrary raw support. Together with (A.95), it also proves measurability of the global risk as a function of $\bar{H}$.

Apply (A.110) in (A.111), use $\sum_{i:\rho_i>0} \rho_i = 1$, and note that zero-mass factors have zero weight and empty bad event. Then

$$R_D(h_{\bar{H}}, c) \le \sum_{i:\rho_i>0} \rho_i \left(\frac{1}{64} + \mathbf{1}_{\mathsf{B}_i}\right) = \frac{1}{64} + W_{\text{bad}},$$

which proves (A.109) without requiring all factors to be good simultaneously. □

Proposition A.14 (Universal numerical PAC closure and one-factor baseline)**.** *Under Assumptions 1, 3, and 4, Propositions A.5, A.11, A.12, and A.13, set*

$$C_{\mathrm{up}} = 65536 = 2^{16}. \tag{A.112}$$

For every $c \in C$, probability measure D on (X, Σ), and integer $n \geq \lceil C_{\mathrm{up}} Q_{\oplus} \rceil$,

$$\Pr_{S \sim D_c^n,\, \bar{A}_n^{\oplus,Q}} \left[R_D(h_{\bar{A}_n^{\oplus,Q}(S)}, c) \leq \frac{1}{16} \right] \geq \frac{15}{16}. \tag{A.113}$$

If $k = 1$, the same threshold entails no shortage and the stronger factor baseline

$$\Pr \left[R_D(h_{\bar{H}_1}, c) \leq \frac{1}{64} \right] \geq 1 - \frac{1}{4096} \tag{A.114}$$

holds directly.

Proof. Write $Q = Q_{\oplus}$. Since $n \geq \lceil 65536Q \rceil \geq 65536Q$, Proposition A.12 gives

$$\mathbb{E}W_{\mathrm{bad}} \leq \frac{128}{65536} + e^{-16} + \frac{1}{4096}. \tag{A.115}$$

The natural exponential satisfies $e = \sum_{m=0}^{\infty} 1/m! > 1 + 1 = 2$, so

$$e^{-16} < 2^{-16} = \frac{1}{65536}. \tag{A.116}$$

Putting the terms in (A.115) over the common denominator 65536 gives

$$\mathbb{E}W_{\mathrm{bad}} < \frac{128 + 1 + 16}{65536} = \frac{145}{65536} < \frac{192}{65536} = \frac{3}{1024}. \tag{A.117}$$

For completeness, nonnegativity gives the pointwise Markov inequality

$$\mathbf{1}\left\{ W_{\mathrm{bad}} > \frac{3}{64} \right\} \leq \frac{W_{\mathrm{bad}}}{3/64}. \tag{A.118}$$

Taking expectations and using (A.117),

$$\Pr\left(W_{\mathrm{bad}} > \frac{3}{64} \right) \leq \frac{64}{3}\mathbb{E}W_{\mathrm{bad}} < \frac{64}{3}\frac{3}{1024} = \frac{1}{16}. \tag{A.119}$$

Proposition A.13 gives the exact event inclusion

$$\left\{ R_D(h_{\bar{H}}, c) > \frac{1}{16} \right\} \subseteq \left\{ W_{\mathrm{bad}} > \frac{1}{16} - \frac{1}{64} \right\} = \left\{ W_{\mathrm{bad}} > \frac{3}{64} \right\}. \tag{A.120}$$

The strict boundary is correct: when $W_{\mathrm{bad}} = 3/64$, risk domination gives risk at most $1/16$, which is success under the closed event. Equations (A.119)– (A.120) give failure probability strictly below $1/16$, hence the stated success probability, over the entire joint law (A.93).

For $k = 1$, one has $X_1 = X$, $\rho_1 = 1$, and $J_1 = n$ for every sample. Since $C_{\mathrm{up}} \geq 1$ and $Q_{\oplus} = q_1$, the threshold implies $n \geq q_1$. The routed input is therefore the first q_1 iid records with no padding. Their quotient images have the iid labeled quotient law for arbitrary raw support, so Proposition A.5 applies directly. Identity (A.111) reduces to $R_D(h_{\bar{H}_1}, c) = r_1(\bar{H}_1)$, proving (A.114). The shortage bound and Markov relaxation are not used in this specialization. ∎

A.6 Public specialization of the quota sum

Lemma A.18 (Endpoint-safe domination of heterogeneous logarithms)**.** *Under Assumptions 2 and 4 and Lemma A.4, define*

$$L_i = \log\left(\frac{es_i}{\varepsilon\delta}\right), \qquad L = \log\left(\frac{eM}{\varepsilon\delta}\right), \quad M = \sum_{i=1}^{k} s_i. \tag{A.121}$$

Then all logarithm arguments are finite and strictly larger than $20e$*, and*

$$2 \leq s_i \leq M, qquad 1 < L_i \leq L, qquad \sum_{i=1}^{k} s_i L_i^2 \leq ML^2. \tag{A.122}$$

Proof. Lemma A.4 gives $s_i \geq 2$ and $M = \sum_{j=1}^{k} s_j$. Since all summands are positive,

$$M - s_i = \sum_{j \neq i} s_j \geq 0, \tag{A.123}$$

where the empty sum is zero when $k = 1$. Hence $2 \leq s_i \leq M$. The same lemma gives $M \geq 2k$, and thus $M \geq 2$.

Assumption 4 gives $0 < \varepsilon\delta < 1/10$. Therefore

$$\frac{es_i}{\varepsilon\delta} > 20e > e, \qquad \frac{eM}{\varepsilon\delta} > 20e > e. \tag{A.124}$$

The natural logarithm is strictly increasing on $(0, \infty)$, so (A.123) gives

$$1 < \log(20e) < L_i \leq L. \tag{A.125}$$

Both logarithms are positive, so squaring preserves their order. After multiplying by the positive heterogeneous weights and summing,

$$\sum_{i=1}^{k} s_i L_i^2 \leq \sum_{i=1}^{k} s_i L^2 = ML^2. \tag{A.126}$$

No balance among the s_i's is used.

The endpoint checks are explicit in (A.124): $\varepsilon = 1/10$ is allowed and the arguments remain strictly above $20e$ for every $\delta < 1$; as either positive parameter approaches zero, the arguments increase. The excluded zero endpoints are never inserted into a logarithm. ☐

Lemma A.19 (Ceiling-resolved heterogeneous quota sum)**.** *Under Assumptions 2 and 4, Lemma A.4, and Lemma A.18, each exact quota satisfies*

$$q_i \leq K_Y \frac{s_i}{\varepsilon} L_i^2 + 1, \tag{A.127}$$

and

$$Q_\oplus \leq \frac{K_Y}{\varepsilon} \sum_{i=1}^{k} s_i L_i^2 + k \leq K_Y \frac{M}{\varepsilon} L^2 + k. \tag{A.128}$$

Proof. If $m = \lceil x \rceil$, then $m - 1 < x \le m$, so $m < x + 1$ and in particular $\lceil x \rceil \le x + 1$. Apply this to each finite positive quantity

$$x_i = K_Y \frac{s_i}{\varepsilon} \log^2\left(\frac{es_i}{\varepsilon\delta}\right) = K_Y \frac{s_i}{\varepsilon} L_i^2. \tag{A.129}$$

Because $q_i = \lceil x_i \rceil$, this proves (A.127), including when x_i is an integer. Summing one $+1$ for each of the exactly k factors gives the first inequality in (A.128); multiplying (A.126) by $K_Y/\varepsilon > 0$ gives the second. No ceiling has been discarded. □

Proposition A.15 (Public quota rate specialization). *Under Assumptions 2 and 4, Lemma A.4, and Lemma A.19, define the universal constant*

$$C_{\text{quota}} = \max\left\{1, K_Y + \frac{1}{20}\right\}. \tag{A.130}$$

Then

$$Q_\oplus \le C_{\text{quota}} \frac{M}{\varepsilon} \log^2\left(\frac{eM}{\varepsilon\delta}\right). \tag{A.131}$$

The constant depends only on the fixed universal source constant K_Y, and the bound has neither a separate factor-count term nor a class, quotient, or support cardinality term.

Proof. Let

$$A = \frac{M}{\varepsilon} L^2 = \frac{M}{\varepsilon} \log^2\left(\frac{eM}{\varepsilon\delta}\right) > 0. \tag{A.132}$$

Lemma A.4 gives $M \ge 2k$, hence

$$k \le \frac{M}{2}. \tag{A.133}$$

Lemma A.18 gives $L > 1$, while Assumption 4 gives $\varepsilon \le 1/10$. Therefore

$$k \le \frac{M}{2} = \frac{\varepsilon}{2L^2} A \le \frac{1}{20} A. \tag{A.134}$$

Combining (A.128) and (A.134),

$$Q_\oplus \le K_Y A + k \le \left(K_Y + \frac{1}{20}\right) A \le C_{\text{quota}} A, \tag{A.135}$$

which proves (A.131).

When every $d_i = 1$, the setting definition gives $s_i = 2$ and $M = 2k$, so (A.133) is tight but the same absorption applies. Heterogeneous factors are handled termwise in (A.126). When $k = 1$, $M = s_1$ and $L_1 = L$; the exact intermediate statement is $q_1 \le K_Y(M/\varepsilon)L^2 + 1$, and (A.134) absorbs only that one ceiling. Thus none of these regimes introduces a cardinality term or changes the deterministic fixed-sample mode. □

A.7 An additive VC lower bound

Lemma A.20 (Unseen labels against unrestricted randomized learners). *Under Assumptions 1 and 2 and Lemma A.2, fix $n \in \mathbb{N}$ and any unrestricted learner kernel A_n with measurable binary decoder $\omega \mapsto h_\omega$. There are points $x_i \in X_i$, targets $c^b \in C$ indexed by $b \in \{0,1\}^k$, and the probability measure*

$$D = \frac{1}{k} \sum_{i=1}^{k} \delta_{x_i} \tag{A.136}$$

such that, for uniform $B \in \{0,1\}^k$, a sample $S \sim D^n_{c^B}$, and output from $A_n(S,\cdot)$,

$$\mathbb{E}_{B,S,A_n} R_D(h_{A_n(S)}, c^B) \geq \frac{1}{2}\left(1 - \frac{1}{k}\right)^n. \tag{A.137}$$

No privacy, properness, or PAC premise is required.

Proof. Lemma A.2 supplies points $x_i \in X_i$ whose set $W = \{x_1, \ldots, x_k\}$ is shattered. The disjoint blocks make them distinct. For every $b \in \{0,1\}^k$, fix one of the finitely many full-product targets supplied by that lemma, so that

$$c^b(x_i) = b_i, qquad 1 \leq i \leq k. \tag{A.138}$$

No measurable selector is involved because only finitely many targets are fixed.

Represent the sample using independent indices $I_1, \ldots, I_n$, each uniform on $[k] = \{1, \ldots, k\}$:

$$S_B(I) = \big((x_{I_t}, B_{I_t})\big)_{t=1}^n. \tag{A.139}$$

For $j \in [k]$, let

$$U_j = \{I_t \neq j \text{ for every } 1 \leq t \leq n\}. \tag{A.140}$$

Then

$$\Pr(U_j) = \left(1 - \frac{1}{k}\right)^n. \tag{A.141}$$

For $a \in \{0,1\}$, the finite-evaluation-cylinder requirement makes

$$E_{j,a} = \{\omega : h_\omega(x_j) = a\} \tag{A.142}$$

measurable. The events $E_{j,0}$ and $E_{j,1}$ partition the entire output space even for improper hypotheses. Fix an index tuple on which U_j occurs and all target bits except b_j. The dataset (A.139) is identical for $b_j = 0$ and $b_j = 1$; call it S. Averaging over the fair bit and integrating all learner randomness through the same kernel gives

$$\begin{aligned} &\frac{1}{2}\Pr_{A_n(S)}[h_{A_n(S)}(x_j) \neq 0] + \frac{1}{2}\Pr_{A_n(S)}[h_{A_n(S)}(x_j) \neq 1] \\ &\quad = \frac{1}{2}A_n(S, E_{j,1}) + \frac{1}{2}A_n(S, E_{j,0}) = \frac{1}{2}. \end{aligned} \tag{A.143}$$

Thus arbitrary output randomization and coupling cannot reveal an unseen target bit.

Under (A.136), the risk is the measurable finite-evaluation sum

$$R_D(h_\omega, c^B) = \frac{1}{k}\sum_{j=1}^k \mathbf{1}\{h_\omega(x_j) \neq B_j\}. \tag{A.144}$$

Dropping the nonnegative losses on seen points, applying (A.143) on U_j, and using (A.141),

$$\begin{aligned} \mathbb{E} R_D(h_{A_n(S)}, c^B) &= \frac{1}{k}\sum_{j=1}^k \Pr[h_{A_n(S)}(x_j) \neq B_j] \\ &\geq \frac{1}{k}\sum_{j=1}^k \Pr[U_j,\ h_{A_n(S)}(x_j) \neq B_j] \\ &= \frac{1}{k}\sum_{j=1}^k \frac{1}{2}\Pr(U_j) = \frac{1}{2}\left(1 - \frac{1}{k}\right)^n. \end{aligned} \tag{A.145}$$

Repeated sample points are already included and can only leave more coordinates unseen. □

Lemma A.21 (Exact PAC-event to expectation conversion)**.** *Let L be measurable with $0 \le L \le 1$. If*

$$\Pr[L \le 1/16] \ge 15/16, \tag{A.146}$$

then

$$\mathbb{E}L \le \frac{31}{256}. \tag{A.147}$$

Consequently,

$$\mathbb{E}L > \frac{31}{256} \implies \Pr[L > 1/16] > 1/16. \tag{A.148}$$

Proof. Put $p = \Pr[L > 1/16]$. On the complementary event $L \le 1/16$, and everywhere $L \le 1$, so

$$\mathbb{E}L \le \frac{1}{16}(1-p) + p = \frac{1}{16} + \frac{15}{16}p. \tag{A.149}$$

Under (A.146), $p \le 1/16$, whence

$$\mathbb{E}L \le \frac{1}{16} + \frac{15}{16}\frac{1}{16} = \frac{31}{256}.$$

The strict contrapositive proves (A.148). The good-event equality boundary is unchanged. □

Proposition A.16 (Additive VC PAC lower certificate)**.** *Under Assumptions 1 and 2, Lemma A.2, Lemma A.20, and Lemma A.21, fix $n \in \mathbb{N}$. If an unrestricted learner A_n satisfies, for every $c \in C$ and every allowed probability measure D',*

$$\Pr_{S\sim(D'_c)^n, A_n}\left[R_{D'}(h_{A_n(S)}, c) \le \frac{1}{16}\right] \ge \frac{15}{16}, \tag{A.150}$$

then

$$n \ge \frac{k}{2}. \tag{A.151}$$

More precisely, if $n < k/2$, the distribution (A.136) and a deterministic target $c^{b_*} \in C$ satisfy

$$\Pr_{S\sim D^n_{c^{b_*}}, A_n}\left[R_D(h_{A_n(S)}, c^{b_*}) > \frac{1}{16}\right] > \frac{1}{16}. \tag{A.152}$$

Proof. Since $n \in \mathbb{N} = \{1, 2, \ldots\}$, for $k = 1$ one has $n \ge 1 \ge k/2$, and for $k = 2$, $n \ge 1 = k/2$. Only $k \ge 3$ can enter the contradiction branch $n < k/2$.

For every integer $m \ge 1$ and $x \in [0, 1]$,

$$(1-x)^m \ge 1 - mx. \tag{A.153}$$

It is equality at $m = 1$. If it holds at m, multiplication by $1 - x \ge 0$ gives

$$(1-x)^{m+1} \ge (1-mx)(1-x) = 1 - (m+1)x + mx^2 \ge 1 - (m+1)x,$$

proving (A.153) by induction.

Suppose $n < k/2$. Lemma A.20 and (A.153) with $x = 1/k$ imply

$$\mathbb{E}_{B,S,A_n} R_D(h_{A_n(S)}, c^B) \ge \frac{1}{2}\left(1 - \frac{1}{k}\right)^n \ge \frac{1}{2}\left(1 - \frac{n}{k}\right) > \frac{1}{4} > \frac{31}{256}. \tag{A.154}$$

For each b, define

$$\ell_b = \mathbb{E}_{S\sim D^n_{c^b}, A_n} R_D(h_{A_n(S)}, c^b). \tag{A.155}$$

The left side of (A.154) is the finite average $2^{-k}\sum_b \ell_b$. Hence some deterministic b_* has

$$\ell_{b_*} > \frac{1}{4} > \frac{31}{256}. \tag{A.156}$$

Lemma A.21, applied contrapositively to the $[0,1]$-valued risk in (A.155), proves (A.152). This contradicts (A.150); therefore $n < k/2$ is impossible. Thus (A.151) holds with the universal constant $a_{\mathrm{VC}} = 1/2$, without privacy, properness, or output-space restriction beyond measurable binary evaluation. □

A.8 An expected-risk Littlestone floor for one factor

For a labeled database $U = ((u_a, y_a))_{a=1}^N$ and a decoded binary hypothesis h, write

$$L_U(h) = \frac{1}{N}\sum_{a=1}^{N} \mathbf{1}\{h(u_a) \neq y_a\}.$$

When $y_a = c(u_a)$, we also write $L_U(h, c)$. In this subsection $[t] = \{1, \ldots, t\}$ and ln denotes the natural logarithm.

Proposition A.17 (Expected empirical threshold obstruction)**.** *There is a universal $a_0 > 0$ with the following property. Let V be a finite totally ordered set of size t, let even $N \geq 72$, and let*

$$B : (V \times \{0,1\})^N \rightsquigarrow \{0,1\}^V$$

be an arbitrary-output randomized algorithm. Suppose that

1. *B is replacement-$(0.1, \eta)$-DP on all labeled databases;*
2. *$\eta \leq 1/(1000N^2 \log_2 N)$; and*
3. *for every threshold target c on V and every N-row database U labeled by c,*

$$\mathbb{E}_B L_U(B(U), c) \leq \frac{1}{8}. \tag{A.157}$$

Then

$$N \geq a_0 \log_2^* t. \tag{A.158}$$

The output may be any bit vector, not necessarily a threshold.

Proof. We adapt the active internal reduction proving the finite-threshold lower bound of Alon et al. (2019); its printed PAC corollary is not used as an expected-risk theorem.

For an N-row database U and $v \in V$, put

$$b_U(v) = \Pr[B(U)(v) = 1].$$

For an increasing database on distinct points, let $\mathrm{ord}_U(v)$ be the number of its instance coordinates at or below v. Call $K \subseteq V$ N-homogeneous if there are $p_0, \ldots, p_N \in [0,1]$ such that every increasing balanced threshold-labeled database U on N distinct points of K and every $v \in K \setminus U_X$ satisfy

$$\left|b_U(v) - p_{\mathrm{ord}_U(v)}\right| \leq \frac{1}{100N}. \tag{A.159}$$

The finite Ramsey lemma of Alon et al. (2019), applied to the $N+1$ output-coordinate marginals rounded to multiples of $1/(100N)$, gives a universal C_R and an N-homogeneous set K with

$$k := |K| \geq \frac{\log_2^{(N)} t}{2^{C_R N \log_2 N}} \tag{A.160}$$

whenever $\log_2^* t > N$. There are at most $(100N+1)^{N+1}$ colors; the cited finite hypergraph Ramsey bound gives the display after changing the fixed logarithm base and enlarging C_R. Only finitely many coordinate marginals are used, so arbitrary improper bit-vector output is admissible.

We next expose the sole utility input to the cited reduction. On its balanced increasing realizable database U, (A.157) gives

$$\frac{7}{8} \leq \mathbb{E}[1 - L_U(B(U), c)] = \frac{1}{N} \sum_{a \leq N/2} (1 - b_U(u_a)) + \frac{1}{N} \sum_{a > N/2} b_U(u_a). \tag{A.161}$$

Thus some positive coordinate has marginal at least $3/4$, and some negative coordinate has marginal at most $1/4$. Replace the selected coordinate by the adjacent unused point in the reduction, retaining its label. Reverse replacement privacy gives, for the high marginal,

$$b_{U'}(u_a) \geq e^{-0.1}(b_U(u_a) - \eta) \geq e^{-0.1}(3/4 - \eta) > 2/3, \tag{A.162}$$

whereas forward privacy gives, for the low marginal,

$$b_{U''}(u_a) \leq e^{0.1} b_U(u_a) + \eta \leq e^{0.1}/4 + \eta < 1/3. \tag{A.163}$$

Both strict estimates follow from $N \geq 72$ and

$$\eta \leq \frac{1}{1000N^2 \log_2 N} < 0.001. \tag{A.164}$$

The removed points lie outside the modified databases, so homogeneity maps (A.162)– (A.163) to entries of the same list: the upper entry is at least $2/3-1/(100N)$, the lower at most $1/3+1/(100N)$, and their ordered index gap is at most N. Telescoping yields

$$\exists j \in \{1, \ldots, N\} : qquad p_j - p_{j-1} \geq \frac{1}{N}\left(\frac{1}{3} - \frac{1}{50N}\right) \geq \frac{1}{4N}. \tag{A.165}$$

This is exactly where the cited printed proof used PAC utility: its PAC premise served only to imply an expected empirical loss at most $1/8$, which is already (A.157).

Suppose $k > 2N + 1$ and put $n = k - N$. The probability-to-distribution lemma of Alon et al. (2019), instantiated with the jump (A.165), chooses a balanced database with one movable row and an interval $J \subseteq K$ of exactly n unused points. For $a \in [n]$, place the movable row at the a-th point of J, call the database U_a, and let P_a be the law on $\{0,1\}^n$ obtained by restricting $B(U_a)$ to J. Every pair $P_a, P_{a'}$ is $(0.1, \eta)$-indistinguishable because the databases differ in one row. With $r = (p_{j-1} + p_j)/2$, homogeneity and the jump give

$$\begin{aligned} \Pr_{z \sim P_a}[z(b) = 1] &\leq r - \frac{1}{10N}, \quad b < a, \\ \Pr_{z \sim P_a}[z(b) = 1] &\geq r + \frac{1}{10N}, \quad b > a. \end{aligned} \tag{A.166}$$

Indeed, the jump $1/(4N)$, minus two homogeneity errors, leaves more than $1/(5N)$ between the sides; taking the midpoint leaves the displayed margin. The output vectors remain arbitrary members of the full cube, and all-input privacy covers every database used here.

The binary lemma and product-composition lemma of Alon et al. (2019) imply

$$n \le 2^{1000N^2(\log_2 N)^2}. \tag{A.167}$$

We include their quantitative contradiction to discharge the additive privacy scale. Set $L = \log_2 N$, suppose the displayed bound fails, and define

$$T = \lfloor 1000N^2L^2 \rfloor - 1, \qquad D = \lceil 50N^2 \ln(6T) \rceil. \tag{A.168}$$

For D independent cube vectors, threshold each coordinate's empirical fraction of ones at r, breaking ties toward zero, and perform T binary-search decisions. Since $n > 2^{T+1}$, each of the 2^T terminal intervals contains at least two indices. Choose in every interval a representative a unequal to every coordinate queried on its path, and let E_a be the event that the search follows that path. These events are disjoint. Under P_a^D, each queried coordinate is at distance at least $1/(10N)$ from r on the side leading toward a. Hoeffding's inequality (Hoeffding, 1963) and a union bound give

$$P_a^D(E_a) \ge 1 - T\exp\left(-\frac{D}{50N^2}\right) \ge \frac{5}{6}. \tag{A.169}$$

For every pair of product laws, the cited product-composition result gives indistinguishability with parameters $(0.1D, D\eta)$. From (A.168) and (A.164),

$$\begin{aligned} D\eta &\le \frac{\ln(6T)}{20L} + \frac{1}{1000N^2L} \\ &\le \frac{\ln(6000) + 2\ln N + 2\ln L}{20L} \\ &\qquad + \frac{1}{1000N^2L} < \frac{1}{5}. \end{aligned} \tag{A.170}$$

The last inequality holds at $N = 72$ and its left side decreases thereafter: $\ln N/L = \ln 2$ is constant, while $1/L$ and $\ln L/L$ decrease for $L \ge \log_2 72 > e$. Thus for any fixed P_b^D, reverse product privacy and (A.169) imply

$$P_b^D(E_a) \ge e^{-0.1D}\left(P_a^D(E_a) - D\eta\right) > \frac{1}{2}e^{-0.1D}. \tag{A.171}$$

The same monotonic calculation gives $\ln(6T) \le 3.4L$, so $0.1D \le 17N^2L + 0.1$. Since

$$(T-1)\ln 2 \ge (1000N^2L^2 - 3)\ln 2 > 17N^2L + 0.1,$$

the disjoint events contradict total mass:

$$1 \ge \sum_a P_b^D(E_a) > 2^T \frac{1}{2} e^{-0.1D} > 1.$$

This proves (A.167) without any further utility input.

Take $C_H = 1001$. If $k \le 2N + 1$, then $k \le 2^{C_H N^2 L^2}$ immediately. Otherwise (A.165), (A.166), and (A.167) give

$$k = N + n \le N + 2^{1000N^2L^2} \le 2^{C_H N^2 L^2}. \tag{A.172}$$

If $\log_2^* t \le N$, the proposition follows after choosing $a_0 \le 1$. Otherwise combine (A.160) and (A.172):

$$\log_2^{(N)} t \le k, 2^{C_R NL} \le 2^{C_1 N^2 L^2} \tag{A.173}$$

for a universal C_1. At most $\log_2^*(2^{C_1 N^2 L^2}) + O(1) = \log_2^* N + O(1)$ further logarithms reduce the right side to at most one. Thus

$$\log_2^* t \le N + \log_2^* N + O(1) \le C_2 N$$

for a universal C_2 and every $N \ge 72$. Taking $a_0 = 1/C_2$, and decreasing it for the finitely many fixed-base endpoints if needed, proves (A.158).

Evenness is used only for the balanced databases. The fixed relabeling between $\{0,1\}$ and $\{-1,+1\}$ preserves loss, replacement adjacency, marginals, and privacy. All cited reductions inspect only finite coordinate marginals or restrictions, so the arbitrary-output scope is unchanged. $\square$

Lemma A.22 (Finite Shelah tasks and the log-star bridge)**.** *Under Assumption 2 and Lemma A.1, fix i and put*

$$r_i = \log_2^* d_i, \qquad t_i = \max\{1, \lfloor \ln d_i \rfloor\}. \tag{A.174}$$

There are fixed quotient points $q_{i,1}, \ldots, q_{i,t_i}$, quotient concepts $\bar{c}_{i,1}, \ldots, \bar{c}_{i,t_i}$, and raw representatives $x_{i,j} \in \kappa_i^{-1}(\{q_{i,j}\})$ such that, for $c_{i,\ell} = \bar{c}_{i,\ell} \circ \kappa_i \in C_i$,

$$\bar{c}_{i,\ell}(q_{i,j}) = c_{i,\ell}(x_{i,j}) = \mathbf{1}\{\ell \le j\}. \tag{A.175}$$

For $m \ge 8$, let N_m be the least even integer at least $9m$. For $\mathbf{j} = (j_1, \ldots, j_{N_m}) \in [t_i]^{N_m}$, define

$$\mu_{i,\mathbf{j}} = \frac{1}{N_m} \sum_{a=1}^{N_m} \delta_{x_{i,j_a}}. \tag{A.176}$$

Then

$$\mathfrak{T}_{i,m} = \{(c_{i,\ell}, \mu_{i,\mathbf{j}}) : \ell \in [t_i],\ \mathbf{j} \in [t_i]^{N_m}\} \tag{A.177}$$

is a finite learner-independent universe of realizable factor tasks. If $r_i \ge 8$, then

$$\log_2^* t_i \ge r_i - 2. \tag{A.178}$$

Proof. Lemma A.1 gives $\mathrm{LD}(\bar{C}_i) = d_i$. Shelah's threshold theorem in Alon et al. (2019) gives a threshold pattern of length $\lfloor \log d_i \rfloor$. If the source logarithm is natural, (A.174) uses no greater length; if it is base two, $\ln d_i \le \log_2 d_i$, so a prefix has the required length. When $\lfloor \ln d_i \rfloor = 0$, nonconstancy supplies a quotient point with both labels available, and a concept labeled one gives the length-one pattern. Quotient pullback and finitely many raw representative choices give (A.175) without a measurable global section.

Every $\mu_{i,\mathbf{j}}$ is a probability measure on the raw measurable space: a Dirac measure is defined on any measurable space, even if its singleton is not itself measurable. The targets are measurable setting concepts. Thus all tasks are legal and realizable, and the family has at most $t_i^{N_m+1}$ members fixed before the learner.

For the shift, let $x = \log_2 d_i$. If $r_i \ge 8$, then $d_i > 1024$ and $x > 10$. Since $x \ln 2 > 2$,

$$t_i = \lfloor x \ln 2 \rfloor \ge \frac{\ln 2}{2} x. \tag{A.179}$$

At $x = 10$, $(\ln 2)x/2 > \log_2 x$; the derivative of the left side is $\ln 2/2$, while that of $\log_2 x$ is $1/(x \ln 2) < \ln 2/2$ for $x \ge 10$. Hence $t_i \ge \log_2 x$. Monotonicity and two applications of the log-star recursion give

$$\log_2^* t_i \ge \log_2^*(\log_2 x) = \log_2^* x - 1 = r_i - 2,$$

proving (A.178). $\square$

Lemma A.23 (Replacement-private resampling and exact risk identity)**.** *Fix the factor and witness of Lemma A.22. Let $m \geq 8$, $\delta > 0$, and let*

$$A : (X_i \times \{0,1\})^m \rightsquigarrow (\Omega, \mathscr{F})$$

be total and replacement-$(0.1, \delta)$-DP on every labeled input, with measurable finite-evaluation cylinders. Define N_m as above and set

$$c_\delta = \frac{\ln 2}{200000}. \tag{A.180}$$

If

$$0 < \delta \leq \frac{c_\delta}{m^2 \ln(m+1)}, \tag{A.181}$$

then the following algorithm $B_{i,m}$ is well defined on every labeled N_m-row database over the raw witness: sample m row indices iid uniformly with replacement, run A on those rows, and output its decoded labels on $(x_{i,1}, \ldots, x_{i,t_i})$. It is replacement- $(\varepsilon'_m, \delta'_m)$-DP with

$$\varepsilon'_m = \frac{0.6m}{N_m} \leq \frac{1}{15}, \qquad \delta'_m = e^{0.6m/N_m} \frac{4m}{N_m} \delta < \delta \leq \frac{1}{1000 N_m^2 \log_2 N_m}. \tag{A.182}$$

For every $(c_{i,\ell}, \mu_{i,\mathbf{j}}) \in \mathfrak{T}_{i,m}$, let

$$U_{i,\ell,\mathbf{j}} = \big((x_{i,j_a}, c_{i,\ell}(x_{i,j_a}))\big)_{a=1}^{N_m}.$$

Then

$$\mathbb{E} L_{U_{i,\ell,\mathbf{j}}}\big(B_{i,m}(U_{i,\ell,\mathbf{j}}), c_{i,\ell}\big) = \mathbb{E}_{S \sim (\mu_{i,\mathbf{j}})^m_{c_{i,\ell}}, A} R_{\mu_{i,\mathbf{j}}}(h_{A(S)}, c_{i,\ell}). \tag{A.183}$$

At $m = 8$, $N_m = 72$.

Proof. Finite-evaluation measurability makes the output vector legal; it can be arbitrary and need not be a threshold. Totality on nonrealizable inputs is inherited from A. By construction,

$$9m \leq N_m \leq 9m + 1 \leq 10m. \tag{A.184}$$

In particular $N_m \geq 2m$. The with-replacement privacy-amplification calculation preceding Lemma 5.9 of Bun et al. (2015) applies to replacement-adjacent databases, including every multiplicity of the changed row. With base privacy 0.1, it gives exactly the two parameters in (A.182). From $N_m \geq 9m$, $\varepsilon'_m \leq 0.6/9 = 1/15$. For $0 \leq y < 1$, the exponential series gives $e^y \leq (1-y)^{-1}$, so

$$e^{0.6m/N_m} \frac{4m}{N_m} \leq e^{1/15} \frac{4}{9} \leq \frac{15}{14} \frac{4}{9} = \frac{10}{21} < 1. \tag{A.185}$$

Thus $\delta'_m < \delta$, and monotonicity also regards the resampled algorithm as $(0.1, \delta'_m)$-DP.

For $m \geq 8$,

$$10m \leq (m+1)^2. \tag{A.186}$$

Equations (A.184)– (A.186) give

$$N_m^2 \log_2 N_m = \frac{N_m^2 \ln N_m}{\ln 2} \leq \frac{100m^2 \, 2\ln(m+1)}{\ln 2}. \tag{A.187}$$

Inverting this display and using (A.180)–(A.181) yields

$$\delta'_m < \delta \leq \frac{\ln 2}{200000 m^2 \ln(m+1)} \leq \frac{1}{1000 N_m^2 \log_2 N_m},$$

completing the privacy claims. At $m = 8$, $N_m = 72$ and

$$\varepsilon'_8 = \frac{0.6 \cdot 8}{72} = \frac{1}{15}, \qquad \delta'_8 = e^{1/15}\frac{4}{9}\delta \le \frac{10}{21}\delta < \delta,$$

so no open endpoint is hidden.

Conditional on $U_{i,\ell,\mathbf{j}}$, the resample is exactly iid from $(\mu_{i,\mathbf{j}})_{c_{i,\ell}}$. For every decoded h, before expectation,

$$\begin{aligned} L_{U_{i,\ell,\mathbf{j}}}(h, c_{i,\ell}) &= \frac{1}{N_m}\sum_{a=1}^{N_m} \mathbf{1}\{h(x_{i,j_a}) \neq c_{i,\ell}(x_{i,j_a})\} \\ &= R_{\mu_{i,\mathbf{j}}}(h, c_{i,\ell}). \end{aligned} \tag{A.188}$$

Taking expectation over sampled indices and learner randomness proves (A.183); repeated rows retain their multiplicity on both sides. ☐

Proposition A.18 (Unrestricted expected-risk factor floor)**.** *Under Assumptions 2 and 4, Lemma A.1, Proposition A.17, Lemma A.22, and Lemma A.23, fix i, $m \ge 8$, $\delta > 0$, and a total arbitrary-output learner*

$$A : (X_i \times \{0,1\})^m \rightsquigarrow (\Omega, \mathscr{F})$$

that is replacement-$(0.1, \delta)$-DP on all labeled inputs and has measurable finite evaluations. For a_0 from Proposition A.17, define

$$a_{\mathrm{L}} = \min\left\{1, \frac{a_0}{20}\right\}, \qquad c_\delta = \frac{\ln 2}{200000}. \tag{A.189}$$

If $\delta \le c_\delta/(m^2 \ln(m+1))$ and A has expected risk at most $1/8$ on every realizable factor task, then

$$m \ge a_{\mathrm{L}} \log_2^* d_i. \tag{A.190}$$

More strongly, for the fixed finite universe $\mathfrak{T}_{i,m}$ of Lemma A.22,

$$m < a_{\mathrm{L}} \log_2^* d_i \quad \Longrightarrow \quad \exists (c, D) \in \mathfrak{T}_{i,m} : \mathbb{E}_{S \sim D_c^m, A} R_D(h_{A(S)}, c) > \frac{1}{8}. \tag{A.191}$$

Proof. Write $r_i = \log_2^* d_i$. If $r_i < 8$, then $m \ge 8 > r_i$, and (A.190) follows from $a_{\mathrm{L}} \le 1$.

Suppose $r_i \ge 8$. If the learner has expected risk at most $1/8$ on every realizable task, it does so on every member of $\mathfrak{T}_{i,m}$. Lemma A.23 constructs an even-N_m arbitrary-output empirical threshold learner satisfying all hypotheses of Proposition A.17. That proposition, Lemma A.22, and $N_m \le 10m$ give

$$10m \ge N_m \ge a_0 \log_2^* t_i \ge a_0(r_i - 2) \ge \frac{a_0}{2} r_i. \tag{A.192}$$

Thus $m \ge (a_0/20) r_i \ge a_{\mathrm{L}} r_i$.

If instead $m < a_{\mathrm{L}} r_i$, the low-r_i case is impossible. In the remaining case, were every task in $\mathfrak{T}_{i,m}$ to have expected risk at most $1/8$, the same calculation (A.192) would contradict the strict hypothesis. Therefore logical negation gives one task with risk strictly greater than $1/8$, proving (A.191). The finite task universe was fixed before A. ☐

A.9 Candidate-wise hard-factor calibration

Lemma A.24 (Low-factor mass calibration)**.** *Under Assumptions 1 and 2, Lemma A.4, Proposition A.16, and Proposition A.18, fix one candidate $n \in \mathbb{N}$ and a learner satisfying the global $(1/16, 1/16)$-PAC guarantee. Put $a_{\mathrm{VC}} = 1/2$, take $a_{\mathrm{L}} \in (0, 1]$ from Proposition A.18, and define the universal constants*

$$R_0 = \left\lceil \frac{32}{a_{\mathrm{L}}} \right\rceil \tag{A.193}$$

and

$$c_{\mathrm{low}} = \min\left\{\frac{a_{\mathrm{VC}}}{128(R_0+2)}, \frac{a_{\mathrm{L}}}{64}\right\} = \min\left\{\frac{1}{256(R_0+2)}, \frac{a_{\mathrm{L}}}{64}\right\} > 0. \tag{A.194}$$

With

$$\pi_i = \omega_i = \frac{s_i}{M}, \qquad H = \{i : r_i > R_0\}, \qquad w_L = \sum_{i \notin H} \pi_i, \tag{A.195}$$

if $n < c_{\mathrm{low}} M$, then

$$w_L < \frac{1}{128}. \tag{A.196}$$

In particular, $H \neq \varnothing$.

Proof. Lemma A.4 gives $s_i \le r_i + 2$. For $i \notin H$, including $r_i = R_0$,

$$s_i \le r_i + 2 \le R_0 + 2. \tag{A.197}$$

The PAC premise and Proposition A.16 give

$$n \ge a_{\mathrm{VC}} k, \qquad k \le \frac{n}{a_{\mathrm{VC}}}. \tag{A.198}$$

Since every $s_i \ge 2$, $M > 0$. Summing (A.197), applying (A.198), and only then using the strict contradiction hypothesis gives

$$\begin{aligned} w_L &= \frac{1}{M}\sum_{i \notin H} s_i \le \frac{(R_0+2)k}{M} \\ &\le \frac{R_0+2}{a_{\mathrm{VC}}}\frac{n}{M} < \frac{R_0+2}{a_{\mathrm{VC}}} c_{\mathrm{low}} \le \frac{1}{128}. \end{aligned} \tag{A.199}$$

Strictness comes from $n < c_{\mathrm{low}} M$, even when the first branch in (A.194) is attained. Because $\sum_i \pi_i = 1$, an empty H would give $w_L = 1$, a contradiction. If every factor is active, the empty low-factor sum is zero. Factors with $r_i = 0$ or $r_i = R_0$ are included in the low set and already paid for by (A.199). □

Lemma A.25 (Strict active-factor budget calibration)**.** *Under Lemma A.4, Proposition A.18, and Lemma A.24, use the exact constants $a_{\mathrm{VC}} = 1/2$, $0 < a_{\mathrm{L}} \le 1$, R_0, c_{low}, and the set H in (A.193)–(A.195). At the same candidate, if $n < c_{\mathrm{low}} M$, then every $i \in H$ satisfies*

$$4n\pi_i < \frac{a_{\mathrm{L}} r_i}{8}, \qquad \lceil 4n\pi_i \rceil < a_{\mathrm{L}} r_i, \qquad 8 < a_{\mathrm{L}} r_i. \tag{A.200}$$

Consequently

$$m_{n,i} = \max\{8, \lceil 4n\pi_i \rceil\} \quad \textit{satisfies} \quad 8 \le m_{n,i} < a_{\mathrm{L}} r_i. \tag{A.201}$$

Proof. Fix $i \in H$. From $r_i > R_0$ and (A.193),

$$a_{\mathrm{L}} r_i > a_{\mathrm{L}} R_0 \geq 32. \tag{A.202}$$

In particular,

$$1 < \frac{a_{\mathrm{L}} r_i}{32}, \qquad 8 < \frac{a_{\mathrm{L}} r_i}{4} < a_{\mathrm{L}} r_i. \tag{A.203}$$

Since $a_{\mathrm{L}} \leq 1$, $R_0 \geq 32$, so $r_i > R_0 \geq 32 > 2$ and

$$r_i + 2 < 2r_i. \tag{A.204}$$

Using $\pi_i = s_i/M$, the strict hypothesis, the accepted comparison $s_i \leq r_i + 2$, and $c_{\mathrm{low}} \leq a_{\mathrm{L}}/64$,

$$\begin{aligned} 4n\pi_i &= 4\frac{n}{M} s_i < 4c_{\mathrm{low}} s_i \\ &\leq 4c_{\mathrm{low}}(r_i + 2) \leq \frac{a_{\mathrm{L}}}{16}(r_i + 2) < \frac{a_{\mathrm{L}} r_i}{8}. \end{aligned} \tag{A.205}$$

This explicitly absorbs the additive $+2$.

For every real x, including an integer, $\lceil x \rceil < x + 1$. Thus (A.205) and the first inequality in (A.203) give

$$\begin{aligned} \lceil 4n\pi_i \rceil &< 4n\pi_i + 1 \\ &< \frac{a_{\mathrm{L}} r_i}{8} + \frac{a_{\mathrm{L}} r_i}{32} = \frac{5}{32} a_{\mathrm{L}} r_i < a_{\mathrm{L}} r_i. \end{aligned} \tag{A.206}$$

Equation (A.203) separately makes the floor eight strictly subcritical. Both entries of the maximum in (A.201) are therefore strictly below $a_{\mathrm{L}} r_i$, including all ceiling and maximum equality cases. ☐

Proposition A.19 (Candidate-wise eligibility of every active factor)**.** *Under Assumption 5, Proposition A.18, and Lemma A.25, use the constants and set fixed by Lemma A.24. At the same fixed candidate, assume $n < c_{\mathrm{low}} M$. Then every $i \in H$ satisfies*

$$m_{n,i} \in \mathbb{N}, \qquad m_{n,i} \geq 8, \qquad m_{n,i} < a_{\mathrm{L}} r_i, \tag{A.207}$$

and

$$0 < \delta \leq \frac{c_\delta}{m_{n,i}^2 \log(m_{n,i} + 1)}. \tag{A.208}$$

The other candidate-budget conjunct remains

$$0 < \delta \leq \frac{1}{n \log(n + 1)}. \tag{A.209}$$

These are exactly the numerical hypotheses required to apply Proposition A.18 at $m = m_{n,i}$ once a total unrestricted replacement-$(0.1, \delta)$-DP factor learner is constructed.

Proof. The budget definition gives $m_{n,i} \in \mathbb{N}$ and $m_{n,i} \geq 8$; Lemma A.25, whose hypotheses use this same candidate and the same strict inequality $n < c_{\mathrm{low}} M$, gives strict subcriticality.

Assumption 5 states at this one candidate that

$$0 < \delta \leq \min\left\{ \frac{1}{n \log(n + 1)}, \min_{1 \leq j \leq k} \frac{c_\delta}{m_{n,j}^2 \log(m_{n,j} + 1)} \right\}.$$

The first coordinate of the minimum gives (A.209). The inner minimum is no larger than its i-th entry, giving (A.208) for every $i \in H$. All denominators are positive because $n \geq 1$ and $m_{n,i} \geq 8$. Equality at either delta cap is allowed; only the subcritical budget inequality is strict. This is a deterministic implication from one finite minimum, not a probabilistic union bound or a uniform-in-candidate schedule. ☐

A.10 Learner-independent hard priors

Fix $i \in H$ and abbreviate $m = m_{n,i}$. The constructions below depend on the fixed class, candidate, factor, and δ, but are made before a factor learner is quantified.

Lemma A.26 (Finite quotient-cell coding of the hard tasks)**.** *Under Assumption 3, Lemma A.1, and Lemma A.22, let*

$$F_i = \{x_{i,1}, \dots, x_{i,t_i}\}, \qquad \mathcal{V}_i = \{0,1\}^{[t_i]}, \qquad \Theta_i = [t_i] \times [t_i]^{N_m}. \tag{A.210}$$

There exist:

1. *a measurable retraction $\tau_i : (X_i, \Sigma_i) \to (F_i, 2^{F_i})$ fixing every $x_{i,j}$;*
2. *for each $v \in \mathcal{V}_i$, a measurable, possibly improper hypothesis $e_i(v) : X_i \to \{0,1\}$ with $e_i(v)(x_{i,j}) = v_j$, such that $v \mapsto e_i(v)$ has measurable finite-evaluation cylinders; and*
3. *for every $\theta = (\ell, \mathbf{j}) \in \Theta_i$, an exact finite labeled law P_θ on $F_i \times \{0,1\}$ and a risk vector $r_\theta : \mathcal{V}_i \to [0,1]$ such that, for*

$$T_\theta = (c_\theta, D_\theta) = \left(c_{i,\ell}, \frac{1}{N_m} \sum_{a=1}^{N_m} \delta_{x_{i,j_a}}\right), \tag{A.211}$$

   one has

$$R_{D_\theta}(e_i(v), c_\theta) = r_\theta(v) \tag{A.212}$$

   for every v, and the iid labeled sample law is the finite mixture indexed by P_θ^m.

All these sets are finite and nonempty, including when $t_i = 1$, and no measurable section of κ_i is used.

Proof. Put

$$G_{i,j} = \kappa_i^{-1}(\{q_{i,j}\}) \in \Sigma_i. \tag{A.213}$$

The quotient points are distinct: if $j < j'$ but $q_{i,j} = q_{i,j'}$, the threshold pattern with $\ell = j+1$ would give labels zero and one at the same quotient point. Hence the cells and their chosen representatives are pairwise distinct.

Define

$$\tau_i(x) = \begin{cases} x_{i,j}, & x \in G_{i,j} \text{ for some } j \in [t_i], \\ x_{i,1}, & x \notin \bigcup_{j=1}^{t_i} G_{i,j}. \end{cases} \tag{A.214}$$

For $j \geq 2$, the fiber of $x_{i,j}$ is $G_{i,j}$; the fiber of $x_{i,1}$ is $X_i \setminus \bigcup_{j=2}^{t_i} G_{i,j}$. All fibers are measurable, so τ_i is measurable into the finite discrete space and fixes each witness point. When $t_i = 1$, it is simply the constant map.

For $v = (v_1, \dots, v_{t_i})$, define

$$e_i(v)(x) = \begin{cases} v_j, & x \in G_{i,j} \text{ for some } j \in [t_i], \\ 0, & x \notin \bigcup_{j=1}^{t_i} G_{i,j}. \end{cases} \tag{A.215}$$

The inverse image of one is a finite union of measurable cells. Since the output space is finite discrete, for arbitrary raw points $z_1, \dots, z_s$ and bits $b_1, \dots, b_s$, the set of v's satisfying $e_i(v)(z_a) = b_a$ for all a is measurable. The extension is allowed to be improper.

For $\theta = (\ell, (j_1, \ldots, j_{N_m}))$, put

$$w_{\theta,j} = \frac{1}{N_m}\big|\{a \in [N_m] : j_a = j\}\big| \tag{A.216}$$

and define

$$P_\theta(x_{i,j}, b) = w_{\theta,j}\mathbf{1}\{b = c_{i,\ell}(x_{i,j})\}. \tag{A.217}$$

These nonnegative weights sum to one. Define

$$\begin{aligned} r_\theta(v) &= \sum_{j=1}^{t_i} w_{\theta,j}\mathbf{1}\{v_j \neq c_{i,\ell}(x_{i,j})\} \\ &= \frac{1}{N_m}\sum_{a=1}^{N_m}\mathbf{1}\{v_{j_a} \neq c_{i,\ell}(x_{i,j_a})\}. \end{aligned} \tag{A.218}$$

Equations (A.211), (A.215), and (A.218) prove (A.212); the iid labeled law is the mixture over $(F_i \times \{0,1\})^m$ with weights P_θ^m.

This finite representation does not assert measurability of raw singletons. Each D_θ is a finite convex combination of Dirac probability measures, valid on any measurable space, while the measurable quotient cells are what make τ_i and e_i legal. □

Proposition A.20 (Exact finite representation of unrestricted private learners). *Under Assumption 3 and Lemma A.26, put*

$$\mathcal{Z}_i = F_i \times \{0,1\}, \qquad \mathcal{U}_i = \mathcal{Z}_i^m, \tag{A.219}$$

and let $\mathsf{Adj}_i^{\rightarrow}$ contain every ordered pair of elements of $\mathcal{U}_i$ differing in exactly one row. Let $\mathcal{P}_i$ be the arrays $p = (p_{u,v})$ satisfying

$$p_{u,v} \geq 0, \qquad \sum_{v\in\mathcal{V}_i} p_{u,v} = 1 \quad (u \in \mathcal{U}_i), \tag{A.220}$$

and, for every $(u, u') \in \mathsf{Adj}_i^{\rightarrow}$ and every $E \subseteq \mathcal{V}_i$,

$$\sum_{v\in E} p_{u,v} \leq e^{0.1}\sum_{v\in E} p_{u',v} + \delta. \tag{A.221}$$

For $\theta \in \Theta_i$, define

$$L_\theta(p) = \sum_{u\in\mathcal{U}_i} P_\theta^m(u)\sum_{v\in\mathcal{V}_i} p_{u,v} r_\theta(v). \tag{A.222}$$

Then every unrestricted total raw-space factor learner on m rows with measurable finite evaluations and all-input replacement- $(0.1, \delta)$-DP induces $p^A \in \mathcal{P}_i$ whose exact expected risk on T_θ is $L_\theta(p^A)$. Conversely, every $p \in \mathcal{P}_i$ extends to such a total raw-space learner A^p, with finite output $\mathcal{V}_i$, decoder e_i, and exact expected risk $L_\theta(p)$ on every T_θ.

Proof. Let $A : (X_i \times \{0,1\})^m \rightsquigarrow (\Omega, \mathscr{F})$ be an arbitrary learner of the stated type, with decoder h_ω. The finite evaluation map

$$\mathrm{ev}_i(\omega) = (h_\omega(x_{i,1}), \ldots, h_\omega(x_{i,t_i})) \in \mathcal{V}_i \tag{A.223}$$

is measurable: its singleton fibers are finite-evaluation cylinders and the inverse image of any $E \subseteq \mathcal{V}_i$ is a finite union of them. For $u \in \mathcal{U}_i$, viewed as a raw labeled input, set

$$p^A_{u,v} = A(u, \mathrm{ev}_i^{-1}(\{v\})). \tag{A.224}$$

The fibers partition Ω, so (A.220) holds. For any ordered adjacent (u, u'), whether realizable or not, apply raw all-input privacy directly to $\mathrm{ev}_i^{-1}(E)$ to obtain (A.221). Because ordered adjacency includes the swapped pair, both directions are retained. Requiring all events is essential: summing atomwise approximate-DP inequalities would multiply δ.

Expanding the finite sample law and grouping outputs by (A.223),

$$\begin{aligned}\mathbb{E}_{S\sim(D_\theta)^m_{c_\theta},A} R_{D_\theta}(h_{A(S)}, c_\theta) \\ = \sum_{u\in\mathcal{U}_i} P_\theta^m(u) \sum_{v\in\mathcal{V}_i} p^A_{u,v} r_\theta(v) = L_\theta(p^A).\end{aligned} \tag{A.225}$$

This is exact for arbitrary improper output because the task risk depends only on the witness evaluations.

Conversely, fix $p \in \mathcal{P}_i$. Extend the retraction to labeled records by

$$\widehat{\tau}_i(x, y) = (\tau_i(x), y) \tag{A.226}$$

and coordinatewise to databases. Define

$$A^p(S, E) = \sum_{v\in E} p_{\widehat{\tau}_i^m(S),v}, \qquad E \subseteq \mathcal{V}_i, \tag{A.227}$$

and decode v as $e_i(v)$. Measurability of $\widehat{\tau}_i^m$ into a finite discrete space makes this a Markov kernel; row stochasticity makes it total on every raw input, including inconsistent and nonrealizable inputs.

For adjacent raw datasets, their preprocessed images are either equal or differ in the same one coordinate. Equal images give identical output laws; otherwise (A.221) applies to every output event, and the swapped ordered pair supplies the reverse direction. Thus there is no composition or added privacy cost. On a task draw, every raw record is a chosen representative fixed by τ_i, so preprocessing is the identity and

$$\mathbb{E}_{S\sim(D_\theta)^m_{c_\theta},A^p} R_{D_\theta}(e_i(A^p(S)), c_\theta) = L_\theta(p). \tag{A.228}$$

This proves both directions with exact privacy and risk. $\square$

Lemma A.27 (Compact finite private-learning game)**.** *Under Proposition A.20, $\mathcal{P}_i$ is a nonempty compact convex polytope in $\mathbb{R}^{\mathcal{U}_i\times\mathcal{V}_i}$. It has a nonempty finite vertex set $\mathcal{W}_i$ and*

$$\mathcal{P}_i = \mathrm{conv}(\mathcal{W}_i). \tag{A.229}$$

Every L_θ is affine and continuous, and

$$g_i(p) = \max_{\theta\in\Theta_i} L_\theta(p) \tag{A.230}$$

is continuous and attains its minimum on $\mathcal{P}_i$.

Proof. The accepted construction and eligibility give $t_i \geq 1$ and $m \geq 8$, and therefore

$$|\mathcal{V}_i| = 2^{t_i} < \infty, \qquad |\mathcal{U}_i| = (2t_i)^m < \infty, \qquad |\Theta_i| = t_i^{N_m+1} < \infty. \tag{A.231}$$

There are finitely many ordered adjacent pairs and finitely many subsets of $\mathcal{V}_i$. Thus (A.220)–(A.221) form a finite system of affine equalities and closed affine inequalities. The set is closed and convex, and stochasticity gives $0 \leq p_{u,v} \leq 1$, hence boundedness.

It is nonempty: fix $v^0 \in \mathcal{V}_i$ and put

$$p^0_{u,v} = \mathbf{1}\{v = v^0\} \quad \text{for every } u. \tag{A.232}$$

For an event containing v^0, privacy reads $1 \le e^{0.1} + \delta$; otherwise both sides are zero. Thus the constant kernel is feasible. A nonempty closed bounded set in this finite-dimensional space is compact. The finite polytope vertex theorem then gives a finite nonempty extreme-point set and (A.229).

Equation (A.222) is a finite linear combination of the coordinates $p_{u,v}$, so every payoff is affine, continuous, and lies in $[0, 1]$. The maximum in (A.230) is over a finite nonempty set, hence continuous, and compactness gives a minimizer. No property of an infinite learner or output space is used for this attainment. □

Proposition A.21 (Strict compact-game value)**.** *Under Propositions A.18 and A.19, Proposition A.20, and Lemma A.27, for every fixed $i \in H$,*

$$\gamma_i = \min_{p\in\mathcal{P}_i} \max_{\theta\in\Theta_i} L_\theta(p) > \frac{1}{8}. \tag{A.233}$$

Thus $\eta_i = \gamma_i - 1/8$ is a well-defined positive finite-game margin, not an additional assumption or universal numerical constant.

Proof. Fix $p \in \mathcal{P}_i$. Proposition A.20 extends it to a total learner A^p defined on all labeled inputs, with all-event replacement-$(0.1, \delta)$-privacy, measurable possibly improper output, and exact task loss $L_\theta(p)$. At this same factor and candidate, Proposition A.19 supplies

$$m \ge 8, \qquad m < a_{\mathrm{L}} \log_2^* d_i, \qquad 0 < \delta \le \frac{c_\delta}{m^2 \log(m+1)}. \tag{A.234}$$

Proposition A.18 therefore gives an indexed task with expected risk strictly greater than 1/8. By exact payoff preservation,

$$g_i(p) = \max_{\theta\in\Theta_i} L_\theta(p) > \frac{1}{8} \quad \text{for every } p \in \mathcal{P}_i. \tag{A.235}$$

Lemma A.27 supplies an actual minimizer p_i^*. Evaluating (A.235) at that point gives

$$\gamma_i = g_i(p_i^*) > \frac{1}{8}.$$

Attainment is essential: taking only an infimum of pointwise strict inequalities would yield merely a non-strict bound. □

Proposition A.22 (Finite learner-independent hard prior)**.** *Under Proposition A.21 and Lemma A.27, for every fixed $i \in H$ there is a probability distribution μ_i on Θ_i such that*

$$\sum_{\theta\in\Theta_i} \mu_i(\theta) L_\theta(p) \ge \gamma_i > \frac{1}{8} \quad \textit{for every } p \in \mathcal{P}_i. \tag{A.236}$$

Its pushforward ν_i under $\theta \mapsto T_\theta$ is a finite prior over realizable finite-support factor tasks, fixed independently of every learner. Every unrestricted total replacement-$(0.1, \delta)$-DP factor learner A on $m_{n,i}$ rows with measurable finite evaluations satisfies

$$\mathbb{E}_{(c_i,D_i)\sim\nu_i} \mathbb{E}_{S_i\sim(D_i)^{m_{n,i}}_{c_i},A} R_{D_i}(h_{A(S_i)}, c_i) > \frac{1}{8}. \tag{A.237}$$

Proof. Write the finite vertex set as $\mathcal{W}_i = \{p^1, \dots, p^J\}$. Every $p \in \mathcal{P}_i$ has a representation $p = \sum_{a=1}^{J} \lambda_a p^a$, $\lambda \in \Delta([J])$, and affinity gives

$$L_\theta(p) = \sum_{a=1}^{J} \lambda_a L_\theta(p^a). \tag{A.238}$$

Apply the finite von Neumann minimax theorem to the real matrix

$$M_{a,\theta} = L_\theta(p^a), \qquad (a, \theta) \in [J] \times \Theta_i. \tag{A.239}$$

Using (A.229), (A.238), and the fact that a linear functional on the task simplex is maximized by a pure task,

$$\begin{aligned} \gamma_i &= \min_{\lambda \in \Delta([J])} \max_{\mu \in \Delta(\Theta_i)} \sum_{a,\theta} \lambda_a \mu(\theta) M_{a,\theta} \\ &= \max_{\mu \in \Delta(\Theta_i)} \min_{\lambda \in \Delta([J])} \sum_{a,\theta} \lambda_a \mu(\theta) M_{a,\theta}. \end{aligned} \tag{A.240}$$

The finite theorem also gives an attaining maximizing mixture μ_i. For any p, choose a representing λ; the last line of (A.240) gives

$$\sum_\theta \mu_i(\theta) L_\theta(p) = \sum_{a,\theta} \lambda_a \mu_i(\theta) M_{a,\theta} \geq \gamma_i. \tag{A.241}$$

Every indexed task is realizable and finitely supported. If two indices encode the same task, pushforward simply adds their masses and leaves all expectations unchanged. Finally, any raw learner in the proposition maps by Proposition A.20 to $p^A \in \mathcal{P}_i$. Combining (A.241), (A.225), and Proposition A.21 gives

$$\mathbb{E}_{(c_i, D_i) \sim \nu_i} \mathbb{E}_{S_i, A} R_{D_i}(h_{A(S_i)}, c_i) \geq \gamma_i > \frac{1}{8}.$$

The finite game, its optimizer, and ν_i are chosen before A, which is the required prior-before-learner order. □

A.11 Uniform control of factor-sample overflow

Lemma A.28 (Optimized binomial exponential envelope)**.** *Let $n \in \mathbb{N}$, $0 < p \leq 1$, $L \sim \mathrm{Bin}(n, p)$, and $\mu = np$. For every real $t > \mu$,*

$$\Pr(L \geq t) \leq B(\mu, t) := \exp\left\{ t - \mu - t \log \frac{t}{\mu} \right\} = e^{-\mu} \left(\frac{e\mu}{t} \right)^t. \tag{A.242}$$

For fixed $t > 0$, $B(\mu, t)$ is strictly increasing on $0 < \mu < t$; for fixed $\mu > 0$, it is strictly decreasing on $t > \mu$. If $t > n$, the probability on the left is zero.

Proof. For $\lambda > 0$, exponential Markov and the exact binomial MGF give

$$\Pr(L \geq t) \leq e^{-\lambda t} \mathbb{E} e^{\lambda L}, \tag{A.243}$$

and

$$\mathbb{E} e^{\lambda L} = (1 - p + pe^\lambda)^n = \big(1 + p(e^\lambda - 1)\big)^n \leq \exp\{\mu(e^\lambda - 1)\}. \tag{A.244}$$

Here $1 + u \leq e^u$ for $u \geq 0$, since $e^u - 1 - u$ vanishes at zero and has nonnegative derivative $e^u - 1$. Because $t > \mu > 0$, choose

$$\lambda = \log(t/\mu) > 0. \tag{A.245}$$

Substitution gives exponent $-t\log(t/\mu)+\mu(t/\mu-1)=t-\mu-t\log(t/\mu)$, proving (A.242). Finite support handles $t>n$.

Finally,

$$\frac{\partial}{\partial\mu}\log B(\mu,t)=-1+\frac{t}{\mu}>0\quad(0<\mu<t),\tag{A.246}$$

and

$$\frac{\partial}{\partial t}\log B(\mu,t)=\log\frac{\mu}{t}<0\quad(t>\mu).\tag{A.247}$$

Since $B>0$, these signs prove the monotonicity claims. □

Lemma A.29 (Exact numerical slack for overflow)**.** *With e the base of the natural logarithm,*

$$\eta_0=e^7\left(\frac{2}{9}\right)^9<\frac{3}{2048}.\tag{A.248}$$

Moreover, $\log 4>1$.

Proof. For $\ell\in\mathbb{N}_0$,

$$(6+\ell)!\geq 6!\,7^{\ell}.\tag{A.249}$$

The exponential and geometric series therefore give

$$\begin{aligned}e&=\sum_{j=0}^{5}\frac{1}{j!}+\sum_{\ell=0}^{\infty}\frac{1}{(6+\ell)!}\\&\leq\frac{163}{60}+\frac{1}{720}\sum_{\ell=0}^{\infty}7^{-\ell}=\frac{163}{60}+\frac{7}{4320}=\frac{11743}{4320}<\frac{87}{32}.\end{aligned}\tag{A.250}$$

The final comparison is exact because

$$11743\cdot 32=375776<375840=87\cdot 4320.\tag{A.251}$$

Thus $e<87/32<4$, and strict monotonicity of the natural logarithm gives $1=\log e<\log 4$.

Using $87=3\cdot 29$ and $9=3^2$,

$$\eta_0<\left(\frac{87}{32}\right)^7\left(\frac{2}{9}\right)^9=\frac{29^7}{2^{26}3^{11}}.\tag{A.252}$$

Exact integer arithmetic gives

$$29^7=17{,}249{,}876{,}309<17{,}414{,}258{,}688=3^{12}2^{15}.\tag{A.253}$$

Substitution into (A.252) yields $\eta_0<3^{12}2^{15}/(2^{26}3^{11})=3/2^{11}$, as claimed. □

Lemma A.30 (Floor-eight overflow at small mean)**.** *Under Lemma A.28, let* $L\sim\mathrm{Bin}(n,p)$, $0<p\leq 1$, $0<\mu=np\leq 2$, *and* $m=\max\{8,\lceil 4\mu\rceil\}$. *Then*

$$\Pr(L>m)\leq e^7(2/9)^9=\eta_0.\tag{A.254}$$

This includes $\mu=2$.

Proof. If $n \le 8$, then $L \le n \le 8 \le m$ pointwise. Otherwise $n \ge 9$, and integer-valued L, m with $m \ge 8$ give

$$\{L > m\} = \{L \ge m+1\} \subseteq \{L \ge 9\}. \tag{A.255}$$

Since $0 < \mu \le 2 < 9$, Lemma A.28 at $t = 9$, followed by its fixed-threshold monotonicity, yields

$$\Pr(L > m) \le B(\mu, 9) \le B(2, 9) = e^{-2}\left(\frac{2e}{9}\right)^9 = e^7\left(\frac{2}{9}\right)^9. \tag{A.256}$$

At $\mu = 2$, this is exactly the transition endpoint. ☐

Lemma A.31 (Factor-four overflow at large mean)**.** *Under Lemmas A.28 and A.29, let* $L \sim \mathrm{Bin}(n, p)$*,* $0 < p \le 1$*,* $\mu = np \ge 2$*, and* $m = \max\{8, \lceil 4\mu \rceil\}$*. Then*

$$\Pr(L > m) \le e^7(2/9)^9 = \eta_0. \tag{A.257}$$

Proof. If $m \ge n$, finite support gives zero probability. Otherwise, because L, m are integers and $m \ge \lceil 4\mu \rceil$,

$$\{L > m\} = \{L \ge m+1\} \subseteq \{L \ge 4\mu + 1\}, \tag{A.258}$$

where

$$m + 1 \ge \lceil 4\mu \rceil + 1 \ge 4\mu + 1. \tag{A.259}$$

The real threshold $4\mu + 1$ exceeds μ, so

$$\Pr(L > m) \le B(\mu, 4\mu + 1). \tag{A.260}$$

Let

$$f(\mu) = \log B(\mu, 4\mu + 1) = 3\mu + 1 - (4\mu + 1)\log\left(4 + \frac{1}{\mu}\right). \tag{A.261}$$

Direct differentiation gives

$$f'(\mu) = 3 + \frac{1}{\mu} - 4\log\left(4 + \frac{1}{\mu}\right). \tag{A.262}$$

For $\mu \ge 2$, $1/\mu \le 1/2$, $4 + 1/\mu > 4$, and Lemma A.29 gives $\log 4 > 1$. Therefore

$$f'(\mu) < 3 + \frac{1}{2} - 4\log 4 < \frac{7}{2} - 4 = -\frac{1}{2} < 0. \tag{A.263}$$

Thus f decreases on $[2, \infty)$, and

$$B(\mu, 4\mu + 1) \le B(2, 9) = e^7(2/9)^9. \tag{A.264}$$

Equations (A.260) and (A.264) prove the claim, with exact agreement at $\mu = 2$. ☐

Lemma A.32 (Finite-support endpoints and one-factor zero overflow)**.** *Let* $L \sim \mathrm{Bin}(n, p)$*,* $p \in [0, 1]$*, and* $m = \max\{8, \lceil 4np \rceil\}$*. Then*

$$\Pr(L > m) = 0 \tag{A.265}$$

whenever $n \le 8$*,* $p = 0$*, or* $p = 1$*. Under Assumptions 1 and 2 and the exact weights of Lemma A.24, this applies with* $p = \pi_i$*, and in particular gives zero overflow when* $k = 1$*.*

Proof. Binomial support gives $L \le n$. Thus $n \le 8$ implies $L \le 8 \le m$. If $p = 0$, then $L = 0$ almost surely. If $p = 1$, then $L = n$ almost surely and

$$m = \max\{8, \lceil 4n \rceil\} = \max\{8, 4n\} \ge n. \tag{A.266}$$

When $k = 1$, $M = s_1$, so $\pi_1 = s_1/M = 1$ and the last case gives $L_1 = n \le m_{n,1}$ almost surely. ☐

Proposition A.23 (Uniform marginal overflow certificate)**.** *Under Assumptions 1 and 2 and Lemma A.24, for the same fixed candidate and every factor, if $L_i \sim \mathrm{Bin}(n, \pi_i)$, then*

$$\Pr[L_i > m_{n,i}] \le \eta_0 = e^7 \left(\frac{2}{9}\right)^9 < \frac{3}{2048}. \tag{A.267}$$

The probability is zero when $\pi_i = 0$, $\pi_i = 1$, or $n \le 8$, and hence is zero in the one-factor case.

Proof. Lemma A.32 handles the stated exact endpoints. Otherwise put $\mu = n\pi_i$. Lemma A.30 handles $0 < \mu \le 2$, while Lemma A.31 handles $\mu > 2$; they agree at $\mu = 2$. Lemma A.29 supplies the strict comparison with $3/2048$. This is a uniform factorwise marginal statement, not a union bound or a claim of independence among counts. □

A.12 One-use hidden-factor simulation

All active priors ν_j are fixed before the arbitrary global learner is quantified. A hidden learner may depend on that already fixed global learner, but none of its hard priors does.

Lemma A.33 (Prior-fixed common ideal mixture experiment)**.** *Under Assumption 1, Proposition A.19, and Proposition A.22, one can fix, before every global learner, a law λ_j on realizable factor tasks for each j, with $\lambda_j = \nu_j$ on H and λ_j a deterministic point mass off H. Independent tasks from these laws determine a measurable full-product target $c^\star \in C$, a probability measure $D^\star$ with $D^\star(X_j) = \pi_j$, and an ideal sample $S^\star \sim (D^\star_{c^\star})^n$. Its route counts satisfy $L_j \sim \mathrm{Bin}(n, \pi_j)$, and every task is chosen before the single call to a fixed global kernel.*

Proof. For each $j \notin H$, fix $c_j^\circ \in C_j$ and $x_j^\circ \in X_j$, and set

$$T_j^\circ = (c_j^\circ, \delta_{x_j^\circ}), \qquad \lambda_j = \delta_{T_j^\circ}. \tag{A.268}$$

Only finitely many choices are made. For $j \in H$, set $\lambda_j = \nu_j$. Proposition A.22 makes each active law finitely supported on realizable finite-support tasks and fixed before the learner.

Draw independently

$$T_j = (c_j, D_j) \sim \lambda_j, \qquad j \in [k]. \tag{A.269}$$

Assumption 1 supplies the unique $c^\star \in C$ restricting to c_j on X_j. Define

$$D^\star(B) = \sum_{j=1}^{k} \pi_j D_j(B \cap X_j), \qquad B \in \Sigma. \tag{A.270}$$

The measurable disjoint blocks and weights summing to one make this a probability measure with $D^\star(X_j) = \pi_j$.

Draw iid route indices

$$I_1, \dots, I_n \in [k], \qquad \Pr(I_r = j) = \pi_j, \tag{A.271}$$

and set

$$\ell_{j,r} = |\{s \le r : I_s = j\}|, \qquad L_j = \ell_{j,n}. \tag{A.272}$$

Conditional on the task vector and route, draw independently over all j and $a \le n \vee m_{n,j}$,

$$U_{j,a} = (X_{j,a}, c_j(X_{j,a})), \qquad X_{j,a} \sim D_j, \tag{A.273}$$

and put

$$S_r^\star = U_{I_r, \ell_{I_r,r}}, \qquad S^\star = (S_1^\star, \dots, S_n^\star). \tag{A.274}$$

Distinct global slots use distinct independent row-array coordinates. For measurable labeled-record events $B_1, \dots, B_n$, conditional on the task vector,

$$\begin{aligned}
&\Pr(S_1^\star \in B_1, \dots, S_n^\star \in B_n \mid (T_j)_j) \\
&= \sum_{j_1,\dots,j_n} \prod_{r=1}^{n} \pi_{j_r} (D_{j_r})_{c_{j_r}}(B_r) \\
&= \prod_{r=1}^{n} \sum_{j=1}^{k} \pi_j (D_j)_{c_j}(B_r) = \prod_{r=1}^{n} D_{c^\star}^\star(B_r).
\end{aligned} \tag{A.275}$$

Thus the sample is iid from the displayed mixture, and (A.271) gives the binomial marginal counts.

All task, support-point, and route indices range over finite sets, so the experiment is a finite mixture of Dirac laws and is measurable without a sigma-field on an infinite task family. After the tasks, route, and arrays are drawn and the sample assembled, make one call

$$\Omega^\star \sim A_n(S^\star, \cdot). \tag{A.276}$$

□

Proposition A.24 (Total measurable one-use hidden-factor kernel)**.** *Under Assumption 3, Lemma A.26, Proposition A.22, and Lemma A.33, fix the task laws, then an arbitrary total global Markov kernel $A_n : Z^n \rightsquigarrow (\Omega, \mathscr{F})$ with measurable finite evaluations, and then $i \in H$. There is a total Markov kernel*

$$K_i^{A_n} : (X_i \times \{0,1\})^{m_{n,i}} \rightsquigarrow (\mathcal{V}_i, 2^{\mathcal{V}_i}) \tag{A.277}$$

with decoder e_i, defined on every labeled factor database. Its overflow branch is input-independent and occurs before any input row is read; off overflow, input row a is inserted only into the a-th requested i-slot. All other tasks and rows are sampled independently of the input before the one call to A_n.

Proof. Write $m_i = m_{n,i}$ and define the finite-evaluation restriction

$$\rho_i : \Omega \to \mathcal{V}_i, \qquad \rho_i(\omega) = (h_\omega(x_{i,1}), \dots, h_\omega(x_{i,t_i})). \tag{A.278}$$

For every $E \subseteq \mathcal{V}_i$, the preimage $\rho_i^{-1}(E)$ is a finite union of finite-evaluation cylinders and is measurable. Fix a fallback $v_i^\circ \in \mathcal{V}_i$.

On arbitrary input

$$s = (z_1, \dots, z_{m_i}) \in (X_i \times \{0,1\})^{m_i}, \tag{A.279}$$

the kernel samples without inspecting s: all tasks $T_j \sim \lambda_j$ for $j \neq i$, the entire route and hence $\mathcal{O}_i = \{L_i > m_i\}$, and, only on $\mathcal{O}_i^c$, the rows needed at non-i slots. On overflow it immediately returns v_i°, without reading s, requesting an unavailable row, assembling a database, or calling A_n.

On $\mathcal{O}_i^c$, for a route slot with $I_r = i$, the occurrence number $\ell_{i,r}$ lies in $[m_i]$. If ξ denotes all realized input-independent auxiliary variables, define

$$\big(\Phi_{i,\xi}(s)\big)_r = \begin{cases} z_{\ell_{i,r}}, & I_r = i, \\ W_r, & I_r \neq i, \end{cases} \tag{A.280}$$

where $W_r \sim (D_{I_r})_{c_{I_r}}$ was already drawn. Each requested input row is used exactly once, and every row z_a with $a > L_i$ is unused. The kernel calls A_n once on this database and returns $\rho_i(\omega)$.

Let Q_i be the finite law of ξ. For $E \subseteq \mathcal{V}_i$, define, without evaluating $\Phi_{i,\xi}$ on overflow,

$$K_{i,\xi}(s,E) = \begin{cases} \mathbf{1}_E(v_i^\circ), & \mathcal{O}_i(\xi), \\ A_n(\Phi_{i,\xi}(s), \rho_i^{-1}(E)), & \mathcal{O}_i(\xi)^c, \end{cases} \tag{A.281}$$

and

$$K_i^{A_n}(s,E) = \int K_{i,\xi}(s,E)\, \mathsf{Q}_i(d\xi). \tag{A.282}$$

For fixed nonoverflow ξ, every coordinate of $\Phi_{i,\xi}$ is an input projection or a constant labeled record. The trace-space inclusion is measurable because $X_i \in \Sigma$; hence the assembly map and the nonoverflow branch are measurable. The overflow branch is constant, and the integral is a finite sum. For each input it is a probability law. It is total on inconsistent and nonrealizable inputs, and Lemma A.26 supplies the measurable possibly improper decoder e_i. □

Proposition A.25 (All-input privacy from one-use insertion)**.** *Under Assumption 4 and Proposition A.24, if A_n is all-input replacement-(ε,δ)-DP, then every $K_i^{A_n}$, $i \in H$, is all-input replacement-(ε,δ)-DP and hence replacement-$(0.1,\delta)$-DP. No realizability premise or privacy composition is used.*

Proof. Fix adjacent factor databases s, s', differing only in row $a \in [m_i]$, and one auxiliary realization ξ. On overflow, both inputs return the same point mass. Off overflow, if $a > L_i$, the row is unused and

$$\Phi_{i,\xi}(s) = \Phi_{i,\xi}(s'). \tag{A.283}$$

If $a \le L_i$, there is a unique requested slot

$$r_i(a) = \min\{r : \ell_{i,r} = a\}. \tag{A.284}$$

The assembly map inserts row a there and nowhere else. Thus the two global datasets are equal or differ in exactly one labeled record, even for inconsistent labels.

For each $E \subseteq \mathcal{V}_i$, global privacy applied once to the measurable event $\rho_i^{-1}(E)$, together with equality in the other branches, gives

$$K_{i,\xi}(s,E) \le e^{\varepsilon} K_{i,\xi}(s',E) + \delta. \tag{A.285}$$

The law Q_i is input-independent. Integrating the same conditional inequality yields

$$\begin{aligned} K_i^{A_n}(s,E) &\le \int \big(e^{\varepsilon} K_{i,\xi}(s',E) + \delta\big)\, \mathsf{Q}_i(d\xi) \\ &= e^{\varepsilon} K_i^{A_n}(s',E) + \delta. \end{aligned} \tag{A.286}$$

Exchanging s, s' supplies the reverse direction. The additive term is one δ, since this is input-independent mixing of one-call kernels, not composition. Finally $\varepsilon \le 0.1$ gives $e^{\varepsilon} \le e^{0.1}$. □

Proposition A.26 (Exact marginal identity coupling)**.** *Under Assumption 1, Lemmas A.6, A.26, and A.33, Propositions A.22, A.23, A.24, and A.25, fix $i \in H$. When $T_i = (c_i, D_i) \sim \nu_i$ and the hidden-kernel input is iid from $(D_i)_{c_i}^{m_i}$, its experiment can be coupled to the common ideal experiment so that, on $\mathcal{O}_i^c$, the task vector, global target, mixture distribution, global dataset, learner-output coordinate, finite restriction, and factor risk are identical. If R_i^{tr} and R_i^{id} are the truncated and ideal risks, then*

$$|R_i^{\mathrm{tr}} - R_i^{\mathrm{id}}| \le \mathbf{1}_{\mathcal{O}_i} \tag{A.287}$$

and

$$|\mathbb{E}R_i^{\mathrm{tr}} - \mathbb{E}R_i^{\mathrm{id}}| \le \Pr(\mathcal{O}_i) \le \eta_0. \tag{A.288}$$

The same holds conditional on any task in $\operatorname{supp}\nu_i$*. This is a factor-marginal coupling, not a joint-factor independence claim.*

Proof. Use the variables of Lemma A.33. Define the factor input

$$S_i^{\mathrm{in}} = (U_{i,1}, \dots, U_{i,m_i}). \tag{A.289}$$

Conditional on T_i, it has law $(D_i)_{c_i}^{m_i}$ and is independent of T_{-i}, the route, and the other-factor arrays, precisely the auxiliary marginal required by the hidden kernel. At a non-i slot use

$$W_r = U_{I_r,\ell_{I_r,r}}. \tag{A.290}$$

On $\mathcal{O}_i^c$, if $I_r = i$,

$$\big(\Phi_{i,\xi}(S_i^{\mathrm{in}})\big)_r = U_{i,\ell_{i,r}} = S_r^\star; \tag{A.291}$$

if $I_r \neq i$, the same follows from (A.290). Hence

$$\Phi_{i,\xi}(S_i^{\mathrm{in}}) = S^\star \quad \text{on } \mathcal{O}_i^c. \tag{A.292}$$

The task vector, target $c^\star$, and mixture $D^\star$ are literally shared. Full-product realizability is used here, but was not used in the privacy proof.

Always sample $\Omega^\star$ from (A.276). Define

$$V_i^{\mathrm{id}} = \rho_i(\Omega^\star), \qquad V_i^{\mathrm{tr}} = \begin{cases} \rho_i(\Omega^\star), & \mathcal{O}_i^c, \\ v_i^\circ, & \mathcal{O}_i. \end{cases} \tag{A.293}$$

The database identity makes this the correct hidden-kernel marginal off overflow; the fallback branch gives the correct marginal on overflow. No random-seed representation of A_n is needed.

Set

$$R_i^{\mathrm{id}} = R_{D_i}(h_{\Omega^\star}|_{X_i}, c_i), \qquad R_i^{\mathrm{tr}} = R_{D_i}(e_i(V_i^{\mathrm{tr}}), c_i). \tag{A.294}$$

Every task in the prior is supported on F_i, and finite coding gives

$$e_i(\rho_i(\omega))(x_{i,a}) = h_\omega(x_{i,a}), \qquad a \in [t_i]. \tag{A.295}$$

Therefore $R_i^{\mathrm{tr}} = R_i^{\mathrm{id}}$ off overflow. Both risks are measurable finite sums in $[0,1]$, so their difference is at most one on overflow, proving (A.287). Expectation and Proposition A.23 give (A.288). The route is task-independent, so conditioning on a fixed supported task does not change the bound. □

Proposition A.27 (Boundary traces and the one-factor specialization)**.** *Under Propositions A.24, A.25, and A.26, the following hold:*

1. *If $L_i = 0$, the factor input is never read and ideal/truncated identity is exact.*
2. *Replacing an unused row changes no global row; replacing a used row changes only its unique requested slot.*
3. *On overflow, output is input-independent and selected before an unavailable row is requested.*
4. *Privacy covers every nonrealizable factor input.*
5. *If $H = \{i\}$, the construction remains valid with no other active prior and fixed tasks on all low factors.*
6. *At the algebraic endpoint $\pi_i = 0$, the construction reads no input and has zero overflow and zero discrepancy.*

7. *If $k = 1$ on the active branch, then $L_1 = n \le m_{n,1}$, the first n input rows are inserted once, and the coupling loss is zero.*

Proof. If $L_i = 0$, the assembly map contains no factor-input coordinate and still equals the ideal database. The used, unused, overflow, and nonrealizable cases are exactly the exhaustive branches in (A.283)–(A.286). If only one factor is active, the product over other active priors is the unique empty-product law, while (A.268) supplies all low-factor tasks. At $\pi_i = 0$, the route gives $L_i = 0$ surely, and the overflow certificate gives probability zero.

For $k = 1$, $\pi_1 = 1$, the route is deterministic, and

$$L_1 = n, \qquad m_{n,1} = \max\{8, \lceil 4n \rceil\} = \max\{8, 4n\} \ge n. \tag{A.296}$$

Hence $\mathcal{O}_1 = \varnothing$. There are no other tasks or rows; input row a is inserted at global slot a for $1 \le a \le n$, and all remaining input rows are unused. The coupling identities hold everywhere, so

$$|\mathbb{E}R_1^{\mathrm{tr}} - \mathbb{E}R_1^{\mathrm{id}}| = 0. \tag{A.297}$$

□

A.13 Product-prior tensorization of exact risk

Fix one lower-bound candidate and let H, w_L, and the active hard priors $(\nu_i)_{i\in H}$ be as in Lemma A.24 and Proposition A.22. For every $j \notin H$, let T_j° be the deterministic realizable task used in Lemma A.33, and set

$$\lambda_j = \begin{cases} \nu_j, & j \in H, \\ \delta_{T_j^\circ}, & j \notin H, \end{cases} \qquad \Lambda = \bigotimes_{j=1}^{k} \lambda_j. \tag{A.298}$$

The finite family of priors, and hence Λ, is fixed before the global learner. After an arbitrary global learner A_n is fixed, write $\mathbb{P}_\star$ for the common experiment of Lemma A.33: draw all factor tasks from Λ, form the full target $c^\star$ and mixture $D^\star$, draw the iid global sample $S^\star$, and finally draw $\Omega^\star \sim A_n(S^\star, \cdot)$. Let

$$R_i^\star = R_{D_i}(h_{\Omega^\star}|_{X_i}, c_i), \qquad R^\star = R_{D^\star}(h_{\Omega^\star}, c^\star). \tag{A.299}$$

These are measurable $[0,1]$-valued random variables under the one common law $\mathbb{P}_\star$; write $\mathbb{E}_\star$ for expectation under that law.

Lemma A.34 (Active marginal floors in one common experiment)**.** *Under Assumption 1, Proposition A.22, Lemma A.33, and Propositions A.24, A.25, and A.26, fix all active priors before an arbitrary global learner and define the common law and risks by* (A.298)–(A.299)*. Then, for every $i \in H$,*

$$\mathbb{E}_\star R_i^\star > \frac{1}{8} - \eta_0. \tag{A.300}$$

Every expectation in (A.300) *is a marginal of the same common experiment; no independence among factor outputs is required.*

Proof. Fix $i \in H$. Propositions A.24 and A.25 construct, from the fixed common task laws and the arbitrary global learner, a total unrestricted replacement-$(0.1, \delta)$-DP factor learner $K_i^{A_n}$ on exactly $m_{n,i}$ rows. This learner lies in the universal learner quantifier of Proposition A.22. Define

$$R_i^{\mathrm{tr}} = R_{D_i}\big(e_i(K_i^{A_n}(S_i)), c_i\big), \qquad (c_i, D_i) \sim \nu_i, \quad S_i \sim (D_i)_{c_i}^{m_{n,i}}. \tag{A.301}$$

The hard-prior conclusion gives the strict inequality

$$\mathbb{E}R_i^{\mathrm{tr}} > \frac{1}{8}. \tag{A.302}$$

Proposition A.26 identifies the ideal counterpart of (A.301) with the i-th marginal of $\mathbb{P}_\star$. Its same-target coupling, together with Proposition A.23, gives

$$\left|\mathbb{E}R_i^{\mathrm{tr}} - \mathbb{E}_\star R_i^\star\right| \le \Pr(\mathcal{O}_i) \le \eta_0. \tag{A.303}$$

Consequently,

$$\mathbb{E}_\star R_i^\star \ge \mathbb{E}R_i^{\mathrm{tr}} - \eta_0 > \frac{1}{8} - \eta_0. \tag{A.304}$$

Thus the ideal/truncated discrepancy is charged exactly once. The finite evaluation map, task restriction, construction of the full target, and raw decoding create no additional residual by Proposition A.26.

The same marginal argument applies to every active i, but each right-hand variable $R_i^\star$ remains the factor risk on the already fixed common law $\mathbb{P}_\star$. The hidden kernels need not be released or run jointly, so this argument never combines inequalities from different ideal experiments. □

Lemma A.35 (Exact disjoint-risk sum in the common experiment)**.** *Under Assumption 1, Lemma A.6, and Lemma A.33, the risks in* (A.299) *satisfy, on every outcome of the common experiment,*

$$R^\star = \sum_{i=1}^{k} \pi_i R_i^\star. \tag{A.305}$$

Consequently,

$$\mathbb{E}_\star R^\star = \sum_{i=1}^{k} \pi_i \mathbb{E}_\star R_i^\star. \tag{A.306}$$

Neither equality requires independence among the restrictions of the global learner output.

Proof. Fix one outcome of the finite task draw and one decoded global output $h_{\Omega^\star}$. Lemma A.33 gives

$$c^\star|_{X_i} = c_i, \qquad D^\star(B) = \sum_{i=1}^{k} \pi_i D_i(B \cap X_i), \tag{A.307}$$

where D_i is supported on X_i. The mistake set is the disjoint union

$$\{x : h_{\Omega^\star}(x) \neq c^\star(x)\} = \bigsqcup_{i=1}^{k} \{x \in X_i : h_{\Omega^\star}(x) \neq c_i(x)\}. \tag{A.308}$$

Applying (A.307) to (A.308), equivalently applying Lemma A.6, yields

$$\begin{aligned} R^\star &= D^\star\{h_{\Omega^\star} \neq c^\star\} \\ &= \sum_{i=1}^{k} \pi_i D_i\{x \in X_i : h_{\Omega^\star}(x) \neq c_i(x)\} \\ &= \sum_{i=1}^{k} \pi_i R_i^\star. \end{aligned} \tag{A.309}$$

This is an equality for the realized joint output, not merely an equality in distribution or an approximation. Arbitrary correlation among the restrictions of $h_{\Omega^\star}$ is present on both sides. Because every summand is measurable and bounded and $k < \infty$, finite linearity of expectation applied to (A.309) proves (A.306). There is no infinite interchange or conditioning conversion. □

Proposition A.28 (Common-prior weighted tensorization)**.** *Under Lemma A.24 and Lemmas A.34 and A.35, the one common product-prior ideal experiment satisfies*

$$\mathbb{E}_\star R_{D^\star}(h_{\Omega^\star}, c^\star) > (1 - w_L)\left(\frac{1}{8} - \eta_0\right). \tag{A.310}$$

The only losses are the low-factor mass w_L and one marginal residual η_0 per unit of active weight.

Proof. All factor risks are nonnegative. Lemma A.35 therefore gives

$$\mathbb{E}_\star R^\star = \sum_{i=1}^{k} \pi_i \mathbb{E}_\star R_i^\star \geq \sum_{i \in H} \pi_i \mathbb{E}_\star R_i^\star. \tag{A.311}$$

Apply (A.302) and (A.303) before summing. Since every active weight is positive,

$$\begin{aligned}\sum_{i \in H} \pi_i \mathbb{E}_\star R_i^\star &\geq \sum_{i \in H} \pi_i \big(\mathbb{E} R_i^{\mathrm{tr}} - \eta_0\big) \\ &> \sum_{i \in H} \pi_i \left(\frac{1}{8} - \eta_0\right).\end{aligned} \tag{A.312}$$

The first line charges factor i's residual as $\pi_i \Pr(\mathcal{O}_i) \leq \pi_i \eta_0$. Hence the complete residual is

$$\sum_{i \in H} \pi_i \eta_0 = \eta_0 \sum_{i \in H} \pi_i, \tag{A.313}$$

not the probability of a union of overflow events and not an unweighted multiple of η_0.

Lemma A.24 gives

$$\sum_{i \in H} \pi_i = 1 - \sum_{i \notin H} \pi_i = 1 - w_L. \tag{A.314}$$

It also gives $H \neq \varnothing$, while every $\pi_i = s_i / M$ is positive. Thus the second line of (A.312) remains strict. Combining (A.311)–(A.314) proves (A.310).

The low-factor tasks were fixed before the learner and remain part of the same target, distribution, sample, and output. Their risks may be arbitrarily correlated with all active risks and may be zero; only their nonnegativity is used. The proof also uses only marginal expectations and never assumes independence of factor outputs or overflow events. □

Proposition A.29 (Exact one-factor active-branch floor)**.** *Under Lemma A.24, Proposition A.22, Lemma A.32, Propositions A.26 and A.27, and Lemma A.35, suppose $k = 1$ on the active fixed-candidate contradiction branch. Then*

$$\mathcal{O}_1 = \varnothing \quad \textit{almost surely}, \qquad \mathbb{E}_\star R^\star > \frac{1}{8}. \tag{A.315}$$

Thus the one-factor baseline has no overflow or coupling loss.

Proof. Lemma A.24 gives $H \neq \varnothing$ on the contradiction branch. When $k = 1$, this forces

$$H = \{1\}, \qquad \pi_1 = 1, \qquad w_L = 0. \tag{A.316}$$

Lemma A.32 gives $L_1 = n$ and

$$m_{n,1} = \max\{8, 4n\} \geq n, \tag{A.317}$$

so $\mathcal{O}_1 = \varnothing$ exactly. Proposition A.27 then shows that the one-use learner inserts the first n factor rows once and that the truncated and ideal risks coincide pointwise:

$$R_1^{\mathrm{tr}} = R_1^\star. \tag{A.318}$$

Proposition A.22 gives $\mathbb{E}R_1^{\mathrm{tr}} > 1/8$. Taking expectations in (A.318) and applying Lemma A.35 with $\pi_1 = 1$ gives

$$\mathbb{E}_\star R^\star = \mathbb{E}_\star R_1^\star = \mathbb{E}R_1^{\mathrm{tr}} > \frac{1}{8}. \tag{A.319}$$

If the sole factor were outside H, then $H = \varnothing$, which Lemma A.24 already excludes on this branch. $\square$

Proof of the product-prior tensorization conclusion. Proposition A.22 fixes every active prior independently of the learner, and Lemma A.33 places their finite product and the deterministic low-factor tasks in one experiment before data or learner randomness. For each active factor, Propositions A.24 and A.25 supply an eligible unrestricted factor learner. Lemma A.34 then transfers that factor's hard-prior floor to the corresponding marginal of the same common experiment, with sole residual η_0.

Lemma A.35 gives the pointwise finite weighted risk identity, and Proposition A.28 combines it with the active-mass identity from Lemma A.24 to obtain (A.310). Equation (A.313) records that each marginal loss is charged once by its weight, with no union bound or independence assumption. Finally, Proposition A.29 gives the exact zero-overflow floor $> 1/8$ when $k = 1$ on the active branch. $\square$

A.14 Candidate-wise unrestricted PAC lower bound

Retain the common finite product-task law Λ from (A.298), and let

$$\mathcal{T} = \operatorname{supp}(\Lambda) \tag{A.320}$$

be its finite set of positive-mass atoms. For $\tau = ((c_i^\tau, D_i^\tau))_{i=1}^k \in \mathcal{T}$, Lemma A.33 gives the full-product target $c^\tau \in C$, characterized by $c^\tau|_{X_i} = c_i^\tau$, and the probability measure

$$D^\tau(B) = \sum_{i=1}^k \pi_i D_i^\tau(B \cap X_i), \qquad B \in \Sigma. \tag{A.321}$$

Because every D_i^τ is supported on X_i,

$$D^\tau(X_i) = \pi_i = \omega_i, \qquad \sum_{i=1}^k \pi_i = 1. \tag{A.322}$$

For a fixed learner and candidate, define

$$L_\tau = R_{D^\tau}(h_{\Omega_\tau}, c^\tau), \qquad S_\tau \sim (D_{c^\tau}^\tau)^n, \quad \Omega_\tau \sim A_n(S_\tau, \cdot). \tag{A.323}$$

The random variable L_τ is measurable and belongs to $[0,1]$. The target, mixture, iid sample, and learner output in (A.323) are precisely the conditional objects of the common experiment. Since $\mathcal{T}$ is finite and each of its atoms has positive mass, Lemma A.35 and finite conditioning give

$$\mathbb{E}_\star R^\star = \sum_{\tau\in\mathcal{T}} \Lambda(\tau)\,\mathbb{E}L_\tau. \tag{A.324}$$

Lemma A.36 (Exact PAC ceiling under finite task averaging)**.** *Under the measurable binary-risk convention, Lemma A.21, Lemma A.33, and Lemma A.35, fix one candidate n, one learner A_n, and define the finite common experiment and atom losses by* (A.320)–(A.324)*. If every deterministic $\tau \in \mathcal{T}$ satisfies*

$$\Pr[L_\tau \le 1/16] \ge \frac{15}{16}, \tag{A.325}$$

then

$$\mathbb{E}L_\tau \le \frac{31}{256} \quad (\tau\in\mathcal{T}), \qquad \mathbb{E}_\star R^\star \le \frac{31}{256}. \tag{A.326}$$

For each fixed atom, one also has the strict implication

$$\mathbb{E}L_\tau > \frac{31}{256} \quad \Longrightarrow \quad \Pr[L_\tau > 1/16] > \frac{1}{16}. \tag{A.327}$$

Proof. Fix $\tau \in \mathcal{T}$ and put

$$p_\tau = \Pr[L_\tau > 1/16]. \tag{A.328}$$

This event is exactly the complement of the event in (A.325); loss equal to $1/16$ remains on the good side. Since $0 \le L_\tau \le 1$,

$$\begin{aligned}\mathbb{E}L_\tau &= \mathbb{E}[L_\tau \mathbf{1}\{L_\tau \le 1/16\}] + \mathbb{E}[L_\tau \mathbf{1}\{L_\tau > 1/16\}]\\ &\le \frac{1}{16}(1-p_\tau) + p_\tau = \frac{1}{16} + \frac{15}{16}p_\tau.\end{aligned} \tag{A.329}$$

Under (A.325), $p_\tau \le 1/16$, and hence

$$\mathbb{E}L_\tau \le \frac{1}{16} + \frac{15}{16}\frac{1}{16} = \frac{16}{256} + \frac{15}{256} = \frac{31}{256}. \tag{A.330}$$

This is also the direct current-notation instance of Lemma A.21. Taking the strict contrapositive of (A.330) proves (A.327). In particular, equality $p_\tau = 1/16$ produces only the required non-strict ceiling.

Average (A.330) with (A.324). Because $\mathcal{T}$ is finite and Λ is a probability law,

$$\mathbb{E}_\star R^\star \le \sum_{\tau\in\mathcal{T}} \Lambda(\tau)\frac{31}{256} = \frac{31}{256}. \tag{A.331}$$

No infinite interchange, measurable selection, or change from taskwise probability to the common prior average occurs. □

Lemma A.37 (Strict rational separation of the tensorized floor)**.** *Under Lemma A.24 and Proposition A.28, fix the same candidate, learner, and common experiment on the branch $n < c_{\mathrm{low}} M_\oplus(C)$. If*

$$w_L < \frac{1}{128}, \qquad \eta_0 < \frac{3}{2048}, \tag{A.332}$$

then

$$\mathbb{E}_\star R^\star > \frac{32131}{262144} > \frac{31}{256}. \tag{A.333}$$

Proof. The strict bounds in (A.332) give

$$1 - w_L > \frac{127}{128} > 0 \tag{A.334}$$

and

$$\frac{1}{8} - \eta_0 > \frac{1}{8} - \frac{3}{2048} = \frac{256-3}{2048} = \frac{253}{2048} > 0. \tag{A.335}$$

Proposition A.28, followed by (A.334) and (A.335), yields

$$\begin{aligned}\mathbb{E}_\star R^\star &> (1 - w_L)\left(\frac{1}{8} - \eta_0\right) \\ &> \frac{127}{128}\frac{253}{2048} = \frac{127 \cdot 253}{128 \cdot 2048} = \frac{32131}{262144}.\end{aligned} \tag{A.336}$$

The final comparison is exact integer arithmetic:

$$\frac{31}{256} = \frac{31 \cdot 1024}{262144} = \frac{31744}{262144}, \qquad 32131 - 31744 = 387 > 0. \tag{A.337}$$

Thus

$$\frac{32131}{262144} - \frac{31}{256} = \frac{387}{262144} > 0. \tag{A.338}$$

Every comparison is strict; no decimal estimate or asymptotic absorption is used. ☐

Proposition A.30 (Deterministic PAC-failure atom)**.** *Under Lemma A.33, Lemma A.35, and Lemma A.36, define* $\mathcal{T}, \Lambda, L_\tau$ *by* (A.320)–(A.324)*. If*

$$\mathbb{E}_\star R^\star > \frac{31}{256}, \tag{A.339}$$

then some positive-mass atom $\tau_\star \in \mathcal{T}$ *has a deterministic full-product target* $c^{\tau_\star} \in C$ *and the legal probability measure*

$$D^{\tau_\star}(B) = \sum_{i=1}^{k} \pi_i D_i^{\tau_\star}(B \cap X_i) \tag{A.340}$$

such that $D^{\tau_\star}(X_i) = \pi_i = \omega_i$ *for every* i*, and*

$$\Pr_{\substack{S \sim (D^{\tau_\star}_{c^{\tau_\star}})^n \\ \Omega \sim A_n(S,\cdot)}} \left[R_{D^{\tau_\star}}(h_\Omega, c^{\tau_\star}) > \frac{1}{16} \right] > \frac{1}{16}. \tag{A.341}$$

Proof. If every atom obeyed $\mathbb{E}L_\tau \le 31/256$, then (A.324) would imply $\mathbb{E}_\star R^\star \le 31/256$, contrary to (A.339). Hence a positive-mass $\tau_\star \in \mathcal{T}$ satisfies

$$\mathbb{E}L_{\tau_\star} > \frac{31}{256}. \tag{A.342}$$

The implication (A.327) in Lemma A.36 gives (A.341), with exactly the same candidate, sample law, learner kernel, and binary risk.

Lemma A.33 constructs $c^{\tau_\star}$ by full Cartesian construction, so it belongs to C, rather than merely to a factorwise relaxation. The same lemma and (A.321)–(A.322) show that (A.340) is a probability measure with the required block masses. Lemma A.35 confirms that the selected loss is the exact global risk of this target and mixture. The mixture is legal on the selected factor measures' actual supports and requires no product-distribution, public-knowledge, quotient-output, finite-cardinality, or properness condition. Finite support describes this witness's provenance, not a restriction on the theorem's distribution family or learner. ☐

Proposition A.31 (Fixed-candidate lower closure)**.** *Under Assumption 5, Lemma A.21, Lemma A.24, Proposition A.19, Proposition A.28, Lemmas A.36 and A.37, and Proposition A.30, fix a candidate $n \in \mathbb{N}$ satisfying the two delta-budget conjuncts in Assumption 5. If an unrestricted measurable replacement-(ε, δ)-DP learner A_n satisfies the universal $(1/16, 1/16)$-PAC guarantee at this candidate, then*

$$n \geq c_{\text{low}} M_{\oplus}(C). \tag{A.343}$$

On the temporary branch $n < c_{\text{low}} M_{\oplus}(C)$, the contradiction is witnessed by the deterministic target and mixture in (A.340)–(A.341).

Proof. Write $M = M_{\oplus}(C)$, and suppose for contradiction that

$$n < c_{\text{low}} M. \tag{A.344}$$

The universal PAC guarantee and (A.344) discharge the local premises of Lemma A.24. Assumption 5 discharges the exact numerical premises of Proposition A.19 at this same n and its exact budgets $(m_{n,i})_i$. In particular, both $\delta \leq 1/[n \log(n+1)]$ and every $\delta \leq c_\delta/[m_{n,i}^2 \log(m_{n,i}+1)]$ remain in force, and equality in either cap is allowed.

The lower chain culminating in Proposition A.28 therefore applies to the same A_n, without changing its output space or restricting it to proper or factorwise hypotheses. Lemma A.37 gives

$$\mathbb{E}_\star R^\star > \frac{32131}{262144} > \frac{31}{256}. \tag{A.345}$$

The universal PAC guarantee applies to every deterministic (c^τ, D^τ) in the same experiment, so Lemma A.36 gives

$$\mathbb{E}_\star R^\star \leq \frac{31}{256}, \tag{A.346}$$

contradicting (A.345). Equivalently, Proposition A.30 extracts from (A.345) a specific atom satisfying (A.341). Thus (A.344) is impossible, and its negation is exactly (A.343).

There is no small-sample exception. At $n = 1$, every denominator in the two candidate delta caps is positive, and the same comparison applies whenever (A.344) is entered. One active factor suffices because its positive weight preserves the strict tensorized lower bound. If $k = 1$, Proposition A.29 strengthens (A.345) to

$$\mathbb{E}_\star R^\star > \frac{1}{8} > \frac{31}{256} \tag{A.347}$$

with zero overflow. Lemma A.36 and Proposition A.30 then recover the exact one-factor strict PAC failure event. The inactive $k = 1$ branch is excluded by Lemma A.24 under the PAC and contradiction premises. Thus the unrestricted one-factor VC and Littlestone baseline is not weakened to an expectation-only statement. □

Proof of the candidate-wise lower conclusion. At the fixed candidate, Lemma A.24 supplies the universal c_{low}, the strict bound $w_L < 1/128$, and a nonempty active set on the branch $n < c_{\text{low}} M_{\oplus}(C)$. Proposition A.19 verifies every exact factor budget and both candidate delta caps at that same candidate. Lemma A.33 supplies the finite common experiment, full-product targets, legal block mixtures, iid samples, and prior-before-learner order, while Lemma A.35 supplies exact global risk.

Proposition A.28 and Lemma A.37 give

$$\mathbb{E}_\star R^\star > \frac{127}{128}\left(\frac{1}{8} - \frac{3}{2048}\right) = \frac{32131}{262144} > \frac{31}{256}. \tag{A.348}$$

Lemma A.36 gives the opposite ceiling under the alleged universal PAC guarantee, and Proposition A.30 removes the averaging device by producing one deterministic full-product task whose strict failure probability exceeds $1/16$. Proposition A.31 composes these results without switching candidate, learner, risk, or probability space and proves $n \geq c_{\text{low}} M_{\oplus}(C)$. □

A.15 Upper, lower, and conditional characterization

Throughout this subsection, write

$$M = M_{\oplus}(C), \qquad C_{\mathrm{up}} = 65536, \qquad C_{\mathrm{quota}} = \max\left\{1, K_Y + \frac{1}{20}\right\}. \tag{A.349}$$

The first quantity is defined by the setting; the two constants are the universal constants in Propositions A.14 and A.15.

Proposition A.32 (Measurable global upper theorem and public quota bound)**.** *Under Assumptions 1, 2, 3, and 4, together with Lemmas A.5 and A.6 and Propositions A.8, A.9, A.14, and A.15, the following holds without Assumption 5. For every integer*

$$n \geq \lceil C_{\mathrm{up}} Q_{\oplus} \rceil, \tag{A.350}$$

the quotient-first routed rule $\bar{A}_n^{\oplus,Q}$ is a measurable Markov kernel into $(\mathcal{H}^{\oplus}, \mathscr{H}^{\oplus})$, is replacement-$(\varepsilon,\delta)$-DP on all labeled inputs, and, for every $c \in C$ and every probability measure D on (X, Σ), satisfies

$$\Pr_{\substack{S \sim D_c^n \\ \bar{H} \sim \bar{A}_n^{\oplus,Q}(S,\cdot)}} \left[R_D(h_{\bar{H}}, c) \leq \frac{1}{16} \right] \geq \frac{15}{16}. \tag{A.351}$$

Consequently,

$$\mathrm{SC}_{\varepsilon,\delta}(C) \leq \lceil C_{\mathrm{up}} Q_{\oplus} \rceil, \qquad Q_{\oplus} \leq C_{\mathrm{quota}} \frac{M}{\varepsilon} \log^2\left(\frac{eM}{\varepsilon\delta}\right). \tag{A.352}$$

Proof. Proposition A.8 supplies, at every fixed n, the tuple-valued Markov kernel in the statement, the measurable decoder, and measurable fixed-c, D risk events. Proposition A.9 supplies the pointwise all-input privacy inequality for every measurable tuple event and every measurable decoded postprocessing. Neither result uses the candidate delta condition.

Proposition A.14 applies under (A.350) and gives (A.351), jointly over the iid sample and all learner randomness and uniformly over every target and arbitrary-support D. Lemma A.6 identifies this as exact raw distributional risk rather than a quotient surrogate, while Lemma A.5 shows that the decoded tuple is a legal unrestricted learner output.

Apply these conclusions at the integer $\lceil C_{\mathrm{up}} Q_{\oplus} \rceil$. The set of successful sample sizes in the definition of $\mathrm{SC}_{\varepsilon,\delta}(C)$ is nonempty and contains that integer. Its least element exists by well-ordering of $\mathbb{N}$, belongs to the same set, and is no larger, which proves the first inequality in (A.352). Proposition A.15 is exactly the second inequality. This establishes the upper clause for every $0 < \delta < 1$ in Assumption 4, without using either lower delta cap. □

Proposition A.33 (Candidate-wise unrestricted lower theorem)**.** *Under Assumptions 1, 2, 3, and 4, fix one integer $n \geq 1$ satisfying Assumption 5. Under Propositions A.30 and A.31, every unrestricted measurable replacement-(ε,δ)-DP learner A_n that satisfies the universal $(1/16, 1/16)$-PAC guarantee at this candidate obeys*

$$n \geq c_{\mathrm{low}} M. \tag{A.353}$$

On the contradicted branch $n < c_{\mathrm{low}} M$, there is a deterministic full-product target $c \in C$ and a legal block-mixture probability measure D such that

$$D(X_i) = \omega_i \qquad (1 \leq i \leq k) \tag{A.354}$$

and

$$\Pr_{\substack{S \sim D_c^n \\ \Omega \sim A_n(S,\cdot)}} \left[R_D(h_{\Omega}, c) > \frac{1}{16} \right] > \frac{1}{16}. \tag{A.355}$$

Proof. The two conjuncts of Assumption 5 are

$$\delta \le \frac{1}{n \log(n+1)} \tag{A.356}$$

and

$$\delta \le \frac{c_\delta}{m_{n,i}^2 \log(m_{n,i}+1)} \qquad (1 \le i \le k). \tag{A.357}$$

They are checked at the same candidate and its exact budgets; neither is discarded or inferred from the other. Proposition A.31 applies verbatim to the same n, A_n, output convention, and exact risk and yields (A.353). Its learner is not restricted to be proper, factorwise, quotient-output, or computationally bounded.

On the contradiction branch, Proposition A.30, as used by Proposition A.31, removes the finite proof prior and supplies (A.354)–(A.355). The event is strict because the PAC success event is the closed event $R_D \le 1/16$. All candidate quantities remain fixed, so there is no asymptotic or uniform-in-candidate inference. □

Proposition A.34 (Conditional sample-complexity sandwich)**.** *Under Proposition A.32, define*

$$n_* = \mathrm{SC}_{\varepsilon,\delta}(C). \tag{A.358}$$

If

$$0 < \delta \le \frac{1}{n_* \log(n_*+1)} \tag{A.359}$$

and

$$\delta \le \frac{c_\delta}{m_{n_*,i}^2 \log(m_{n_*,i}+1)} \qquad (1 \le i \le k) \tag{A.360}$$

both hold, then Proposition A.33 applies at this integer and

$$c_{\mathrm{low}} M \le n_* \le \lceil C_{\mathrm{up}} Q_\oplus \rceil \le C_{\mathrm{up}} C_{\mathrm{quota}} \frac{M}{\varepsilon} \log^2\left(\frac{eM}{\varepsilon\delta}\right). \tag{A.361}$$

If either (A.359) or any inequality in (A.360) fails, the upper conclusion of Proposition A.32 remains valid, but no lower bound at n_* is asserted.

Proof. Proposition A.32 shows that the defining set for sample complexity is nonempty. Therefore $n_* \in \mathbb{N}$ is finite and attained by an unrestricted private learner satisfying the universal PAC guarantee. Equations (A.359) and (A.360) are exactly Assumption 5 at this attained candidate. Proposition A.33 therefore gives the first inequality in (A.361); the second comes from Proposition A.32.

Every q_i is an integer, so $Q_\oplus \in \mathbb{N}$, and $C_{\mathrm{up}} = 65536$ is an integer. Hence

$$\lceil C_{\mathrm{up}} Q_\oplus \rceil = C_{\mathrm{up}} Q_\oplus. \tag{A.362}$$

Multiplying the quota bound in Proposition A.15 by the positive constant C_{up} gives

$$C_{\mathrm{up}} Q_\oplus \le C_{\mathrm{up}} C_{\mathrm{quota}} \frac{M}{\varepsilon} \log^2\left(\frac{eM}{\varepsilon\delta}\right), \tag{A.363}$$

which proves the last inequality in (A.361) without an untracked ceiling term. The order is essential: identify n_*, check both numerical conditions there, and only then invoke the lower result. Failure of that check cannot affect the independently proved arbitrary-δ upper clause. □

Proposition A.35 (Exact one-factor quotient-first upper reduction)**.** *Under the assumptions and results of Proposition A.32, suppose $k = 1$. Then the routed rule is exactly the measurable quotient-first factor kernel on an unpadded quota-length iid prefix, followed by the same measurable quotient decoder. More precisely,*

$$X_1 = X, \qquad M = s_1, \qquad Q_\oplus = q_1, \qquad J_1(S) = n \quad (S \in Z^n), \tag{A.364}$$

and

$$q_1 = \left\lceil K_Y \frac{M}{\varepsilon} \log^2\left(\frac{eM}{\varepsilon\delta}\right)\right\rceil \leq C_{\mathrm{quota}} \frac{M}{\varepsilon} \log^2\left(\frac{eM}{\varepsilon\delta}\right). \tag{A.365}$$

At every theorem sample size there is zero shortage and zero quotient-to-raw risk residual, and the same learner directly satisfies

$$\Pr\left[R_D(h_{\bar{H}_1}, c) \leq \frac{1}{64}\right] \geq \frac{4095}{4096}. \tag{A.366}$$

Proof. When $k = 1$, the sole block is the whole domain, giving $X_1 = X$ and $J_1(S) = n$ pointwise. The definitions give $M = s_1$, $Q_\oplus = q_1$, and the equality in (A.365). Since $C_{\mathrm{up}} \geq 1$, the threshold $n \geq \lceil C_{\mathrm{up}} q_1 \rceil$ implies $n \geq q_1$. Thus the routed tuple has one coordinate and its input is the first q_1 real quotient records, with no padding. By construction that coordinate law is precisely the factor kernel $\bar{A}_1^{\mathrm{Yan}}$, rather than a new or relaxed learner.

Proposition A.8 gives its measurable tuple form, and Lemma A.5 gives the same quotient decoder. Proposition A.9 uses only the sole factor privacy cost, so there is no cross-factor composition. Proposition A.14 bypasses weighted shortage and the Markov relaxation in this specialization and gives (A.366) directly. Lemma A.6 makes this exact raw risk for every arbitrary-support D, with zero quotient or reference residual. Finally, Proposition A.15 gives the inequality in (A.365). Thus this is the same measurable quotient-first factor learner at the same quota order, with only the setting-required measurable totalization and decoding made explicit. □

Proposition A.36 (Exact one-factor unrestricted lower reduction)**.** *Under Assumption 5 at one fixed candidate and the assumptions and results of Proposition A.33, suppose $k = 1$. Then*

$$\omega_1 = 1, \qquad M = s_1 = 1 + \log^*(d_1 + 1), \qquad m_{n,1} = \max\{8, 4n\}. \tag{A.367}$$

The lower mixture requests exactly n rows from factor one, so

$$n \leq m_{n,1}, \qquad \Pr[\textit{factor-one overflow}] = 0. \tag{A.368}$$

Proposition A.31 therefore yields

$$n \geq c_{\mathrm{low}}\big(1 + \log^*(d_1 + 1)\big), \tag{A.369}$$

and, on the contradicted branch, the exact strict PAC-failure event for the same arbitrary improper learner. Both conditions (A.356)–(A.357) remain in force.

Proof. When $k = 1$, the definitions give $M = s_1$, $\omega_1 = s_1/M = 1$, and $\lceil 4n\omega_1 \rceil = 4n$, proving (A.367). Every one of the n global mixture rows belongs to factor one, so the one-use simulation requests exactly n factor rows. Since

$$m_{n,1} = \max\{8, 4n\} \geq 4n \geq n, \tag{A.370}$$

the overflow event is empty, which proves (A.368) exactly rather than by a tail estimate.

The one-factor part of Proposition A.31 preserves both source branches: bounded iterated-log complexity is covered by the ordinary unrestricted VC floor, while the high-complexity branch uses the unrestricted expected-risk Littlestone interface. That proposition also records that no low-factor mass is lost at weight one and converts the strict expected-risk floor back to the event $R_D > 1/16$. Substituting $M = s_1$ proves (A.369). The learner remains arbitrary, improper, joint, and computationally unrestricted; no factorwise, quotient-coded, finite-output, or efficiency condition is introduced. ☐

Proof of the conditional characterization. Proposition A.32 composes the measurable kernel, all-input privacy, arbitrary-distribution PAC, exact-risk, and public quota interfaces. It proves a finite attained $n_* = \mathrm{SC}_{\varepsilon,\delta}(C)$ for every $0 < \delta < 1$ permitted by Assumption 4, without invoking Assumption 5.

Proposition A.33 separately retains both

$$\delta \le \frac{1}{n \log(n+1)} \quad \text{and} \quad \delta \le \frac{c_\delta}{m_{n,i}^2 \log(m_{n,i}+1)} \quad (1 \le i \le k)$$

at one fixed candidate, and exports the exact unrestricted lower theorem and deterministic strict PAC-failure witness. No claim is made at a candidate where either condition fails. Proposition A.34 is the only point where the scopes meet: it checks both conditions at the attained n_* before applying the lower result. Equations (A.362) and (A.363) give

$$O\left(\frac{M_\oplus(C)}{\varepsilon} \log^2 \frac{eM_\oplus(C)}{\varepsilon\delta}\right)$$

with universal hidden constant $C_{\mathrm{up}}C_{\mathrm{quota}}$. If the candidate check fails, only the lower side is inactive.

Proposition A.35 proves the upper baseline by object identity: one block, one quota, the same quotient-first factor kernel, no padding, exact pullback risk, and the stronger direct factor utility. Proposition A.36 proves the lower baseline by object and budget identity: factor weight one, all n rows routed to it, budget at least $4n$, zero overflow, and the unrestricted VC/Littlestone dichotomy converted to the strict PAC event.

The simultaneous universal constants are $C_{\mathrm{up}} = 65536$, $C_{\mathrm{quota}} = \max\{1, K_Y + 1/20\}$, and $c_{\mathrm{low}} > 0$, independent of the class, factors, candidate, distribution, and privacy parameters. The resulting two-sided theorem remains conditional on the canonical finite product and countable evaluation coding, and the lower sample-complexity comparison remains conditional on its candidate check. It therefore does not assert a characterization for all finite-Littlestone classes. ☐

A.16 Proof of the Main Theorem

Proof of Theorem 3.1. Proposition A.32, with the constants in (A.349), proves that the specified routed rule is a measurable Markov kernel, is private on all labeled inputs, and satisfies (3.1) at every stated sample size. The same proposition proves the first inequality in (3.2), and its application of Proposition A.15 proves the second. That proof uses only Assumptions 1, 2, 3, and 4; in particular, it is valid for every $0 < \delta < 1$ in the theorem.

At a fixed candidate satisfying Assumption 5, Proposition A.33 gives (3.3). Its deterministic witness is exactly (3.4), with block masses $D(X_i) = \omega_i$, and its learner quantifier is the unrestricted measurable one from Section 2.

Proposition A.32 also makes $n_* = \mathrm{SC}_{\varepsilon,\delta}(C)$ finite and attained. Under (3.5) and (3.6), Proposition A.34 applies at that same integer and gives all three inequalities in (3.7). Its exact ceiling calculation (A.362) and rate bridge (A.363) establish the displayed universal constant $C_{\mathrm{up}}C_{\mathrm{quota}}$.

The proposition also proves that failure of either candidate check leaves the upper clause intact and activates no lower conclusion.

Finally, Proposition A.35 proves the object identities, absence of padding, exact risk pullback, and stronger utility (3.8) when $k = 1$. Proposition A.36 proves $\omega_1 = 1$, the exact budget, zero overflow, the rate (3.9), and the strict failure event for the same unrestricted learner. These four cited propositions retain the probability, horizon, exact-risk, and constant-dependence modes stated at the end of the theorem, completing every clause. □